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Prophylactic versus selective use of surfactant for preventing morbidity and mortality in preterm infants at risk of respiratory distress syndrome (Review)

Beijers RJHCG, Alonso-Fernández S, Soll RF, Rojas-Reyes MX, supported by the Cochrane Neonatal Group

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[Intervention Review]

Prophylactic versus selective use of surfactant for preventing morbidity and mortality in preterm infants at risk of respiratory distress syndrome

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ABSTRACT

Background

Respiratory distress syndrome (RDS), or hyaline membrane disease, is a common condition in preterm infants (< 37 weeks' gestation) and a leading cause of neonatal morbidity and mortality. The risk is highest in extremely preterm (< 28 weeks) and very preterm (28 to < 31 weeks) infants due to immature lung and cardiovascular development.

RDS results from a deficiency or dysfunction of pulmonary surfactant, which lines the alveoli to reduce surface tension, prevent atelectasis, and protect the lungs. Surfactant is primarily composed of dipalmitoylphosphatidylcholine (DPPC), other phospholipids, and four proteins that support its function, recycling, and innate lung defense.

Surfactant replacement therapy improves lung compliance, reduces the need for ventilator support, and decreases the risk of pneumothorax, death, and the combined outcome of death or bronchopulmonary dysplasia. Its widespread use has substantially improved survival among preterm infants without increasing long-term neurological or developmental disability.

Various surfactant preparations, including animal-derived, synthetic, and protein or peptide-containing formulations have been evaluated. Surfactant can be administered prophylactically immediately after birth or selectively once RDS develops. Both strategies are effective, with theoretical advantages and disadvantages. Prophylactic surfactant may prevent respiratory insufficiency, reduce the need for ventilator support, and distribute surfactant more evenly in fluid-filled lungs, lowering the risk of lung injury. Selective treatment targets only infants with clinical RDS, avoiding unnecessary therapy, potential risks, and costs for those who would not benefit.

Administration methods include endotracheal tube, intubation with rapid extubation, thin catheter, laryngeal mask, hypopharyngeal deposition, and, more recently, aerosolized or nebulized approaches, though the effectiveness of the latter remains unproven.

For this review, a prophylactic strategy refers to intubation and bolus surfactant administration immediately after birth, while selective therapy refers to administration once evidence of RDS is present. The effect of surfactant may differ in infants stabilized early on continuous positive airway pressure (CPAP) and those whose mothers received a complete course of antenatal corticosteroids. In these infants, the benefits of prophylactic surfactant appear less pronounced than in neonates who did not receive early CPAP or antenatal steroids.

In this update, we explored these factors in subgroup analyses, as in the previous version of the review. Additionally, we examined how the threshold of FiO₂ (fraction of inspired oxygen) used to initiate selective treatment, as well as the method of surfactant administration, might influence the effect of the surfactant replacement strategies.

Objectives

To compare the effect of prophylactic surfactant administration versus selective surfactant administration on morbidity and mortality in preterm infants at risk of respiratory distress syndrome (RDS).

Search methods

We searched CENTRAL, MEDLINE, Embase, and CINAHL on 31 January 2025. To identify any studies not captured by our search of bibliographical databases, we also searched clinical trial registers, conference proceedings, and reference lists of included studies and surfactant reviews.

Selection criteria

We included randomized controlled trials (RCTs) and quasi-RCTs comparing the effects of prophylactic surfactant administration in preterm infants at risk of RDS versus surfactant treatment of preterm infants with established RDS.

Data collection and analysis

We used standard Cochrane methods. Our main outcomes were mortality, neurodevelopmental disability, and complications of preterm birth including pneumothorax and chronic lung disease.

We performed meta-analysis and expressed our results using mean difference (MD), standardized mean difference (SMD), or risk ratio (RR), with 95% confidence intervals (CIs). We used GRADE to assess the certainty of the evidence.

Main results

We identified 10 relevant individually randomized controlled trials (involving 3151 preterm infants). All trials were conducted in North America and Europe. Eight were conducted in the 1990s, and two were published more recently, at a time of greater use of antenatal steroids and nasal continuous positive airway pressure (CPAP). Nine trials were at risk of performance and detection bias.

When all studies are considered, prophylactic surfactant probably results in little to no difference in the risk of chronic lung disease (CLD) at 36 weeks' postmenstrual age compared to selective use of surfactant (RR 1.13, 95% CI 1.00 to 1.28; $I^2 = 0\%$; RD 0.04, 95% CI 0.00 to 0.08; NNTH 25 95% CI 13 to > 1000; $I^2 = 0\%$; 5 trials, 1874 infants; moderate-certainty evidence). There may be little to no difference between the prophylactic and selective approaches in moderate to severe neurodevelopmental impairment (RR 0.83, 95% CI 0.57 to 1.21; 1 trial, 976 infants; I^2 not applicable; low-certainty evidence). Prophylactic surfactant may result in a slight reduction in pneumothorax compared to selective surfactant administration (RR 0.76, 95% CI 0.56 to 1.04; $I^2 = 12\%$; 8 trials, 3094 infants; low-certainty evidence).

In studies in which infants were stabilized on CPAP, the use of prophylactic surfactant, compared with selective surfactant administration, may make little to no difference to the risk of chronic lung disease and probably slightly increases mortality, while the evidence remains uncertain regarding the effect of prophylactic surfactant on all other outcomes, including pneumothorax. In contrast, in settings without routine use of CPAP, prophylactic surfactant reduces mortality and may reduce pneumothorax slightly, with very uncertain evidence for CLD.

In our view, these findings support managing neonates at risk of respiratory distress syndrome with initial stabilization using CPAP and then selective surfactant administration.

We were not able to draw conclusions about our subgroup analyses conducted according to receipt of antenatal steroids, threshold used to apply selective treatment, and method of surfactant administration, which provided evidence of low to very low certainty. There were insufficient data to carry out subgroup analysis by gestational age.

Authors' conclusions

Studies of prophylactic surfactant administration in infants at risk of developing RDS that were conducted prior to widespread use of maternal prenatal steroids and routine early stabilization on CPAP demonstrated a decreased risk of mortality and slight reduction in pneumothorax compared with selective surfactant use in infants with established RDS. However, larger trials that reflect current neonatal care practices do not support this finding. Instead, they show that, compared with selective surfactant administration in infants stabilized on CPAP, prophylactic surfactant likely results in little to no difference in the risk of chronic lung disease and slightly increases mortality.

PLAIN LANGUAGE SUMMARY

For babies born early, is it better to give surfactant (a lung coating) preventively to babies at risk of breathing difficulties or only to those babies who show signs of breathing difficulty?

Key messages

- Current care for preterm infants (babies born early) who are at risk of respiratory distress syndrome (lung problems causing breathing difficulties) usually involves an intervention called continuous positive airway pressure (CPAP). In these infants, giving surfactant (a coating for the lungs to make breathing easier) as a preventive measure may make little or no difference to the risk of chronic lung disease (needing long-term oxygen support), and probably slightly increases the risk of death, compared with giving surfactant only to those babies who actually develop breathing problems.
- In contrast, for preterm infants not receiving CPAP, giving surfactant preventively reduces the risk of death and may slightly reduce the risk of pneumothorax (lung injury), compared to giving surfactant only if and when breathing problems develop.
- In our view, giving all preterm infants CPAP at birth and then giving surfactant only to those who have symptoms of breathing problems may be the best approach to managing the risk of respiratory distress syndrome.

What is a preterm infant?

A preterm infant is a baby who is born early, having spent less than 37 weeks in the womb.

What is surfactant?

Surfactant is a mixture of fats and proteins produced by the body that coats the inside of the alveoli (tiny air sacs in the lungs) and helps the lungs work properly. Some preterm infants are born before their lungs produce enough surfactant, which makes it difficult for their lungs to stay open and get enough oxygen. This is called respiratory distress syndrome (RDS). Surfactant replacement therapy can be used to replace the missing surfactant.

What did we want to find out?

We wanted to know whether to give surfactant to preterm babies, preventively, that is, immediately after birth before any breathing problems occur ('prophylactic surfactant'), or only if the baby has symptoms ('selective surfactant').

Studies conducted 30 years ago suggested that preterm infants who were intubated (breathing tube placed in the airway, down the throat, and into the lungs) and were treated with prophylactic surfactant (before breathing problems occurred) had fewer RDS complications such as chronic lung disease or pneumothorax.

Newer studies, however, suggest that a better treatment approach may be using breathing support (i.e. continuous positive airway pressure (CPAP)) through a mask or nose prongs placed on the baby's face, and applying surfactant only if breathing problems develop. This avoids putting tubes into the windpipe and giving medicines that may be unnecessary.

What did we do?

We searched for studies that tested the prophylactic versus selective use of surfactant. We combined results for infant mortality (death) within the first 28 days of life (the neonatal period) and the development of more severe lung disease that may require long-term oxygen supplementation (chronic lung disease). We looked at whether results varied if preterm babies were given CPAP to support their breathing at birth or if their mothers had received medication, such as steroids, which help the baby's lungs mature during the prenatal period.

What did we find?

We identified 10 relevant studies that involved 3151 babies.

When considering all studies together, prophylactic surfactant probably results in little to no difference in the risk of chronic lung disease (CLD) compared to selective use of surfactant (5 studies, 1874 babies) and may slightly reduce the risk of pneumothorax (8 studies, 3094 babies). It may make little to no difference to the risk of moderate-to-severe neurodevelopmental impairment (problems with brain development) (1 study, 976 babies).

Studies where infants received CPAP showed that prophylactic use of surfactant probably slightly increases the risk of mortality compared to its selective use as a treatment once RDS symptoms have developed, and it may make little to no difference to the risk of CLD. The evidence for pneumothorax is very uncertain.

In contrast, studies where infants did not receive CPAP showed that prophylactic use of surfactant reduces mortality risk compared with giving surfactant selectively after RDS develops, and it may reduce the risk of pneumothorax slightly. The evidence for CLD is very uncertain.

In our view, the review findings favor early stabilization of breathing with CPAP for all preterm infants at risk of developing RDS and then selective use of surfactant for preterm infants showing RDS symptoms.

What are the limitations of the evidence?

People running the studies and families of babies taking part may have known which treatment was being given, which may have influenced measurements.

Not all studies provided information about everything of interest. For example, some studies did not measure longer-term infant development.

The studies did not provide enough reliable information for us to understand the effects of the age of the babies (born before 28 weeks or closer to their due date), mothers receiving steroids before giving birth, or different methods of giving the surfactant.

How up to date is this evidence?

This review updates a previous 2012 Cochrane review. The evidence is current to 31 January 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS

Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS

Patient or population: preterm infants at risk of RDS

Setting: single - and multi-center randomized controlled trials conducted in the USA, Italy, Canada, France, The Netherlands, Switzerland, Finland, Bulgaria and Romania

Intervention: prophylactic surfactant administration

Comparison: selective surfactant administration

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with selective surfactant administration	Risk with prophylactic surfactant administration				
Chronic lung disease in all infants (defined by need of supplemental oxygen at 36 weeks' postmenstrual age)	319 per 1000	361 per 1000 (319 to 408)	RR 1.13 (1.00 to 1.28)	1874 (5 RCTs)	⊕⊕⊕⊖ Moderate ^a	
- Chronic lung disease in subgroup 1: studies without routine application of CPAP	51 per 1000	80 per 1000 (37 to 174)	RR 1.56 (0.72 to 3.40)	362 (3 RCTs)	⊕⊖⊖⊖ Very low ^{a,b}	
- Chronic lung disease in subgroup 2: studies with routine application of CPAP	379 per 1000	425 per 1000 (376 to 478)	RR 1.12 (0.99 to 1.26)	1512 (2 RCTs)	⊕⊕⊖⊖ Low ^{a,c}	
- Mortality at 36 weeks post menstrual age or prior to hospital discharge (any cause) in subgroup 1: studies without routine application of CPAP	227 per 1000	164 per 1000 (127 to 212)	RR 0.72 (0.56 to 0.93)	1030 (4 RCTs)	⊕⊕⊕⊕ High	
- Mortality at 36 weeks post menstrual age or prior to hospital discharge (any cause) in subgroup 2: studies with routine application of CPAP	116 per 1000	149 per 1000 (117 to 188)	RR 1.28 (1.01 to 1.62)	1748 (2 RCTs)	⊕⊕⊕⊖ Moderate ^c	
Moderate-severe neurodevelopmental impairment (defined as any one of the following: cerebral palsy or gross motor disability; developmental delay; visual impairment; hearing impairment at 18 to 22 months)	109 per 1000	91 per 1000 (62 to 132)	RR 0.83 (0.57 to 1.21)	976 (1 RCT)	⊕⊕⊖⊖ Low ^{a,b}	

Pneumothorax in all infants	55 per 1000	42 per 1000 (31 to 57)	RR 0.76 (0.56 to 1.04)	3094 (8 RCTs)	⊕⊕○○ Low ^{a,d}
- Pneumothorax in subgroup 1: studies without routine application of CPAP	9 per 1000	7 per 1000 (3 to 10)	RR 0.75 (0.35 to 1.05)	2663 (7 RCTs)	⊕⊕○○ Low ^{a,d}
- Pneumothorax in subgroup 2: studies with routine application of CPAP	340 per 1000	302 per 1000 (132 to 682)	RR 0.89 (0.39 to 2.01)	431 (1 RCT)	⊕○○○ Very low ^{a,e}

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_447217643826257849.

^a Downgraded one level due to serious risk of bias: no blinding of participants and personnel and no/unclear blinding in outcome assessment in all of the studies

^b Downgraded two levels due to imprecision: 95% CI is consistent with the possibility for no effect including important benefit and harm, estimation is based on fewer than 300 events

^c Downgraded one level due to imprecision: lower limit of 95% CI is close to null effect, estimation is based on fewer than 300 events.

^d Downgraded one level due to imprecision: upper limit of 95% CI is close to null effect, estimation is based on fewer than 300 events.

^e Downgraded two levels due to imprecision: 95% CI is consistent with the possibility of no effect including important benefit and harm. Information based on a single trial with fewer than 300 events.

BACKGROUND

Description of the condition

Respiratory distress syndrome (RDS), also known as hyaline membrane disease, is a common respiratory condition affecting preterm infants (defined as neonates born before 37 weeks' gestation) and continues to be a leading cause of neonatal morbidity and mortality worldwide (Jacobs 2015). Preterm infants are categorized according to gestational age as extremely preterm (< 28 weeks), very preterm (28 to < 31 weeks), moderate (32 to < 33 weeks) and late preterm (34 to < 37 weeks). Respiratory distress syndrome is associated with a deficiency or dysfunction of pulmonary surfactant; therefore, it occurs predominantly in very preterm infants, who have immature lung structures, an immature cardiovascular system, and a highly compliant chest wall (Avery 1959; Wright 1987).

The pulmonary surfactant is predominantly dipalmitoylphosphatidylcholine (DPPC), with lesser amounts of other phospholipids, including phosphatidylglycerol (PG), phosphatidylethanolamine, and phosphatidylinositol. In addition, pulmonary surfactant also contains neutral lipids and four distinct surfactant proteins (SP-A, SP-B, SP-C, and SP-D). These proteins play an important role in surfactant secretion, its recycling, and cooperative functioning with other surfactant proteins and phospholipids (Possmayer 1990; Schürch 1992), and innate host defense of the lung (Weaver 1991; Wright 1997). The physiologic functions of a surfactant include the ability to lower surface tension and the ability to rapidly adsorb, spread, and reform a monolayer in the dynamic conditions associated with the respiratory cycle (Jobe 1993). Surfactant lines the alveolar surface and prevents atelectasis at end-expiration and a potential lung injury.

Description of the intervention

Surfactant replacement therapy has been shown to improve lung compliance, reduce the need for respiratory support, and improve clinical outcomes in very preterm newborns (Jobe 1993; Soll 1992).

Clinical trials have evaluated a wide range of surfactant preparations, including animal-derived products, synthetic formulations, and peptide- or protein-containing surfactants. These preparations are administered either to prevent respiratory distress syndrome, through prophylactic or delivery-room strategies, or to treat it, using very early selective or rescue approaches (Ardell 2015; Pfister 2007; Pfister 2009; Seger 2009; Soll 1997; Soll 1998; Soll 2010).

The surfactant could be administered via endotracheal tube, intubation, surfactant administration and rapid extubation (InSure), thin catheter administration, laryngeal mask airway, and hypopharyngeal deposition. More recent proposals include aerosolized or nebulized surfactants, but these approaches have not yet been proven to be effective methods of surfactant administration (Fortas 2022).

Using either approach to treatment, surfactant therapy has been proven to reduce the need for ventilator support and decrease the risk of pneumothorax, death, and the combined outcome of death or bronchopulmonary dysplasia (Engle 2008; Polin 2014). However, the relative risk of bronchopulmonary dysplasia in preterm infants of less than 30 weeks' gestation who received

prophylactic surfactant compared with those who received later selective treatment is unclear (Rojas-Reyes 2012; Soll 2001).

The widespread use of surfactant therapy has been credited with significant improvements in survival among preterm infants (Horbar 1993; Schwartz 1994), without evidence of an increased incidence of neurological or developmental disability in children born preterm who were followed through infancy and school age (D'Angio 2002; Palta 2000).

For the purpose of this review, we defined a "prophylactic strategy" for surfactant administration as a strategy that mandated intubation and bolus administration of surfactant immediately after birth (either prior to the first breath or after a brief period of resuscitation). "Selective" surfactant therapy was defined as administration of surfactant once the infant had evidence of RDS, also known in the literature as "treatment" or "rescue" surfactant.

How the intervention might work

Although both prophylactic surfactant administration and selective surfactant treatment of infants with established RDS are successful treatment strategies, there are theoretical advantages and disadvantages to each approach. Prophylactic administration offers the theoretical advantage of replacing a surfactant before the onset of respiratory insufficiency, decreasing the need for ventilator support and avoiding barotrauma that may result from even short periods of assisted ventilation (Nilsson 1978). Surfactant may be distributed more homogeneously when given immediately at birth into lungs still filled with fluid, leading to improvement in response and decreasing the risk of lung injury (Jobe 1984). Reserving surfactant treatment for infants with established RDS offers the advantage of treating only those with clinical disease, thereby avoiding the potential risks and costs of treating infants who would not benefit from therapy because they do not have surfactant deficiency.

Potential differences in the intervention effect of surfactant administration have been reported among infants who were stabilized early on continuous positive airway pressure (CPAP) after birth, as well as among those whose mothers received a complete course of antenatal corticosteroids (Rojas-Reyes 2012). In these patients, the benefits of prophylactic surfactant administration on pulmonary outcomes appear to be less clear than in neonates who did not receive CPAP or were not exposed to antenatal corticosteroids.

Why it is important to do this review

Meta-analyses of the original randomized controlled trials of surfactants for the treatment and prevention of RDS were first published in *Effective Care of the Newborn* (Soll 1992). Since then, multiple systematic reviews have been published in the *Cochrane Database of Systematic Reviews*, including reviews of protein-free synthetic surfactants for the prevention and treatment of RDS (Soll 1998; Soll 2010), reviews of animal-derived surfactants for the prevention and treatment of RDS (Seger 2009; Soll 1997), and reviews that compare different treatment strategies (Bahadue 2012; Rojas-Reyes 2012; Soll 2009; Stevens 2007). This is an update of a Cochrane review first published in 2001 (Soll 2001) and previously updated in 2012 (Rojas-Reyes 2012).

Since the original trials of the 1990s, the increased utilization of antenatal steroids, more gentle resuscitation in the delivery room,

and the routine use of early delivery room continuous positive airway pressure (CPAP) may have changed the balance of risks and benefits of using aggressive universal prophylactic intubation and treatment of infants at high risk of RDS (Finer 2004; Gittermann 1997; Lindner 1999; Rojas 2009; Sahni 1998).

In our initial review (Soll 2001), we reported that prophylactic administration of surfactant decreases the risk of air leak and neonatal mortality in the absence of routine application of CPAP. However, in the updated review (Rojas-Reyes 2012), which included more recent studies, we found a reduction in the risk of chronic lung disease or death in infants who were stabilized on CPAP and only treated with surfactant selectively if symptoms worsened, when compared to prophylactic use of surfactant. When combined with the studies from the pre-CPAP era, the beneficial effects seen in the original trials were no longer demonstrated (Rojas-Reyes 2012).

Given the potential differences in the intervention effect of surfactant administration among infants who were stabilized early on continuous positive airway pressure (CPAP) after birth, as well as among those whose mothers received a complete course of antenatal corticosteroids, we explored the influence of these factors in subgroup analyses, as in the previous version of this review. In this update, we additionally examined the potential impact of the threshold used to initiate selective treatment and the method of surfactant administration on the intervention effect.

OBJECTIVES

To compare the effect of prophylactic surfactant administration versus selective surfactant administration on morbidity and mortality in preterm infants at risk of respiratory distress syndrome (RDS).

METHODS

Criteria for considering studies for this review

Types of studies

We included randomized controlled trials (RCTs), quasi-RCTs, and cluster trials comparing prophylactic administration of surfactant to preterm infants at risk for respiratory distress syndrome (RDS) (i.e. surfactant given prior to the first breath or immediately after stabilization in the delivery room) versus selective surfactant treatment of preterm infants with established RDS.

We did not include trials of cross-over design trials because they are not suitable for addressing this clinical question.

Types of participants

We included preterm infants who were at risk of RDS. We defined preterm infants as infants born before 37 completed weeks' gestation or with low birth weight (less than 2500 g).

Gestational age

Gestational age is defined as the duration of pregnancy measured in weeks and days from the first day of the mother's last menstrual period to the time of birth. According to the gestational age at birth, preterm infants are classified as follows.

- Late preterm infants – defined as infants 34⁰ to 37⁶ weeks gestational age

- Moderate preterm infants – defined as infants born at 32⁰ to 33⁶ weeks gestational age
- Very preterm infants – defined as infants born at 28⁰ to 31⁶ weeks gestational age
- Extremely preterm infants – defined as infants born at less than 28⁰ weeks gestational age

Birth weight

Given that studies in a variety of settings may not have reliable gestational age data and may therefore use birth-weight categories, we adopted the following widely accepted categorization for inclusion:

- Low birth weight (LBW) infants – defined as infants with birth weight < 2500 g
- Very low birth weight (VLBW) infants – defined as infants with birth weight < 1500 g
- Extremely low birth weight (ELBW) infants – defined as infants with birth weight < 1000 g

Infants with birth weights 2000 g to 2500 g were grouped with the late preterm infants; infants 1500 to 2000 g were grouped with the moderate preterm infants; infants 1000 to 1499 g were grouped with the very preterm infants; and infants less than 1000 g were grouped with the extremely preterm infants.

For those studies that reported on a broad population of infants (including both preterm infants < 34 weeks' gestation, late preterm infants, and term infants), we included studies that had a preponderance of preterm infants (defined as having a mean gestational age of < 34 weeks or reporting greater than 80% of infants as preterm < 34 weeks gestational age).

Types of interventions

1. Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP)
2. Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by gestational age)
3. Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by receipt of antenatal steroids)
4. Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by the threshold used to apply selective treatment)
5. Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by the method of surfactant administration)

Any surfactant product was eligible (synthetic surfactant, protein-containing synthetic surfactant, animal-derived surfactant). The control group could be managed with or without the early application of CPAP.

Types of outcome measures

Our outcomes of interest are listed below. We included studies irrespective of whether they reported these outcomes. For health outcomes measured during the neonatal period, we used postmenstrual age (PMA) to express the timing of outcome

assessment. PMA is defined as gestational age plus chronological age after birth, usually expressed in weeks.

Primary outcomes

1. Chronic lung disease (use of supplemental oxygen at 36 weeks' PMA) (Shennan 1988)
2. Mortality at 36 weeks' PMA or prior to hospital discharge (from any cause)
3. Moderate to severe neurodevelopmental impairment as reported by study authors, using validated tools, including the following domains: cerebral palsy or gross motor disability; developmental delay more than two standard deviations (SDs) below population mean on a standardized test of development; visual impairment (visual acuity less than 6/60); hearing impairment (requiring amplification using validated tools). We will perform separate analyses for children aged 18 to 24 months and three to five years.
4. Pneumothorax diagnosed by radiological findings (as reported by authors)

Secondary outcomes

1. Pulmonary interstitial emphysema diagnosed by radiological findings (as reported by authors)
2. Any air leak syndrome by radiological findings (as reported by authors)
3. Any pulmonary hemorrhage (as reported by authors)
4. Bronchopulmonary dysplasia at 28 to 30 days of age or use of supplemental oxygen at 28 to 30 days of age
5. Bronchopulmonary dysplasia or death prior to 28 to 30 days of age
6. Chronic lung disease or death (use of supplemental oxygen or death at 36 weeks' PMA)
7. Neonatal mortality (mortality < 28 days of age) from any cause
8. Patent ductus arteriosus (PDA) requiring treatment (cyclo-oxygenase inhibitors or surgical ligation) diagnosed clinically and/or echographically
9. Any culture-proven bacterial sepsis (as reported by authors)
10. Any culture-proven fungal sepsis (as reported by authors)
11. Necrotizing enterocolitis (defined as Bell Stage II or greater) (Bell 1978)
12. Intraventricular hemorrhage on brain ultrasound (IVH any grade) (Papile 1978)
13. Severe intraventricular hemorrhage (and severe IVH grade 3 to 4) (Papile 1978)
14. Periventricular leukomalacia (De Vries 1992)
15. Retinopathy of prematurity (all stages), (International Classification for Retinopathy of Prematurity (ICROP) Stage I to V) (International Committee ROP 2005)
16. Severe retinopathy of prematurity (ICROP stage 2 or greater) (International Committee ROP 2005)

See [Appendix 1](#) for definitions of preterm infants and other definitions related to outcomes and control variables.

Search methods for identification of studies

Electronic searches

Search strategies were written and conducted by Information Specialist Ivan Solá from the Iberoamerican Cochrane Center from 2011 to January 2023. As he was unavailable to update searches in 2024, Michelle Fiander translated the original strategies written for PubMed and Embase.com, to OVID syntax; and the Cochrane Library strategy to Cochrane Register of Studies (CRS) syntax. Given these changes to the strategies, we ran the searches in 2024 and 2025 from 2010 onwards and deduplicated the results against the records retrieved in the 2023 search. We ran searches without limits by publication type or publication status.

We searched the following sources on 31 January 2025.

- Cochrane Central Register of Controlled Trials (CENTRAL) via the Cochrane Register of Studies
- Ovid MEDLINE, 1946 to 30 January 2025
- Ovid EMBASE 1974 to 30 January 2025
- CINAHL Complete, EbscoHost, 1986 to 30 January 2025
- National Library of Medicine (<https://clinicaltrials.gov/>)
- World Health Organization, International Clinical Trials Registry Platform (<https://trialsearch.who.int>)

See [Appendix 2](#) for full search strategies for each database.

Searching other resources

We checked the reference lists of included studies and of other systematic reviews investigating surfactant use. We searched for retractions and errata for included studies using The Retraction Watch Database (<https://retractiondatabase.org/>) and PubMed.

Data collection and analysis

We identified and included a study by a review author (RS) (Dunn 2011). Two review authors (RB and SA or MXR and RB) independently undertook the following for this study: screening and selection, data extraction, risk of bias assessment, and certainty of the evidence assessments for meta-analyses to which the study contributed.

If we encountered incomplete information in a study reference or report, we planned to contact study authors to request additional information.

Selection of studies

Our selection of studies was guided by our inclusion criteria, which are outlined in [Criteria for considering studies for this review](#).

Two review authors (RB and SA) independently screened search results, and then the full texts of references that we retained after title/abstract review. Disagreements were resolved by discussion between them. We included studies measuring outcomes of interest in this review, even if they did not report data for them.

We documented the reasons for excluding studies during the full-text review in [Characteristics of excluded studies](#). We provided any information we were able to obtain about ongoing studies in [Characteristics of ongoing studies](#). We described studies with insufficient information to allow data extraction in [Characteristics of studies awaiting classification](#). We will assess any study awaiting

assessment for inclusion in the next update of this review if data become available. We grouped multiple reports of the same study under a single study ID so that each study, rather than each report,

is the unit of interest in the review. We recorded the selection process in sufficient detail to complete a PRISMA flow diagram ([Page 2021](#); [Page 2021a](#)). See [Figure 1](#).

Figure 1. PRISMA figure - process of study selection

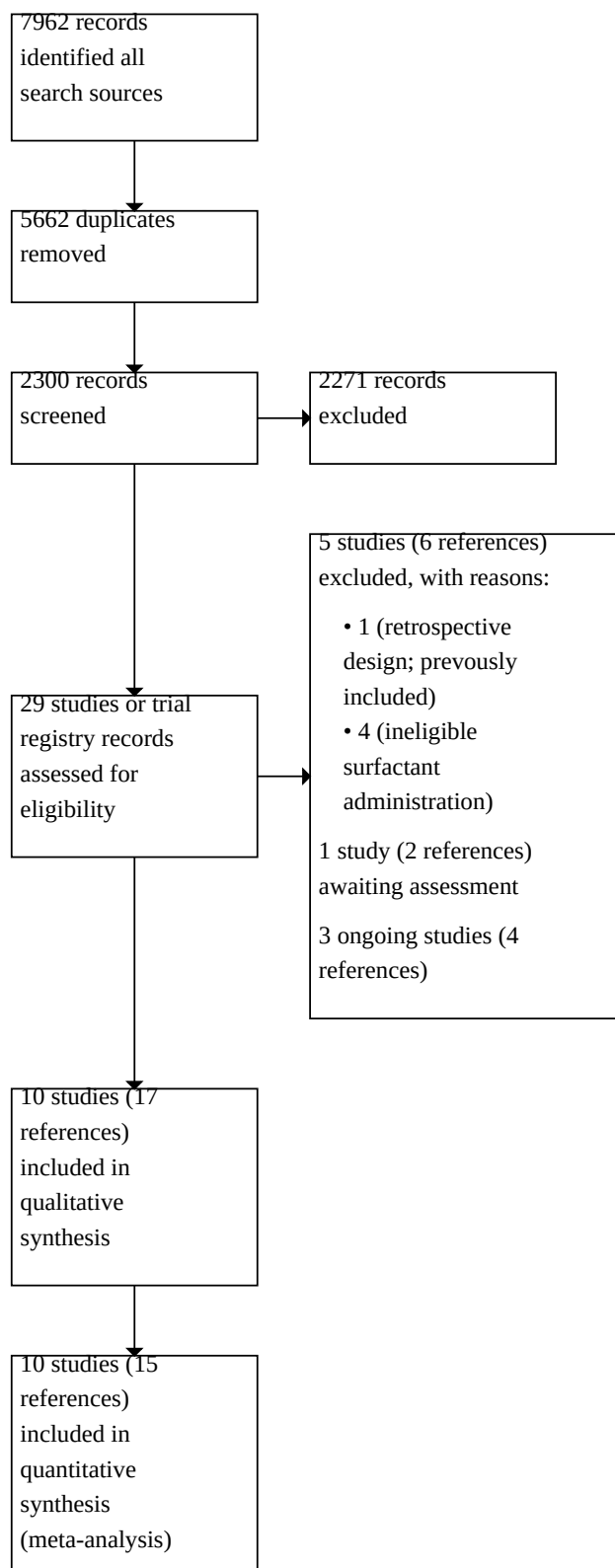


Figure 1. (Continued)

(meta-analysis)

Data extraction and management

Two review authors (RB and SA) independently extracted data, resolving any discrepancies through discussion and, when necessary, consulting a third review author (RS or MXR). The extracted data were subsequently entered into Cochrane Review Manager software (RevMan) (RevMan 2025). We extracted data using a form developed for this review, piloted on a sample of three studies. Extracted information included administrative details (study contacts, declarations of interest, funding, location, and single- or multi-center status); study methodology (trial design, participant selection criteria, randomization methods, interventions and comparators, and outcome measures); data for synthesis (population characteristics such as gestational age at randomization, number of participants per study arm, time from birth to surfactant administration, number of events, and withdrawals per arm); and all information necessary to perform the planned subgroup analyses. If questions arose regarding the data or methodology in a study, we planned to contact study authors for additional information.

We reassessed data from previously included studies to identify data for the new subgroup analyses added in this update. For the study involving review author RS (Dunn 2011), data were checked by review authors RB, SA, and MXR.

Assessment of risk of bias in included studies

Two review authors (RB and SA) independently reassessed the risk of bias (low, high, or unclear) in all included trials. We used the Cochrane risk of bias tool RoB 1 for the following domains (Higgins 2011).

- Sequence generation (selection bias)
- Allocation concealment (selection bias)
- Blinding of participants and personnel (performance bias)
- Blinding of outcome assessment:
 - (detection bias): hospital outcomes
 - (detection bias): follow-up outcomes
- Incomplete outcome data (attrition bias)
- Selective reporting (reporting bias)
- Any other bias

We resolved disagreements by discussion or by consulting a third review author (MXR). See Appendix 3 for a more detailed description of the risk of bias for each domain.

Measures of treatment effect

We performed statistical analyses using RevMan (RevMan 2025). For categorical data, we used the risk ratio (RR) and risk difference (RD). For statistically significant outcomes, we calculated the number needed to treat for an additional beneficial outcome (NNTB) or the number needed to treat for an additional harmful outcome (NNTH). For continuous data, we calculated mean differences (MDs)

between treatment groups. We reported 95% confidence intervals (CIs) for all outcomes.

Unit of analysis issues

The unit of analysis was the participating infant in individually randomized trials, and an infant was considered only once in the analysis.

Cluster-randomized trials

Had we identified cluster-RCTs, we would have considered the participating neonatal unit or section of a neonatal unit or hospital as the unit of analysis. We would have analyzed them using an estimate of the intracluster correlation coefficient (ICC) derived from the trial (if possible), or from a similar trial, or from a study with a similar population, as described in Section 16.3.6 of the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins 2024). If we used ICCs from a similar trial or from a study with a similar population, we would have reported this and conducted sensitivity analysis to investigate the effect of variation in the ICC.

Had we identified both cluster-randomized trials and individually randomized trials, we would have combined the results from both if there was little heterogeneity between the study designs, and the interaction between the effect of the intervention and the choice of randomization unit was considered to be unlikely (Higgins 2024).

Other unit of analysis issues

Other unit-of-analysis issues included multiple pregnancies or the inclusion of more than two treatment groups. We consulted a statistical expert for support when necessary.

Dealing with missing data

For the included studies, we noted levels of attrition in the [Characteristics of included studies](#) tables.

We performed the analysis according to the intention-to-treat principle, analyzing all available data for participants in the groups to which they were originally randomized, regardless of whether they received the allocated intervention. If, in the original reports, participants were not analyzed in the group to which they were randomized, and there was sufficient information in the trial report, we attempted to restore them to the correct group.

We used the number of women randomized as the denominator for maternal health outcomes and the number of infants randomized as the denominator for infant health outcomes.

When data were missing and could not be derived, we handled them as follows. We imputed missing standard deviations (SDs) where possible using the coefficient of variation (CV) or calculated from other available statistics (standard errors, confidence intervals, t values, or P values). We analyzed data assumed to be missing at random without imputation. If this assumption could not be made, we imputed missing outcomes

with replacement values, assuming all had a poor outcome, and we conducted sensitivity analyses to assess any impact on the direction or magnitude of effect.

Assessment of heterogeneity

We estimated the treatment effects of individual trials and examined heterogeneity among trials by inspecting the forest plots and quantifying the impact of heterogeneity using the I^2 statistic. We graded the degree of heterogeneity as follows.

1. Less than 25%: no heterogeneity
2. 25% to 49%: low heterogeneity
3. 50% to 75%: moderate heterogeneity
4. More than 75%: substantial heterogeneity

If we noted moderate statistical heterogeneity ($I^2 > 50\%$), we explored the possible causes (e.g. differences in study quality, participants, intervention regimens, or outcome assessments).

Assessment of reporting biases

We assessed reporting bias by comparing each study's stated primary and secondary outcomes with the reported outcomes. Where study protocols were available, we compared these to the full publications to determine the likelihood of reporting bias. Studies using interventions in a potentially eligible infant population but not reporting on any of the primary and secondary outcomes were documented in the [Characteristics of included studies](#) table. We used funnel plots to screen for publication bias where there were sufficient studies (> 10) reporting the same outcome. If publication bias was suggested by our visual assessment of significant asymmetry in the funnel plot, we incorporated this in our assessment of the certainty of evidence.

Data synthesis

If we identified multiple studies that we considered to be sufficiently similar, we performed meta-analysis using RevMan ([RevMan 2025](#)). As described in [Measures of treatment effect](#), for categorical outcomes, we calculated the typical estimates of RR and RD, and for continuous outcomes, we calculated the MD, each with its 95% CI.

For continuous data, where outcomes were measured differently in different studies, we reported data as standardized mean differences (SMD) with 95% confidence intervals (CIs).

We used the fixed-effect model for combining data where it was reasonable to assume that studies were estimating the same underlying treatment effect, that is, where trials were examining the same intervention, and the trials' populations and methods were sufficiently similar. When there was clinical heterogeneity sufficient to expect that the underlying treatment effects differed between trials, or when moderate statistical heterogeneity was detected ($I^2 \geq 50\%$), we used the random-effects model to produce an overall summary if we considered an average treatment effect across trials to be clinically meaningful. We treated the random-effects summary as the average of the range of possible treatment effects, and we discussed the clinical implications of treatment effects differing between trials.

For the assessment of the magnitude of effect, as a rough guide, we considered relative risks (RRs) lower than 0.75 or greater than

1.25 to be indicative of a clinically important effect for dichotomous outcomes.

For continuous outcomes, we interpreted the magnitude of effect according to Cohen's criteria for standardized mean differences (SMDs), where an effect size of approximately 0.2 was considered small, 0.5 moderate, and 0.8 or greater large. Values below 0.2 were considered trivial or not clinically important.

If we judged meta-analysis to be inappropriate, we analyzed and interpreted individual trials separately. If there was evidence of clinical heterogeneity, we tried to explain this based on the different study characteristics and subgroup analyses.

Subgroup analysis and investigation of heterogeneity

Given the potential differences in the intervention effect related to prophylactic surfactant administration compared to selective surfactant administration that we discuss in the [Background](#) section of this review, we investigated the following subgroups to explore the influence of particular factors on the intervention's effects. If heterogeneity in the pooled effect estimates was absent or low (i.e. $\leq 49\%$), we considered it unnecessary to perform subgroup analysis. When subgroup comparisons were possible, we conducted stratified meta-analysis and a formal statistical test for interaction to examine subgroup differences that could account for effect heterogeneity (e.g. Cochran's Q test) ([Borenstein 2013](#); [Higgins 2024](#)).

1. Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by the use of CPAP)
 - a. Studies that did not routinely place control infants on CPAP (without routine application of CPAP or non-invasive respiratory support)
 - b. Studies that routinely placed control infants on CPAP (with the routine application of CPAP or non-invasive respiratory support)
2. Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by gestational age)
 - a. Studies reporting on infants ≤ 28 weeks gestation
 - b. Studies reporting on infants > 28 weeks gestation
3. Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by receipt of antenatal steroids)
 - a. Studies in which $< 50\%$ of mothers received antenatal steroids
 - b. Studies in which $\geq 50\%$ of mothers received antenatal steroids
4. Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by the threshold used to apply selective treatment)
 - a. Studies that treated infants in the selective group at $FiO_2 > 0.3$
 - b. Studies that treated infants in the selective group at $FiO_2 > 0.4$
 - c. Studies that treated infants in the selective group at $FiO_2 > 0.5$
 - d. Studies that treated infants in the selective group at $FiO_2 > 0.6$
 - e. Studies that treated infants in the selective group based on other indications

5. Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by the method of surfactant administration)
 - a. Studies that treated infants with intubation, surfactant administration and ongoing ventilation
 - b. Studies that treated infants with intubation, surfactant administration and rapid extubation
 - c. Studies that treated infants with thin catheter, surfactant administration
 - d. Studies that treated infants with laryngeal mask, surfactant administration
 - e. Studies that treated infants with hypopharyngeal, surfactant administration

When we found fewer than five studies per category, we interpreted tests for subgroup differences in effects with caution, given the potential of confounding.

Sensitivity analysis

We performed sensitivity analysis in the following situations.

- There was unexplained moderate to high heterogeneity in a meta-analysis. We explored this by removing the outlying study/studies causing heterogeneity (if feasible) and comparing the result.
- A study with a high risk of (material) bias was included in a meta-analysis with studies at low risk of bias. We removed the high-risk study (or studies) to compare the result.

Summary of findings and assessment of the certainty of the evidence

We used the GRADE approach, as outlined in the GRADE Handbook (Schünemann 2013), to assess the certainty of the evidence for the main comparison (prophylactic surfactant compared to selective surfactant for preventing or treating RDS in preterm infants) and the following four critical outcomes.

1. Chronic lung disease (use of supplemental oxygen at 36 weeks' postmenstrual age (PMA)) (Shennan 1988)
2. Mortality prior to hospital discharge (from any cause)
3. Moderate-severe neurodevelopmental impairment as reported by authors (using validated tools).
4. Pneumothorax

Two review authors (RB and MXR) independently assessed the certainty of the evidence found for each of these outcomes to result in one of four grades, as follows.

- High: we are very confident that the true effect lies close to that of the estimate of the effect.
- Moderate: we are moderately confident in the effect estimate. The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
- Low: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
- Very low: we have very little confidence in the effect estimate. The true effect is likely to be substantially different from the estimate of the effect.

We considered evidence from randomized controlled trials (RCTs) to be of high certainty and downgraded the certainty of the evidence by one level for serious concerns or by two levels for very serious concerns based on the following domains: risk of bias, inconsistency of results (substantial heterogeneity across studies), indirectness of the evidence (similarity of the population, intervention, comparator, and outcomes [PICO] to the review question), imprecision of effect estimates (95% confidence intervals and optimal information size), and publication bias. Our assessment of imprecision considered the magnitude of the effect according to the type of outcome measure, as described in [Measures of treatment effect](#). We considered estimates based on fewer than 300 outcome events as potentially imprecise because the available information may be insufficient to provide a stable and reliable estimate of effect. This threshold is consistent with the concept of the optimal information size, whereby a small number of events increases the likelihood of random error and wide confidence intervals.

We used [GRADEpro GDT](#) to create a summary of findings table to report the key results and our certainty about them. We justified all decisions to downgrade the certainty of the evidence using footnotes and made comments to aid the reader's understanding of the review where necessary.

Even though no statistically significant interaction effect was observed, we presented the CPAP subgroup analyses in the summary of findings table because of our particular interest in illustrating the effects of the same intervention strategies in these clinically relevant populations.

RESULTS

Description of studies

See [Characteristics of included studies](#); [Characteristics of excluded studies](#); [Characteristics of studies awaiting classification](#); [Characteristics of ongoing studies](#)

Results of the search

The searches identified 7962 records. After removal of 5662 duplicates, 2300 records remained for screening. We excluded 2271 records based on title and abstract. We assessed 29 full-text articles or trial registry records for eligibility. Of these, 11 studies (16 references) were already included in the previous version of the review. One additional reference was identified as a new follow-up report of an already included study (for a total of 17 references). We excluded five studies (six references), classified one study (two references) as awaiting assessment, and identified three ongoing studies (four references). Details of the study selection process are provided in [Figure 1](#).

We excluded one study that had been included in the previous version of the review following reassessment of its design (Iarŭkova 1999). We determined that this was a retrospective study comparing prophylactic versus rescue surfactant (curosurf 200 mg/kg) and therefore did not meet the inclusion criteria.

We did not identify any new studies for inclusion in this update.

Included studies

This 2026 review includes 10 individually randomized controlled trials (involving 3151 newborns) that evaluated the influence of the

timing of surfactant administration for the prevention or treatment of neonatal respiratory distress syndrome (RDS) on mortality and other neonatal outcomes (Bevilacqua 1996; Bevilacqua 1997; Dunn 1991; Dunn 2011; Egberts 1993; Kattwinkel 1993; Kendig 1991; Merritt 1991; SUPPORT 2010; Walti 1995). Details of each trial are given in the [Characteristics of included studies](#) table. Data reported in this review are published data only for all trials. It was not necessary for us to contact trial authors to retrieve any additional or missing data.

Four trials were conducted in the USA (Dunn 2011; Kattwinkel 1993; Kendig 1991; SUPPORT 2010), one in Italy (Bevilacqua 1996), one in Canada (Dunn 1991), one in France (Walti 1995), and three were multi-site studies conducted in two or more countries (Bevilacqua 1997 in Italy, Bulgaria, and Romania; Egberts 1993 in the Netherlands and Switzerland; and Merritt 1991 in the USA and Finland).

Sample sizes (indicated by infants included in the final analysis) of the included studies varied widely, ranging from 93 (Bevilacqua 1997) to 1310 (SUPPORT 2010).

All participants were preterm infants less than 32 weeks of gestational age: SUPPORT 2010 included infants with a gestational age between 24 and 28 weeks, while the other nine studies included infants with a gestational age between 24 and 32 weeks. All studies attempted to exclude infants who had known major congenital anomalies. Five studies excluded infants with prolonged ruptured membranes greater than two to three weeks (Bevilacqua 1997; Dunn 1991; Egberts 1993; Merritt 1991; Walti 1995).

In all trials, infants were randomly assigned before birth to the study arms. In seven studies, participants were divided into two groups and randomized to receive "prophylactic surfactant" (administration at the time of birth) or surfactant treatment (several hours after delivery when the infant met predefined criteria for established RDS) (Bevilacqua 1996; Bevilacqua 1997; Egberts 1993; Kattwinkel 1993; Kendig 1991; SUPPORT 2010; Walti 1995). Two studies included a placebo arm with installation of air given at birth (Dunn 1991; Merritt 1991). Only one trial, in addition to the prophylactic and selective surfactant arms, included a third active arm consisting of intubation-surfactant treatment and rapid extubation to nasal CPAP (Dunn 2011).

In all the trials but one, animal-derived surfactant extracts were used. Three studies utilized surfactant obtained by calf lung lavage (Dunn 1991; Kattwinkel 1993; Kendig 1991). Two studies used animal-derived surfactants (Dunn 2011; SUPPORT 2010). Four studies utilized the porcine surfactant curosurf (Bevilacqua 1996; Bevilacqua 1997; Egberts 1993; Walti 1995). Merritt 1991 utilized a surfactant obtained from human amniotic fluid.

Three of the trials published additional long-term follow-up studies. Vaucher 1993 followed up the infants enrolled in the trial of Merritt 1991 at one year of age; Sinkin 1998 followed up infants born at one of the participating centers of the Kendig 1991 trial for eight years; Vaucher 2012 followed up the participants of the SUPPORT 2010 trial, assessing their neurodevelopmental impairment at 18 to 22 months of corrected age.

With respect to potential differences in the baseline risk of the participants, which may result in a differential effect of surfactant

interventions, we considered the following characteristics of the included studies of interest for our proposed subgroups.

Treatment of mothers with antenatal steroids

Of the 10 included RCTs, eight did not report on the treatment of mothers with antenatal steroids (Bevilacqua 1996; Bevilacqua 1997; Dunn 1991; Egberts 1993; Kattwinkel 1993; Kendig 1991; Merritt 1991; Walti 1995); and in two RCTs, 50% or more of mothers received antenatal steroids (Dunn 2011; SUPPORT 2010).

FiO₂ threshold used to administer surfactant in the selective group

One study used the FiO₂ > 0.3 as the threshold for treating infants in the selective treatment group (Kattwinkel 1993). Three RCTs used the FiO₂ > 0.4 as the threshold for treating infants in the selective group (Bevilacqua 1997; Dunn 2011; Kendig 1991). Two RCTs used the FiO₂ > 0.5 as the threshold for treating infants in the selective group (Merritt 1991; SUPPORT 2010). One RCT used the FiO₂ > 0.6 as the threshold for treating infants in the selective group (Egberts 1993). Three RCTs treated infants in the selective group based on other indications (Bevilacqua 1996; Dunn 1991; Walti 1995).

Method of surfactant administration

Eight RCTs treated infants with intubation and ongoing ventilation (Dunn 1991; Dunn 2011; Egberts 1993; Kattwinkel 1993; Kendig 1991; Merritt 1991; SUPPORT 2010; Walti 1995); and two RCTs treated infants with intubation surfactant administration and rapid extubation (Bevilacqua 1996; Bevilacqua 1997). We identified no RCTs comparing the effect of prophylactic surfactant administration with surfactant treatment of established respiratory distress in preterm infants that had used thin catheters, laryngeal masks, or hypopharyngeal for surfactant administration.

Excluded studies

Following full-text screening, we excluded five trials for the following reasons.

- One study was a retrospective study comparing prophylactic surfactant (curosurf 200 mg/kg) versus rescue surfactant treatment (curosurf 200 mg/kg) (Iarŭkova 1999). This study had been included in previous versions of the review but was excluded in this update following reassessment of its design. It does not seem that its exclusion would have altered the conclusions of the previous review. It would have resulted in a very slight reduction in the precision of the effect estimate for bronchopulmonary dysplasia in comparison 1, the only outcome reported by the study. The study contributed minimal weight to the meta-analysis.
- In two studies, the treatment group received surfactant as "very early surfactant" once the first RDS signs appeared, instead of prophylactically (Lefort 2003; Rojas 2009).
- One trial did not administer prophylactic treatment to all infants at high-risk of RDS; instead, it included only infants who did not require acute resuscitation and showed no signs of respiratory distress (Morley 2008).
- In another trial, the comparison group received surfactant as an "early surfactant" instead of selective administration of surfactant (Sandri 2010).

Ongoing studies

We identified two ongoing studies ([DRKS00028086](#); [NCT04715373](#)). At the time of preparing this review for publication, the [DRKS00028086](#) updated registration record indicates that the trial is still recruiting, and the [NCT04715373](#) registry record indicates it is completed, but the results have not been posted. See [Characteristics of ongoing studies](#) for more details.

Studies awaiting classification

We categorized one study as awaiting classification. This study was published online on 11 December 2023 ([Murphy 2024](#)).

Risk of bias in included studies

The overall risk of bias assessment for each study, including all domain evaluations and justifications for judgment, is displayed in the risk of bias section ([Characteristics of included studies](#)), on the right side of all forest plots, and in [Figure 2](#);

Figure 2. Risk of bias summary

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding of participants and personnel (performance bias): All outcomes	Blinding of outcome assessment (detection bias): Hospital outcomes	Blinding of outcome assessment (detection bias): Follow-up outcomes	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)	Other bias
Bevilacqua 1996	?	+	-	?		+	?	+
Bevilacqua 1997	?	+	-	?		+	?	+
Dunn 1991	?	+	-	?		+	?	+
Dunn 2011	+	+	-	-		+	+	?
Egberts 1993	+	+	-	?		+	?	+
Kattwinkel 1993	?	+	-	?		?	?	+
Kendig 1991	+	+	-	?	?	+	?	+
Merritt 1991	+	+	+	+	+	+	?	+
SUPPORT 2010	?	+	-	?	+	+	+	+
Walti 1995	?	+	-	?		?	?	+

Allocation

Random sequence generation

We judged six studies to have an unclear risk of bias for the following reasons: two studies reported stratification by gestational age (GA) but did not clearly describe the method used to randomly assign participants within strata (Bevilacqua 1996; Bevilacqua 1997). One study reported stratification by GA and antenatal corticosteroids but did not describe the method used to randomly assign participants within these strata (Dunn 1991). The same applied to the SUPPORT 2010 study, which stratified by postmenstrual age (PMA) and center. In two studies (Kattwinkel 1993; Walti 1995), the methods used for randomization were not described.

We judged the remaining four studies to be at low risk of bias.

Allocation concealment

We judged all 10 studies to have a low risk of concealment bias. In all 10 studies, sealed envelopes with randomly allocated treatment assignments were provided to participating centers. Three studies used block randomization (Dunn 2011; Egberts 1993; Kendig 1991); and only Merritt 1991 used a variation of the biased-coin method for allocation sequence generation.

Blinding

Performance bias: all outcomes

We judged eight studies to have a high risk of bias because no blinding was described, and it was unlikely that personnel were blinded. We judged one study at high risk of bias because it was an open-label protocol (Kattwinkel 1993).

We judged Merritt 1991 to have a low risk of bias as the health care workers were unaware of the actual surfactant administration.

Detection bias: hospital outcomes

We judged Merritt 1991 to have a low risk of bias because the pediatric radiologists and pathologists were unaware of treatment allocation.

We judged Dunn 2011 to have a high risk of bias as no blinding was performed.

We judged the remaining eight studies to have an unclear risk of bias because it was unclear whether the outcome assessors were blinded.

Detection bias: follow-up outcomes

Only three studies reported on whether outcome assessors were blinded to follow-up outcomes.

Both Merritt 1991 and SUPPORT 2010 reported that staff were unaware of the treatment (low risk of bias). Walti 1995 did not describe whether the assessors were blinded to the follow-up outcomes (unclear risk of bias).

The degree of blinding of outcome assessment was variable across the studies. Only in Merritt 1991 was the assessment of outcome wholly blinded. Other studies included blinded radiologic evaluation and developmental follow-up.

Incomplete outcome data

We judged eight of the 10 studies at low risk of bias. Two studies had unclear risk of attrition bias: Walti 1995 excluded 11% of infants initially enrolled, and the study by Kattwinkel 1993 did not include a number of infants because one center was dropped due to a high incidence of protocol deviations.

For the study of Merritt 1991, only data from the intention-to-treat analysis are used in this review. Merritt 1991 reported an intention-to-treat analysis of specific major outcomes, but for various other complications of preterm birth, data were reported only for those infants with proven lung immaturity.

Selective reporting

We identified trial registrations for two studies (Dunn 2011; SUPPORT 2010: low risk of bias); methods and inclusion criteria in the registration aligned with those in the study reports. For the other eight studies, no register or protocol was available (unclear risk of bias).

Other potential sources of bias

No other sources of bias were identified in the 10 studies included (low risk of bias), but one study was stopped prematurely by the Data Safety Monitoring Committee based on declining enrollment and the low likelihood of demonstrating an effect based on the predefined sample size (Dunn 2011); we therefore deemed it at unclear risk for other potential sources of bias.

Effects of interventions

See: **Summary of findings 1** Summary of findings table - Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS

The results of the 10 included trials are presented below according to the outcome reported and the proposed subgroups (Bevilacqua 1996; Bevilacqua 1997; Dunn 1991; Dunn 2011; Egberts 1993; Kattwinkel 1993; Kendig 1991; Merritt 1991; SUPPORT 2010; Walti 1995).

We conducted subgroup analysis based on the use of CPAP, with all 10 included trials. Only two trials routinely stabilized infants on CPAP in the selective surfactant group (hereafter 'control') (Dunn 2011; SUPPORT 2010). Thanks to the comprehensive reporting of the included studies for this subgroup, we performed analysis of all primary and secondary outcomes.

We did not conduct subgroup analysis based on gestational age at birth (< 28 weeks versus ≥ 28 weeks), as initially proposed, as only three of the 10 included studies reported outcomes by gestational age (Dunn 2011; Egberts 1993; SUPPORT 2010). We judged such an analysis to be inappropriate as it would be based on imprecise estimates and would likely result in low or very low certainty evidence, thereby limiting our ability to draw reliable conclusions.

For the subgroups based on receipt of antenatal steroids, the threshold used to initiate selective surfactant treatment, and the method of surfactant administration, we conducted analyses only for critical and important outcomes as these were the only outcomes reported in the included studies. We were unable to perform the analysis for moderate to severe neurodevelopmental

impairment because this outcome was not reported in the included studies.

Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by the use of CPAP)

Primary outcomes

See [Summary of findings 1](#).

1.1 Chronic lung disease (CLD) (use of supplemental oxygen at 36 weeks' postmenstrual age)

Five studies reported on CLD at 36 weeks' postmenstrual age ([Bevilacqua 1997](#); [Dunn 1991](#); [Dunn 2011](#); [Egberts 1993](#); [SUPPORT 2010](#)). Of these studies, [Dunn 2011](#) and [SUPPORT 2010](#) routinely placed control infants on CPAP.

The meta-analysis of all five studies suggests that prophylactic surfactant administration likely results in little to no difference in the risk of CLD at 36 weeks' postmenstrual age compared to selective surfactant administration (RR 1.13, 95% CI 1.00 to 1.28; $I^2 = 0\%$; RD 0.04, 95% CI 0.00 to 0.08; number needed to treat for an additional harmful outcome (NNTH) 25, 95% CI 13 to > 1000 ; $I^2 = 0\%$; 5 trials, 1874 infants; moderate-certainty evidence; [Analysis 1.1](#)). The certainty of the evidence was moderate; we downgraded it from high due to the risk of bias in the included studies.

Subgroups based on routine use of CPAP are reported below in 1.1.1 and 1.1.2 and in [Analysis 1.1](#). The test for subgroup differences indicates that there is no subgroup effect ($I^2 = 0\%$, $P = 0.40$), suggesting that routinely placing control infants on CPAP does not modify the prophylactic surfactant effect on chronic lung disease ([Analysis 1.1](#)).

1.1.1. CLD in studies without routine application of continuous positive airway pressure to infants in control groups

Three studies did not routinely place control infants on CPAP ([Bevilacqua 1997](#); [Dunn 1991](#); [Egberts 1993](#)).

The meta-analysis of these three studies revealed very uncertain results that do not allow us to draw conclusions about the effect of prophylactic surfactant on the risk of CLD compared with selective surfactant (RR 1.56, 95% CI 0.72 to 3.40; $I^2 = 0\%$; RD 0.03, 95% CI -0.02 to 0.08; $I^2 = 0\%$; 3 trials, 362 infants; very low certainty evidence; [Analysis 1.1](#)). No heterogeneity was noted. We downgraded the certainty of evidence to very low due to the risk of bias in the included studies and very serious imprecision in the effect estimates.

1.1.2. CLD in studies with routine application of continuous positive airway pressure to infants in control groups

Two studies routinely placed control infants on CPAP and reported on CLD ([Dunn 2011](#); [SUPPORT 2010](#)). These studies reported an increased risk of CLD with the prophylactic administration of surfactant when compared with the selective use of surfactant in infants with respiratory failure and early stabilization on CPAP.

Meta-analysis of the two studies revealed that prophylactic surfactant administration may result in little to no difference in the risk of CLD in comparison with selective administration in infants placed on CPAP early (RR 1.12, 95% CI 0.99 to 1.26; $I^2 = 0\%$; RD 0.04, 95% CI -0.00 to 0.09; $I^2 = 0\%$; 2 trials, 1512 infants; low-

certainty evidence; [Analysis 1.1](#)). No heterogeneity was noted. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

1.2 Mortality at 36 weeks' PMA or prior to hospital discharge

Six studies reported mortality at 36 weeks' PMA or prior to discharge ([Dunn 2011](#); [Egberts 1993](#); [Kendig 1991](#); [Merritt 1991](#); [SUPPORT 2010](#); [Walti 1995](#)). Two of the six studies, [Dunn 1991](#) and [SUPPORT 2010](#), routinely placed control infants on CPAP.

We report subgroups based on routine use of CPAP below in 1.2.1 and 1.2.2. The test for subgroup differences indicates that there is substantial heterogeneity ($I^2 = 90.5\%$, $P = 0.001$), suggesting that routinely placing control infants on CPAP modifies the prophylactic surfactant effect on mortality ([Analysis 1.2](#)). Therefore, we have not reported an overall estimation of the effect that includes all studies.

1.2.1. Mortality at 36 weeks' PMA in studies without routine application of continuous positive airway pressure to infants in control groups

Four studies did not routinely place control infants on CPAP ([Egberts 1993](#); [Kendig 1991](#); [Merritt 1991](#); [Walti 1995](#)).

Meta-analysis of these four studies demonstrated that prophylactic surfactant reduces neonatal mortality in infants who are not routinely placed on CPAP (RR 0.72, 95% CI 0.56 to 0.93; $I^2 = 53\%$; RD -0.06, 95% CI -0.11 to -0.02; $I^2 = 13\%$; number needed to treat for an additional beneficial outcome (NNTB) 17, 95% CI 9 to 50; 4 trials, 1030 infants; high-certainty evidence; [Analysis 1.2](#)).

1.2.2. Mortality at 36 weeks' PMA in studies with routine application of continuous positive airway pressure to infants in control groups

Two studies routinely placed control infants on CPAP and reported on mortality at 36 weeks' PMA ([Dunn 2011](#); [SUPPORT 2010](#)).

Meta-analysis of these two studies shows that mortality at 36 weeks' PMA probably increases slightly with the use of prophylactic surfactant when compared with selective use of surfactant in infants with respiratory failure stabilized on CPAP (RR 1.28, 95% CI 1.01 to 1.62; $I^2 = 0\%$; RD 0.03, 95% CI 0.00 to 0.06; $I^2 = 0\%$; NNTH 33, 95% CI 17 to > 1000 ; 2 trials, 1748 infants; moderate-certainty evidence; [Analysis 1.2](#)). No heterogeneity was present. We downgraded the certainty of the evidence to moderate due to serious imprecision in the effect estimates.

1.3 Moderate to severe neurodevelopmental impairment

One study ([SUPPORT 2010](#)) reported neurodevelopmental impairment at 18 to 22 months' corrected age in a follow-up publication ([Vaucher 2012](#)). Infants in this study were routinely placed on CPAP.

Moderate to severe neurodevelopmental impairment

Only the [SUPPORT 2010](#) study reported the individual neurodevelopmental impairment outcomes of cerebral palsy, significant mental developmental delay, legal blindness, and hearing deficit in enrolled infants at 18 to 22 months' corrected age, in a follow-up publication ([Vaucher 2012](#)). Infants were routinely placed on CPAP.

Neurodevelopmental impairment was observed in 43 (9.1%) of the 473 toddlers of the prophylactic group and in 55 (10.9%) of the 503

toddlers of the selective group (RR 0.83, 95% CI 0.57 to 1.21; RD -0.02, 95% CI -0.06 to 0.02; 1 trial, 976 children; I^2 not applicable; low-certainty evidence; [Analysis 1.3](#)). This study's results suggest that prophylactic surfactant administration may result in little to no difference in neurodevelopmental impairment compared to selective surfactant administration. We downgraded the certainty of the evidence to low due to the risk of bias in the included study and serious imprecision in the effect estimate. No subgroup analysis could be performed because only one study assessed this outcome; no studies were identified that reported on this outcome without routine application of CPAP.

Components of moderate-severe neurodevelopmental impairment

Only the [SUPPORT 2010](#) study presented components of moderate-severe neurodevelopmental impairment in enrolled infants at 18 to 22 months' corrected age, in a follow-up publication ([Vaucher 2012](#)). It was therefore not possible to do meta-analysis or subgroup analysis.

Cerebral palsy

In [SUPPORT 2010](#), cerebral palsy was observed in 19 (4.0%) of the 479 toddlers of the prophylactic group and in 21 (4.1%) of the 511 toddlers of the selective group (RR reported in the [Vaucher 2012](#) publication was adjusted accordingly; RR 0.93, 95% CI 0.51 to 1.72; I^2 not applicable; 1 trial, 990 children; very low certainty evidence). These results are very uncertain and do not allow us to determine the effects of prophylactic surfactant on cerebral palsy compared with selective surfactant administration. We downgraded the certainty of the evidence to very low due to risk of bias and very serious imprecision in the effect estimate.

Significant mental developmental delay

In [SUPPORT 2010](#), mental developmental delay was observed in 36 (7.6%) of the 472 toddlers of the prophylactic group and in 36 (7.2%) of the 502 toddlers of the selective group (RR reported in the [Vaucher 2012](#) publication was adjusted accordingly; RR 0.95, 95% CI 0.61 to 1.50; I^2 not applicable; 1 trial, 974 children; very low certainty evidence). The results are very uncertain and do not allow us to determine the effects of prophylactic surfactant on the risk of significant mental developmental delay when compared with selective surfactant administration. We downgraded the certainty of the evidence to very low due to risk of bias and very serious imprecision in the effect estimate.

Visual impairment

In [SUPPORT 2010](#), legal blindness was observed in 7 (1.5%) of the 479 toddlers of the prophylactic group and in 4 (0.8%) of the 511 toddlers of the selective group, (adjusted RR reported by [Vaucher 2012](#) 0.53, 95% CI 0.16 to 1.78; I^2 not applicable; 1 trial, 990 children; very low certainty evidence). The results are uncertain and do not allow us to determine the effect of prophylactic surfactant on this outcome when compared with selective surfactant administration. We downgraded the certainty of the evidence to very low due to risk of bias and very serious imprecision in the effect estimate.

Hearing impairment

In [SUPPORT 2010](#), hearing deficit was observed in 7 (1.5%) of the 479 toddlers of the prophylactic group and in 17 (3.3%) of the 511 toddlers of the selective group (adjusted RR reported in [Vaucher 2012](#): 2.27, 95% CI 0.96 to 5.37; I^2 not applicable; 1 trial, 990 children; very low certainty evidence). The results are very uncertain and do

not allow us to determine the effect of prophylactic surfactant on the risk of hearing deficit compared with selective surfactant. We downgraded the certainty of the evidence to very low due to risk of bias and very serious imprecision in the effect estimate.

1.4. Pneumothorax

Eight studies reported on pneumothorax ([Bevilacqua 1996](#); [Dunn 1991](#); [Dunn 2011](#); [Egberts 1993](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Merritt 1991](#); [Walti 1995](#)). Only [Dunn 2011](#) placed infants in the control groups on CPAP.

The meta-analysis including all eight studies, regardless of whether control infants were routinely placed on CPAP, suggests that prophylactic surfactant administration may result in a slight reduction in pneumothorax compared to selective surfactant administration (RR 0.76, 95% CI 0.56 to 1.04, I^2 = 12%; RD -0.01, 95% CI -0.03 to 0.00, I^2 = 26%; NNTB 100, 95% CI 33 to > 1000; 8 trials, 3094 infants; low-certainty evidence; [Analysis 1.4](#)). No important heterogeneity was noted in the RR estimation. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

Subgroups based on routine use of CPAP are reported below in 1.4.1 and 1.4.2. The test for subgroup differences indicates that there is no subgroup effect (I^2 = 0%, P = 0.70), suggesting that routinely placing control infants on CPAP does not modify the prophylactic surfactant effect on pneumothorax ([Analysis 1.4](#)).

1.4.1. Pneumothorax in studies without routine application of continuous positive airway pressure to infants

Seven studies did not routinely place control infants on CPAP ([Bevilacqua 1996](#); [Dunn 1991](#); [Egberts 1993](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Merritt 1991](#); [Walti 1995](#)).

The meta-analysis of these seven trials suggests that prophylactic surfactant administration may result in a slight reduction in pneumothorax in comparison with selective administration of surfactant in infants with respiratory failure (RR 0.75, 95% CI 0.53 to 1.05; I^2 = 23%; RD -0.01, 95% CI -0.03 to 0.00; I^2 = 38%; 7 trials, 2663 infants; low-certainty evidence; [Analysis 1.4](#)). No important heterogeneity was noted. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

1.4.2. Pneumothorax in studies with routine application of continuous positive airway pressure to infants in control groups

Only [Dunn 2011](#), a study that routinely used CPAP for infants in the control group, reported on pneumothorax.

The results of this study are very uncertain and do not allow us to determine the effects of prophylactic surfactant on the risk of pneumothorax compared with selective surfactant administration in this population (RR 0.89, 95% CI 0.39 to 2.01; RD -0.01, 95% CI -0.05 to 0.04; I^2 not applicable; 1 trial, 431 infants; very low certainty evidence; [Analysis 1.4](#)). We downgraded the certainty of the evidence to very low due to risk of bias and very serious imprecision in the effect estimates.

Secondary outcomes

1.5. Pulmonary interstitial emphysema

Six studies reported on the risk of pulmonary interstitial emphysema (Bevilacqua 1996; Dunn 1991; Egberts 1993; Kattwinkel 1993; Merritt 1991; Walti 1995). All six did not routinely place control infants on CPAP.

The meta-analysis of these six trials suggests that prophylactic surfactant administration may result in little to no difference in the risk of pulmonary interstitial emphysema when compared with selective surfactant administration (RR 0.78, 95% CI 0.56 to 1.08; $I^2 = 41\%$; RD -0.01, 95% CI -0.03 to 0.00; $I^2 = 77\%$; 6 trials, 2185 infants; low-certainty evidence; Analysis 1.5). There was low heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

No subgroup analyses were performed because none of the studies routinely placed control infants on CPAP.

1.6. Any air leak syndrome

Two studies reported any air leak (Bevilacqua 1997; SUPPORT 2010); one of them, SUPPORT 2010, placed control infants on CPAP.

The meta-analysis of these two studies suggests that prophylactic surfactant may result in little to no difference in the risk of any air leak syndrome when compared with selective surfactant administration (RR 1.13, 95% CI 0.78 to 1.63; $I^2 = 0\%$; RD 0.01, 95% CI -0.02 to 0.03; $I^2 = 0\%$; 2 trials, 1409 infants; low-certainty evidence; Analysis 1.6). No heterogeneity was noted. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

Subgroups based on routine use of CPAP are reported below in 1.6.1 and 1.6.2. The test for subgroup differences indicates that there is no heterogeneity ($I^2 = 0\%$, $P = 0.55$), suggesting that routinely placing control infants on CPAP does not modify the prophylactic surfactant effect on any air leak.

1.6.1. Air leak syndrome in studies without routine application of continuous positive airway pressure to infants in control groups

Bevilacqua 1997 did not routinely place control infants on CPAP.

Results of this study are very uncertain and do not allow us to determine the effects of prophylactic surfactant on the risk of any air leak compared with selective surfactant administration (RR 1.57, 95% CI 0.49 to 5.01; RD 0.05, 95% CI -0.08 to 0.18; I^2 not applicable; 1 trial, 93 infants; very low certainty evidence; Analysis 1.6). We downgraded the certainty of the evidence to very low due to risk of bias and very serious imprecision in the effect estimates.

1.6.2. Air leak syndrome in studies with routine application of continuous positive airway pressure to infants in control groups

Only SUPPORT 2010 reported on air leaks in infants with respiratory failure and early stabilization on CPAP.

This study suggests that prophylactic administration may result in little to no difference in air leak when compared with selective surfactant administration (RR 1.08, 95% CI 0.73 to 1.60; RD 0.01, 95% CI -0.02 to 0.03; I^2 not applicable; 1 trial, 1316 infants; low-certainty evidence; Analysis 1.6). We downgraded the certainty of

the evidence to low due to the risk of bias in the included study and serious imprecision in the effect estimates.

1.7. Any pulmonary hemorrhage

Four studies reported on pulmonary hemorrhage (Bevilacqua 1997; Dunn 2011; Kattwinkel 1993; Walti 1995); one of them, Dunn 2011, placed control infants on CPAP.

The meta-analysis of these four studies suggests that prophylactic surfactant may result in little to no difference in the risk of pulmonary hemorrhage when compared to selective administration of surfactant (RR 1.05, 95% CI 0.49 to 2.22; $I^2 = 10\%$; RD 0.00, 95% CI -0.01 to 0.01; $I^2 = 29\%$; 4 trials, 2023 infants; very low-certainty evidence; Analysis 1.7). Low heterogeneity was noted. We downgraded the certainty of the evidence to very low for risk of bias in the included studies and very serious imprecision in the effect estimates.

Subgroups based on routine use of CPAP are reported in 1.7.1 and 1.7.2. The test for subgroup differences indicates that there is low heterogeneity ($I^2 = 36.9$, $P = 0.21$), suggesting that routinely placing control infants on CPAP does not modify the prophylactic surfactant effect on pulmonary hemorrhage.

1.7.1. Pulmonary hemorrhage in studies without routine application of continuous positive airway pressure to infants in control groups

Three studies did not routinely place control infants on CPAP (Bevilacqua 1997; Kattwinkel 1993; Walti 1995). None of them reported an effect of prophylactic surfactant on the risk of pulmonary hemorrhage.

The meta-analysis of these three trials does not allow us to determine the effects of prophylactic surfactant on the risk of pulmonary hemorrhage when compared with selective surfactant administration because the evidence is very uncertain (RR 0.73, 95% CI 0.28 to 1.87; $I^2 = 2\%$; RD -0.00, 95% CI -0.01 to 0.01; $I^2 = 0\%$; 3 trials, 1592 infants; very low certainty evidence; Analysis 1.7). No heterogeneity was noted. We downgraded the certainty of the evidence to very low due to the risk of bias in the included studies and very serious imprecision in the effect estimates.

1.7.2. Pulmonary hemorrhage in studies with routine application of continuous positive airway pressure to infants in control groups

Only Dunn 2011 reported on this outcome.

The results of this study suggest that prophylactic surfactant may result in little to no difference in the risk of air leak when compared with selective surfactant administration (RR 2.12, 95% CI 0.54 to 8.39; RD 0.02, 95% CI -0.01 to 0.04; 1 trial; I^2 not applicable; 431 infants; low-certainty evidence; Analysis 1.7). We downgraded the certainty of the evidence to low due to the risk of bias in the included study and serious imprecision in the effect estimates.

1.8. Bronchopulmonary dysplasia (at 28 to 30 days of age or use of supplemental oxygen at 28 to 30 days of age)

Nine studies reported on bronchopulmonary dysplasia at 28 to 30 days of age (Bevilacqua 1996; Bevilacqua 1997; Dunn 1991; Dunn 2011; Egberts 1993; Kattwinkel 1993; Kendig 1991; Merritt 1991; Walti 1995). Only Dunn 2011 routinely placed control infants on CPAP.

The meta-analysis of all nine studies revealed that prophylactic surfactant administration likely results in little to no difference in the risk of bronchopulmonary dysplasia when compared with selective surfactant administration (RR 1.02, 95% CI 0.91 to 1.14; $I^2 = 49\%$; RD 0.00, 95% CI -0.02 to 0.03; $I^2 = 57\%$; 9 trials, 3163 infants; moderate-certainty evidence; [Analysis 1.8](#)). There was low heterogeneity. We downgraded the certainty of the evidence to moderate due to the risk of bias in the included studies.

Subgroups based on routine use of CPAP are reported below in 1.8.1 and 1.8.2. The test for subgroup differences indicates that there is moderate heterogeneity ($I^2 = 66.4\%$, $P = 0.08$), suggesting that there may be a difference in bronchopulmonary dysplasia in trials conducted with or without routine administration of CPAP.

1.8.1. Bronchopulmonary dysplasia in studies without routine application of continuous positive airway pressure to infants in control groups

Eight studies that did not routinely place control infants on CPAP reported on bronchopulmonary dysplasia at 28 to 30 days of age ([Bevilacqua 1996](#); [Bevilacqua 1997](#); [Dunn 1991](#); [Egberts 1993](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Merritt 1991](#); [Walti 1995](#)). All of them except [Dunn 1991](#) reported no difference in the risk of bronchopulmonary dysplasia associated with the strategy of surfactant administration.

Meta-analysis of the eight studies that did not routinely place control infants on CPAP showed that prophylactic surfactant may make little to no difference to the risk of bronchopulmonary dysplasia (RR 0.95, 95% CI 0.82 to 1.11; $I^2 = 46\%$; RD -0.01, 95% CI -0.03 to 0.02; $I^2 = 44\%$; 8 trials, 2761 infants; low-certainty evidence; [Analysis 1.8](#)). There was low heterogeneity. We downgraded the certainty of the evidence to low due to risk of bias in the included studies and serious imprecision in the effect estimates.

1.8.2. Bronchopulmonary dysplasia in studies with routine application of continuous positive airway pressure to infants in control groups

[Dunn 2011](#) was the only study with routine application of CPAP that reported on the risk of bronchopulmonary dysplasia.

Results from this study showed that prophylactic use of surfactant may result in little to no difference in the risk of bronchopulmonary dysplasia compared to selective administration (RR 1.16, 95% CI 0.99 to 1.36; RD 0.09, 95% CI -0.01 to 0.18; 1 trial, 402 infants; I^2 not applicable; low-certainty evidence; [Analysis 1.8](#)). We downgraded the certainty of the evidence to low due to the risk of bias in the included study and serious imprecision in the effect estimates.

1.9. Bronchopulmonary dysplasia or death prior to 28 to 30 days of age

Eight studies reported on the combined outcome of bronchopulmonary dysplasia or death at 28 days ([Bevilacqua 1996](#); [Bevilacqua 1997](#); [Dunn 1991](#); [Dunn 2011](#); [Egberts 1993](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Walti 1995](#)). Only [Dunn 2011](#) placed control infants on CPAP.

Meta-analysis including all eight studies regardless of whether control infants were routinely placed on CPAP shows that prophylactic surfactant administration may not reduce the risk of bronchopulmonary dysplasia or death (RR 0.89, 95% CI 0.81 to 0.98; $I^2 = 77\%$; RD -0.04, 95% CI -0.06 to -0.01; $I^2 = 73\%$; NNTB 25 95% CI 17 to 100; 8 trials, 3033 infants; low-certainty evidence; [Analysis 1.9](#)). There was substantial heterogeneity, most of it explained by

the a priori subgroup analysis described below. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious inconsistency.

Subgroups based on routine use of CPAP are reported below in 1.9.1 and 1.9.2. The test for subgroup differences indicates substantial heterogeneity ($I^2 = 90.3\%$, $P = 0.001$), suggesting that routinely placing infants on CPAP modifies the intervention's effect on bronchopulmonary dysplasia or death prior to 28 to 30 days of age.

1.9.1. Studies without routine application of continuous positive airway pressure to infants in control groups

Seven studies that did not routinely place control infants on CPAP reported on bronchopulmonary dysplasia or death at 28 days ([Bevilacqua 1996](#); [Bevilacqua 1997](#); [Dunn 1991](#); [Egberts 1993](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Walti 1995](#)). Three studies reported a reduction in the risk of bronchopulmonary dysplasia or death in association with prophylactic surfactant administration ([Bevilacqua 1996](#); [Kattwinkel 1993](#); [Walti 1995](#)). [Dunn 1991](#) reported an increase in the risk of bronchopulmonary dysplasia or death in the prophylactic administration group. The other three studies did not find differences between treatment groups on this outcome ([Bevilacqua 1997](#); [Egberts 1993](#); [Kendig 1991](#)).

Meta-analysis of these seven studies demonstrates that prophylactic surfactant administration may not reduce the risk of bronchopulmonary dysplasia or death at 28 days when compared to selective surfactant administration (RR 0.82, 95% CI 0.73 to 0.92; $I^2 = 72\%$; RD -0.05, 95% CI -0.08 to -0.02; $I^2 = 71\%$; NNTB 20 95% CI 13 to 50; 7 trials, 2613 infants; low-certainty evidence; [Analysis 1.9](#)). There was moderate heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious inconsistency.

1.9.2. Studies with routine application of continuous positive airway pressure to infants in control groups

[Dunn 2011](#) was the only study with routine application of CPAP that reported on this outcome.

[Dunn 2011](#) suggests that the risk of BPD may not increase with the prophylactic administration of surfactant when compared with early stabilization on CPAP with selective use of surfactant in infants with respiratory failure (RR 1.12, 95% CI 0.96 to 1.31; RD 0.07, 95% CI -0.02 to 0.16; I^2 not applicable; 1 trial, 420 infants; low-certainty evidence; [Analysis 1.9](#)). We downgraded the certainty of the evidence due to the risk of bias in the included studies and serious imprecision in the effect estimates.

1.10. Chronic lung disease or death (use of supplemental oxygen or death at 36 weeks' PMA)

Four studies reported information for the combined outcome of CLD or death ([Dunn 2011](#); [Dunn 1991](#); [SUPPORT 2010](#); [Walti 1995](#)). Of these four trials, [Dunn 2011](#) and [SUPPORT 2010](#) routinely placed control infants on CPAP.

The meta-analysis including all four studies, regardless of whether control infants were routinely placed on CPAP, suggests that prophylactic surfactant administration may result in little to no difference in the risk of CLD or death compared with selective surfactant administration (RR 1.09, 95% CI 0.99 to 1.20; $I^2 = 31\%$; RD 0.04, 95% CI -0.00 to 0.08; $I^2 = 41\%$; 4 trials, 2122 infants; low-

certainty evidence; [Analysis 1.10](#)). There was low heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

Subgroups based on routine use of CPAP are reported below in 1.10.1 and 1.10.2. The test for subgroup differences indicates that there is moderate heterogeneity ($I^2 = 52.0\%$, $P = 0.15$), suggesting that routinely placing control infants on CPAP may modify the prophylactic surfactant effect on CLD or death ([Analysis 1.10](#)).

1.10.1. CLD or death in studies without routine application of continuous positive airway pressure to infants in control groups

[Dunn 1991](#) and [Walti 1995](#) did not routinely place control infants on CPAP.

The meta-analysis of the two studies suggests that prophylactic surfactant administration may not reduce the risk of CLD or death when compared to selective surfactant administration (RR 0.87, 95% CI 0.63 to 1.21; $I^2 = 47\%$; RD -0.04, 95% CI -0.13 to 0.05; $I^2 = 55\%$; 2 trials, 378 infants; moderate-certainty evidence; [Analysis 1.10](#)). There was low heterogeneity. We downgraded the certainty of the evidence to moderate due to the risk of bias in the included studies. Using thresholds of $RR < 0.75$ or > 1.25 to define a clinically important effect, we did not downgrade the certainty of the evidence for imprecision.

1.10.2. CLD or death in studies with routine application of continuous positive airway pressure to infants in control groups

Both trials with the routine application of CPAP reported a concerning trend toward increased risk of CLD or death in the prophylactic surfactant administration group ([Dunn 2011](#); [SUPPORT 2010](#)).

The meta-analysis of these two studies shows that prophylactic surfactant administration probably results in little to no difference in the risk of CLD or death compared to selective use of surfactant in infants with respiratory failure and early stabilization on CPAP (RR 1.12, 95% CI 1.02 to 1.24; $I^2 = 0\%$; RD 0.06, 95% CI 0.01 to 0.10; $I^2 = 0\%$; NNTB 17, 95% CI 10 to 100; 2 trials, 1744 infants; moderate-certainty evidence; [Analysis 1.10](#)). No heterogeneity was noted. We downgraded the certainty of the evidence to moderate due to the risk of bias in the included studies.

1.11 Neonatal mortality (< 28 days of age) from any cause

Nine studies reported on neonatal mortality ([Bevilacqua 1996](#); [Bevilacqua 1997](#); [Dunn 1991](#); [Dunn 2011](#); [Egberts 1993](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Merritt 1991](#); [Walti 1995](#)). One of them, [Dunn 2011](#), routinely placed control infants on CPAP.

The meta-analysis that included all nine studies shows that prophylactic surfactant administration reduces the risk of neonatal mortality when compared with selective surfactant administration (RR 0.71, 95% CI 0.58 to 0.87; $I^2 = 35\%$; RD -0.03, 95% CI -0.05 to -0.01; $I^2 = 69\%$; NNTB 33, 95% CI 20 to 100; 9 trials, 3191 infants; high-certainty evidence; [Analysis 1.11](#)). There was low heterogeneity. We judged the certainty of the evidence to be high.

Subgroups based on routine use of CPAP are reported below in 1.11.1 and 1.11.2. The test for subgroup differences indicates the presence of heterogeneity ($I^2 = 47.2\%$, $P = 0.17$), suggesting

that routinely placing control infants on CPAP might modify the prophylactic surfactant effect on neonatal mortality.

1.11.1. Neonatal mortality in studies without routine application of continuous positive airway pressure to infants in control groups

Eight studies that did not routinely place control infants on CPAP reported on neonatal mortality ([Bevilacqua 1996](#); [Bevilacqua 1997](#); [Dunn 1991](#); [Egberts 1993](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Merritt 1991](#); [Walti 1995](#)). Three trials noted a decrease in neonatal mortality associated with prophylactic surfactant administration ([Bevilacqua 1997](#); [Kattwinkel 1993](#); [Kendig 1991](#)).

The meta-analysis of studies that did not routinely place control infants on CPAP shows that the use of prophylactic surfactant reduces the risk of neonatal mortality compared to selective surfactant (RR 0.69, 95% CI 0.56 to 0.85; $I^2 = 34\%$; RD -0.04, 95% CI -0.06 to -0.02; $I^2 = 75\%$; NNTB 25, 95% CI 17 to 50; 8 trials, 2761 infants; high-certainty evidence; [Analysis 1.11](#)). There was low heterogeneity when estimating the RR. We judged the certainty of the evidence to be high.

1.11.2. Neonatal mortality in studies with routine application of continuous positive airway pressure to infants in control groups

One study routinely placed control infants on CPAP ([Dunn 2011](#)).

The results of this study suggest that prophylactic surfactant administration may increase the risk of neonatal mortality slightly when compared with selective surfactant administration in infants with respiratory failure and early stabilization on CPAP (RR 1.32, 95% CI 0.53 to 3.28; RD 0.01, 95% CI -0.03 to 0.05; I^2 not applicable; 1 trial, 430 infants; low-certainty evidence; [Analysis 1.11](#)). We downgraded the evidence to low due to very serious imprecision in the effect estimates.

1.12. Patent ductus arteriosus (PDA) that has been treated with cyclo-oxygenase inhibitor or surgery

Eight studies reported on the incidence of patent ductus arteriosus ([Bevilacqua 1996](#); [Dunn 1991](#); [Dunn 2011](#); [Egberts 1993](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Merritt 1991](#); [Walti 1995](#)). Of them, only [Dunn 2011](#) placed control infants on CPAP.

The meta-analysis including all eight studies regardless of whether control infants were routinely placed on CPAP demonstrates that prophylactic surfactant administration probably results in little to no difference in the risk of patent ductus arteriosus when compared with selective surfactant administration in infants presenting with respiratory failure (RR 0.98, 95% CI 0.89 to 1.08; $I^2 = 33\%$; RD -0.01, 95% CI -0.04 to 0.03; $I^2 = 31\%$; 8 trials, 3093 infants; moderate-certainty evidence; [Analysis 1.12](#)). There was low heterogeneity. We downgraded the certainty of the evidence to moderate due to the risk of bias in the included studies.

Subgroups based on routine use of CPAP are reported below in 1.12.1 and 1.12.2. The test for subgroup differences indicates no heterogeneity ($I^2 = 0\%$, $P = 0.94$), suggesting that routinely placing infants on CPAP does not modify the prophylactic surfactant effect on patent ductus arteriosus.

1.12.1. Patent ductus arteriosus in studies without routine application of continuous positive airway pressure to infants in control groups

Seven trials did not routinely place control infants on CPAP (Bevilacqua 1996; Dunn 1991; Egberts 1993; Kattwinkel 1993; Kendig 1991; Merritt 1991; Walti 1995).

The meta-analysis of the seven studies that did not routinely place control infants on CPAP shows that prophylactic surfactant administration probably results in little to no difference in the risk of patent ductus arteriosus compared to selective use of surfactant (RR 0.98, 95% CI 0.88 to 1.10; $I^2 = 42\%$; RD -0.01, 95% CI -0.04 to 0.03; $I^2 = 41\%$; 7 trials, 2663 infants; moderate-certainty evidence; Analysis 1.12). There was low heterogeneity. We downgraded the certainty of the evidence to moderate due to the risk of bias in the included studies.

1.12.2. Patent ductus arteriosus in studies with routine application of continuous positive airway pressure to infants in control groups

Dunn 2011 was the only study with routine application of CPAP that reported on this outcome.

Results of this study suggest that prophylactic surfactant administration may result in little to no difference in risk of patent ductus arteriosus when compared with selective use of surfactant (RR 0.97, 95% CI 0.79 to 1.20; RD -0.01, 95% CI -0.11 to 0.08; I^2 not applicable; 1 trial, 430 infants; low-certainty evidence; Analysis 1.12). We downgraded the certainty of the evidence to low due to the risk of bias in the included study and serious imprecision in the effect estimates.

1.13. Any culture-proven bacterial sepsis

Five studies reported proven bacterial sepsis (Bevilacqua 1996; Bevilacqua 1997; Dunn 2011; Kattwinkel 1993; Merritt 1991). Of them, only Dunn 2011 placed control infants on CPAP.

The meta-analysis that included all five studies regardless of whether control infants were routinely placed on CPAP suggests that prophylactic surfactant administration may result in little to no difference in the risk of proven bacterial sepsis when compared to selective surfactant administration (RR 1.06, 95% CI 0.80 to 1.40; $I^2 = 25\%$; RD 0.00, 95% CI -0.02 to 0.03; $I^2 = 23\%$; 5 trials, 2182 infants; low-certainty evidence; Analysis 1.13). There was low heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

Subgroups based on routine use of CPAP are reported below in 1.13.1 and 1.13.2. The test for subgroup differences indicates substantial heterogeneity ($I^2 = 71.6\%$, $P = 0.06$), suggesting that routinely placing infants on CPAP can modify the prophylactic surfactant effect on any culture-proven bacterial sepsis.

1.13.1. Bacterial sepsis in studies without routine application of continuous positive airway pressure to infants in control groups

Four studies are in this subgroup (Bevilacqua 1996; Bevilacqua 1997; Kattwinkel 1993; Merritt 1991).

The meta-analysis of these four studies suggests that prophylactic surfactant administration may result in little to no difference in the risk of proven bacterial sepsis when compared with selective surfactant administration (RR 0.91, 95% CI 0.66 to 1.25; $I^2 = 0\%$; RD -0.01, 95% CI -0.03 to 0.02; $I^2 = 0\%$; 4 trials, 1757 infants; low-

certainty evidence; Analysis 1.13). No heterogeneity was present. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

1.13.2. Bacterial sepsis in studies with routine application of continuous positive airway pressure to infants in control groups

Dunn 2011 was the only study with routine application of CPAP that reported on sepsis.

The results of this study are very uncertain and do not allow us to determine the effects of prophylactic surfactant administration on the risk of bacterial sepsis when compared with selective administration of surfactant in infants and early stabilization on CPAP (RR 1.70, 95% CI 0.96 to 3.03; RD 0.05, 95% CI -0.00 to 0.11; I^2 not applicable; 1 trial, 425 infants; very low certainty evidence; Analysis 1.13). We downgraded the certainty of the evidence to very low due to the risk of bias in the included study and very serious imprecision in the effect estimates.

1.14. Any culture-proven fungal sepsis

Dunn 2011 was the only study with routine application of CPAP reporting on fungal sepsis.

The results of this study are very uncertain regarding the effect of prophylactic surfactant administration on the risk of any culture-proven fungal sepsis, in comparison with selective surfactant administration (RR 3.23, 95% CI 0.34 to 30.84; RD 0.01, 95% CI -0.01 to 0.03; I^2 not applicable; 1 trial, 426 infants; very low-certainty evidence). We downgraded the certainty of the evidence to very low due to risk of bias and very serious imprecision in the effect estimate. No subgroup analysis could be performed because there was only one study; no studies were identified that reported on this outcome without routine application of CPAP.

1.15. Necrotizing enterocolitis (NEC defined as Bell Stage II or greater)

Eight studies reported on the risk of NEC (Bevilacqua 1996; Dunn 1991; Dunn 2011; Kattwinkel 1993; Kendig 1991; Merritt 1991; SUPPORT 2010; Walti 1995). Two of them, Dunn 2011 and SUPPORT 2010, routinely placed control infants on CPAP.

The meta-analysis including all eight studies, regardless of whether control infants were routinely placed on CPAP, suggests that prophylactic surfactant administration may result in little to no difference in the risk of NEC when compared with selective surfactant administration (RR 0.90, 95% CI 0.73 to 1.10; $I^2 = 0\%$; RD -0.01, 95% CI -0.02 to 0.01; $I^2 = 0\%$; 8 trials, 4237 infants; low-certainty evidence; Analysis 1.15). No heterogeneity was present. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

Subgroups based on routine use of CPAP are reported below in 1.15.1 and 1.15.2. The test for subgroup differences indicates low heterogeneity ($I^2 = 42.2\%$, $P = 0.19$), suggesting that routinely placing infants on CPAP does not modify the prophylactic surfactant effects on NEC.

1.15.1. NEC in studies without routine application of continuous positive airway pressure to infants in control groups

Six studies in this subgroup reported on NEC (Bevilacqua 1996; Dunn 1991; Kattwinkel 1993; Kendig 1991; Merritt 1991; Walti 1995).

None of these studies reported an effect of prophylactic surfactant on the incidence of NEC.

The meta-analysis of these six studies suggests that prophylactic surfactant administration may result in little to no difference in the risk of NEC when compared with the selective surfactant administration (RR 1.05, 95% CI 0.76 to 1.44; $I^2 = 0\%$; RD 0.00, 95% CI -0.02 to 0.02; $I^2 = 0\%$; 6 trials, 2516 infants; low-certainty evidence; [Analysis 1.15](#)). No heterogeneity was noted. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

1.15.2. NEC in studies with routine application of continuous positive airway pressure to infants in control groups

[Dunn 2011](#) and [SUPPORT 2010](#) routinely placed control infants on CPAP. Neither trial reported an effect of prophylactic surfactant on NEC.

The meta-analysis of these two studies suggests that prophylactic surfactant administration may result in little to no difference in the risk of NEC when compared with selective use of surfactant in infants with respiratory failure and early stabilization on CPAP (RR 0.79, 95% CI 0.60 to 1.04; $I^2 = 0\%$; RD -0.02, 95% CI -0.05 to 0.00; $I^2 = 0\%$; 2 trials, 1721 infants; low-certainty evidence; [Analysis 1.15](#)). No heterogeneity was present. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

1.16. Intraventricular hemorrhage (IVH any grade)

Nine of the included studies reported on the incidence of intraventricular hemorrhage ([Bevilacqua 1996](#); [Bevilacqua 1997](#); [Dunn 1991](#); [Dunn 2011](#); [Egberts 1993](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Merritt 1991](#); [Walti 1995](#)). One of them, [Dunn 2011](#), routinely placed control infants on CPAP.

The meta-analysis that included all nine studies regardless of whether control infants were routinely placed on CPAP suggests that prophylactic surfactant administration likely results in little to no difference in the risk of IVH when compared with selective use of surfactant administration (RR 0.94, 95% CI 0.85 to 1.04; $I^2 = 0\%$; RD -0.02, 95% CI -0.05 to 0.01; $I^2 = 7\%$; 9 trials, 3077 infants; moderate-certainty evidence; [Analysis 1.16](#)). No heterogeneity was noted. We downgraded the certainty of the evidence to moderate due to the risk of bias in the included studies.

Subgroups based on routine use of CPAP are reported below in 1.16.1, and 1.16.2. The test for subgroup differences indicates no heterogeneity ($I^2 = 0\%$, $P = 0.51$), suggesting that routinely placing infants on CPAP does not modify the effect of prophylactic surfactant on intraventricular hemorrhage.

1.16.1. IVH in studies without routine application of continuous positive airway pressure to infants in control groups

Eight studies did not routinely place control infants on CPAP ([Bevilacqua 1996](#); [Bevilacqua 1997](#); [Dunn 1991](#); [Egberts 1993](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Merritt 1991](#); [Walti 1995](#)).

The meta-analysis of these eight studies revealed that prophylactic surfactant administration likely results in little to no difference in the risk of IVH when compared with selective surfactant administration (RR 0.93, 95% CI 0.83 to 1.03; $I^2 = 0\%$; RD -0.02, 95% CI -0.06 to 0.01; $I^2 = 14\%$; 8 trials, 2656 infants; moderate-

certainty evidence; [Analysis 1.16](#)). No heterogeneity was noted. We downgraded the certainty of the evidence to moderate for risk of bias in the included studies.

1.16.2. IVH in studies with routine application of continuous positive airway pressure to infants in control groups

[Dunn 2011](#) was the only study with routine application of CPAP that reported on any grade of IVH.

The results of this study suggest that prophylactic surfactant administration may result in little to no difference in the risk of IVH when compared with selective use of surfactant in infants stabilized early on CPAP (RR 1.05, 95% CI 0.73 to 1.50; RD 0.01, 95% CI -0.07 to 0.09; I^2 not applicable; 1 trial, 421 infants; low-certainty evidence; [Analysis 1.16](#)). We downgraded the certainty of the evidence to low due to the risk of bias in the included study and serious imprecision in the effect estimates.

1.17. Severe intraventricular hemorrhage (grades 3 and 4 IVH)

Ten studies reported severe IVH grades 3 and 4 ([Bevilacqua 1996](#); [Bevilacqua 1997](#); [Dunn 1991](#); [Dunn 2011](#); [Egberts 1993](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Merritt 1991](#); [SUPPORT 2010](#); [Walti 1995](#)). [Dunn 2011](#) and [SUPPORT 2010](#) routinely placed control infants on CPAP.

The meta-analysis that included all 10 studies regardless of whether control infants were routinely placed on CPAP revealed that prophylactic surfactant administration may result in no difference in the risk of IVH in comparison with selective administration of surfactant, (RR 0.87, 95% CI 0.74 to 1.04; $I^2 = 46\%$; RD -0.01, 95% CI -0.03 to 0.00; $I^2 = 61\%$; 10 trials, 4347 infants; low-certainty evidence; [Analysis 1.17](#)). There was moderate heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious inconsistency.

Subgroups based on routine use of CPAP are reported below in 1.16.1 and 1.16.2. The test for subgroup differences indicates no heterogeneity ($I^2 = 0\%$, $P = 0.95$), suggesting that routinely placing infants on CPAP does not modify the prophylactic surfactant effect on severe intraventricular hemorrhage.

1.17.1. Severe IVH in studies without routine application of continuous positive airway pressure to infants in control groups

Eight trials did not routinely place control infants on CPAP ([Bevilacqua 1996](#); [Bevilacqua 1997](#); [Dunn 1991](#); [Egberts 1993](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Merritt 1991](#); [Walti 1995](#)).

The meta-analysis of these eight studies shows very uncertain results that do not allow us to determine the effects that prophylactic surfactant administration may have on the risk of IVH in comparison with selective administration of surfactant (RR 0.87, 95% CI 0.70 to 1.08; $I^2 = 46\%$; RD -0.01, 95% CI -0.04 to 0.01; $I^2 = 60\%$; 8 trials, 2656 infants; very low certainty evidence; [Analysis 1.17](#)). There was moderate heterogeneity. We downgraded the certainty of the evidence for risk of bias in the included studies, serious inconsistency, and serious imprecision in the effect estimates.

1.17.2. Severe IVH in studies with routine application of continuous positive airway pressure to infants in control groups

[SUPPORT 2010](#) and [Dunn 2011](#) routinely placed control infants on CPAP.

The meta-analysis of these two studies shows very uncertain results that do not allow us to determine the effects of prophylactic surfactant administration on the risk of IVH compared with selective use of surfactant in infants stabilized early on CPAP (RR 0.88, 95% CI 0.67 to 1.16; $I^2 = 73\%$; RD -0.01, 95% CI -0.04 to 0.02; $I^2 = 83\%$; 2 trials, 1691 infants; very low-certainty evidence; [Analysis 1.17](#)). There was moderate heterogeneity. We downgraded the certainty of the evidence to very low due to the risk of bias in the included studies, serious inconsistency, and serious imprecision in the effect estimates.

1.18. Periventricular leukomalacia (PVL)

Five studies reported on PVL ([Dunn 2011](#); [Egberts 1993](#); [Kattwinkel 1993](#); [Merritt 1991](#); [Walti 1995](#)). One of them, [Dunn 2011](#), routinely placed control infants on CPAP.

The meta-analysis that included all five studies regardless of whether control infants were routinely placed on CPAP suggests that prophylactic surfactant administration may result in little to no difference in the risk of PVL when compared to selective surfactant administration in infants with respiratory failure (RR 1.23, 95% CI 0.88 to 1.72; $I^2 = 0\%$; RD 0.01, 95% CI -0.01 to 0.03; $I^2 = 0\%$; 5 trials, 2091 infants; low-certainty evidence; [Analysis 1.18](#)). No heterogeneity was present. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

Subgroups based on routine use of CPAP are reported in 1.18.1 and 1.18.2. The test for subgroup differences indicates no heterogeneity ($I^2 = 0\%$, $P = 0.55$), suggesting that routinely placing infants on CPAP does not modify the prophylactic surfactant effect on PVL.

1.18.1. PVL in studies without routine application of continuous positive airway pressure to infants in control groups

Four studies did not routinely place control infants on CPAP ([Egberts 1993](#); [Kattwinkel 1993](#); [Merritt 1991](#); [Walti 1995](#)). None of these trials reported an effect of prophylactic surfactant on the incidence of PVL.

The meta-analysis of these four studies suggests that prophylactic surfactant administration may result in an increase in the risk of PVL when compared to selective administration (RR 1.26, 95% CI 0.89 to 1.77; $I^2 = 3\%$; RD 0.01, 95% CI -0.01 to 0.04; $I^2 = 0\%$; 4 trials, 1695 infants; low-certainty evidence; [Analysis 1.18](#)). No heterogeneity was present. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

1.18.2. PVL in studies with routine application of continuous positive airway pressure to infants in control groups

[Dunn 2011](#) was the only study of this subgroup reporting on PVL.

The results of this study are very uncertain regarding the effect of prophylactic surfactant administration on the risk of PVH when compared with selective use of surfactant in infants stabilized early on CPAP (RR 0.72, 95% CI 0.12 to 4.28; RD -0.00, 95% CI -0.03 to 0.02; I^2 not applicable; 1 trial, 396 infants; very low-certainty evidence; [Analysis 1.18](#)). We downgraded the certainty of the evidence to low due to the risk of bias in the included study and very serious imprecision in the effect estimates.

1.19. Retinopathy of prematurity (ROP all ICROP stages)

Eight studies reported on the incidence of all stages of retinopathy (only in survivors) ([Bevilacqua 1996](#); [Dunn 1991](#); [Dunn 2011](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Merritt 1991](#); [SUPPORT 2010](#); [Walti 1995](#)). [Dunn 2011](#) and [SUPPORT 2010](#) routinely placed control infants on CPAP.

The meta-analysis that included all studies regardless of whether control infants were routinely placed on CPAP suggests that prophylactic surfactant administration may have little to no effect on ROP compared with selective surfactant administration (RR 1.03, 95% CI 0.89 to 1.19; $I^2 = 63\%$; RD 0.00, 95% CI -0.02 to 0.03; $I^2 = 66\%$; 8 trials, 3574 infants; very low-certainty evidence; [Analysis 1.19](#)). There was moderate heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious inconsistency. Using thresholds of $RR < 0.75$ or > 1.25 to define a clinically important effect, we did not downgrade the certainty of the evidence for imprecision.

Subgroups based on routine use of CPAP are reported in 1.19.1 and 1.19.2. The test for subgroup differences indicates moderate heterogeneity ($I^2 = 66.0\%$, $P = 0.09$), suggesting that routinely placing infants on CPAP might modify the prophylactic surfactant effect on ROP.

1.19.1. ROP in studies without routine application of continuous positive airway pressure to infants in control groups

Six studies did not routinely place control infants on CPAP ([Bevilacqua 1996](#); [Dunn 1991](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Merritt 1991](#); [Walti 1995](#)). None of these studies reported an effect of prophylactic surfactant on the incidence of ROP.

Results from the meta-analysis of these six studies are very uncertain and do not allow us to determine the effect of prophylactic surfactant administration on the risk of ROP (RR 1.17, 95% CI 0.95 to 1.45; $I^2 = 63\%$; RD 0.02, 95% CI -0.01 to 0.04; $I^2 = 69\%$; 6 trials, 2215 infants; very low-certainty evidence; [Analysis 1.19](#)). There was moderate heterogeneity. We downgraded the certainty of the evidence for risk of bias in the included studies, serious inconsistency, and serious imprecision in the effect estimates.

1.19.2. ROP in studies with routine application of continuous positive airway pressure to infants in control groups

[Dunn 2011](#) and [SUPPORT 2010](#) routinely placed control infants on CPAP and reported on this outcome.

The meta-analysis of these two studies suggests that prophylactic surfactant administration may have little to no effect on ROP when compared with selective use of surfactant in infants stabilized early on CPAP (RR 0.91, 95% CI 0.75 to 1.11; $I^2 = 42\%$; RD -0.02, 95% CI -0.06 to 0.02; $I^2 = 69\%$; 2 trials, 1359 infants; low-certainty evidence; [Analysis 1.19](#)). There was low heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

1.20. Severe retinopathy of prematurity (ICROP stages 2 or greater)

Five studies reported on ROP stages 2 or greater ([Bevilacqua 1996](#); [Dunn 1991](#); [Dunn 2011](#); [Kattwinkel 1993](#); [SUPPORT 2010](#)). Of these five trials, [Dunn 2011](#) and [SUPPORT 2010](#) routinely placed control infants on CPAP.

The meta-analysis that included all studies regardless of whether control infants were routinely placed on CPAP suggests that prophylactic surfactant administration may not reduce the risk of stage 2 to 4 retinopathy when compared with selective surfactant administration in infants with respiratory failure (RR 0.97, 95% CI 0.73 to 1.28; $I^2 = 0\%$; RD -0.00, 95% CI -0.02 to 0.02; $I^2 = 0\%$; 5 trials, 2738 infants; low-certainty evidence; [Analysis 1.20](#)). No important heterogeneity was noted. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

Subgroups based on routine use of CPAP are reported in 1.20.1 and 1.20.2. The test for subgroup differences indicates no heterogeneity ($I^2 = 0\%$, $P = 1.00$), suggesting that routinely placing infants on CPAP does not modify the prophylactic surfactant effects on ROP stages 2 or greater.

1.20.1. ROP Stages 2 or greater in studies without routine application of continuous positive airway pressure to infants in control groups

Three studies that did not routinely place control infants on CPAP reported on this outcome ([Bevilacqua 1996](#); [Dunn 1991](#); [Kattwinkel 1993](#)). None of these trials reported an effect of prophylactic surfactant on ROP stages 2 or greater.

The meta-analysis of these trials shows very uncertain results that do not allow us to determine the effects of prophylactic surfactant administration on the risk of ROP stages 2 or more when compared to selective surfactant administration (RR 0.97, 95% CI 0.46 to 2.04; $I^2 = 0\%$; RD -0.00, 95% CI -0.01 to 0.01; $I^2 = 0\%$; 3 trials, 1379 infants; very low certainty evidence; [Analysis 1.20](#)). No heterogeneity was present. We downgraded the certainty of the evidence to very low due to the risk of bias in the included studies and very serious imprecision in the effect estimates.

1.20.2. ROP Stages 2 to 4 in studies with routine application of continuous positive airway pressure to infants in control groups

[Dunn 2011](#) and [SUPPORT 2010](#) reported on ROP stages 2 or more. Neither study reported an effect of prophylactic surfactant on ROP stages 2 or greater.

The meta-analysis of these two studies suggests that prophylactic surfactant administration may result in little to no difference in the risk of stage 2 to 4 retinopathy when compared with early stabilization on CPAP with selective use of surfactant in infants with respiratory failure (RR 0.97, 95% CI 0.72 to 1.30; $I^2 = 39\%$; RD -0.00, 95% CI -0.04 to 0.03; $I^2 = 32\%$; 2 trials, 1359 infants; low-certainty evidence; [Analysis 1.20](#)). No important heterogeneity was noted. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by receipt of antenatal steroids)

Of the 10 included studies, only two enrolled a population of infants in whom over 50% of mothers had received antenatal steroids (defined as one complete course of steroids) ([Dunn 2011](#); [SUPPORT 2010](#)). Five trials enrolled a population of infants in whom less than 50% of mothers had received antenatal steroids ([Bevilacqua 1996](#); [Bevilacqua 1997](#); [Dunn 1991](#); [Egberts 1993](#); [Walti 1995](#)). Three trials did not report on the treatment of mothers with antenatal steroids ([Kattwinkel 1993](#); [Kendig 1991](#); [Merritt 1991](#)). Analyses for

this subgroup are based on the seven studies that reported on the use of antenatal steroids.

Critical and important outcomes

2.1 Chronic lung disease (use of supplemental oxygen at 36 weeks' postmenstrual age)

Five studies on antenatal steroid use reported on CLD at 36 weeks' postmenstrual age ([Bevilacqua 1997](#); [Dunn 1991](#); [Dunn 2011](#); [Egberts 1993](#); [SUPPORT 2010](#)).

Subgroups based on maternal treatment with antenatal steroids are reported below in 2.1.1 and 2.1.2. The test for subgroup differences indicates that there is no subgroup effect ($I^2 = 0\%$, $P = 0.40$; [Analysis 2.1](#)), suggesting that treatment with antenatal steroids does not modify the effect of compared interventions on chronic lung disease.

2.1.1. CLD in studies in which < 50% of mothers received antenatal steroids

There were three studies in which < 50% of mothers received antenatal steroids ([Bevilacqua 1996](#); [Dunn 1991](#); [Egberts 1993](#)).

The meta-analysis of these three studies shows very uncertain results that do not allow us to determine the effects that prophylactic surfactant may have on the risk of CLD compared with selective surfactant administration in this population of infants (RR 1.56, 95% CI 0.72 to 3.40; $I^2 = 0\%$; RD 0.03, 95% CI -0.02 to 0.08; $I^2 = 0\%$; 3 trials, 362 infants; very low-certainty evidence; [Analysis 2.1](#)). No heterogeneity was noted. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and very serious imprecision in the effect estimates.

2.1.2. CLD in studies in which $\geq 50\%$ of mothers received antenatal steroids

Two studies reported the treatment of $\geq 50\%$ of mothers with antenatal steroids ([Dunn 2011](#); [SUPPORT 2010](#)). These studies independently reported an increased risk of CLD with the prophylactic administration of surfactant when compared with the selective use of surfactant in infants with respiratory failure.

The meta-analysis of these two studies reveals that prophylactic surfactant administration probably results in little to no difference in the risk of CLD when compared with the selective administration in infants whose mothers mostly received antenatal steroids (RR 1.12, 95% CI 0.99 to 1.26; $I^2 = 0\%$; RD 0.04, 95% CI -0.00 to 0.09; $I^2 = 0\%$; 2 trials, 1512 infants; low-certainty evidence; [Analysis 2.1](#)). No heterogeneity was noted. We downgraded the certainty of evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

2.2 Mortality at 36 weeks' PMA or prior to hospital discharge

Four studies on the antenatal use of steroids reported mortality at 36 weeks' PMA or prior to discharge ([Dunn 2011](#); [Egberts 1993](#); [SUPPORT 2010](#); [Walti 1995](#)).

Subgroups based on maternal treatment with antenatal steroids are reported below (see 2.2.1 and 2.2.2). The test for subgroup differences indicates there is important heterogeneity ($I^2 = 89.4\%$, $P = 0.002$; [Analysis 2.2](#)), suggesting that the treatment of mothers with antenatal steroids modifies the prophylactic surfactant effect on mortality of infants with respiratory failure.

2.2.1. Mortality at 36 weeks' PMA in studies that treated < 50% of mothers with antenatal steroids

Two studies treated < 50% of mothers with antenatal steroids (Egberts 1993; Walti 1995).

The meta-analysis of the two studies in this subgroup suggests that prophylactic surfactant administration probably reduces the risk of mortality at 36 weeks' PMA in this population of infants (RR 0.59, 95% CI 0.38 to 0.91; $I^2 = 0\%$; RD -0.09, 95% CI -0.17 to -0.02; $I^2 = 0\%$; NNTB 11 9% CI 6 to 50; 2 trials, 403 infants; moderate-certainty evidence; Analysis 2.2). No heterogeneity was noted. We downgraded the certainty of evidence to moderate due to the risk of bias in the included studies.

2.2.2. Mortality at 36 weeks' PMA in studies that treated $\geq 50\%$ of mothers with antenatal steroids

Two studies treated $\geq 50\%$ of mothers with antenatal steroids (Dunn 2011; SUPPORT 2010).

The meta-analysis of these two studies suggests that the risk of mortality at 36 weeks' PMA may increase slightly with the use of prophylactic surfactant in comparison with the selective administration in infants with respiratory failure whose mothers mostly received antenatal steroids (RR 1.28, 95% CI 1.01 to 1.62; $I^2 = 0\%$; RD 0.03, 95% CI 0.00 to 0.06; NNTB 33, 95% CI 17 to > 1000; $I^2 = 0\%$; 2 trials, 1748 infants; low-certainty evidence; Analysis 2.2). No heterogeneity was noted. We downgraded the certainty of evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

2.3. Pneumothorax

Five studies on antenatal steroid use reported on pneumothorax (Bevilacqua 1996; Dunn 1991; Dunn 2011; Egberts 1993; Walti 1995).

Subgroups based on maternal treatment with antenatal steroids are reported below in 2.3.1 and 2.3.2. The test for subgroup differences indicates that there is no subgroup effect ($I^2 = 0\%$, $P = 0.48$; Analysis 2.3), suggesting that treatment with antenatal steroids may not modify the prophylactic surfactant effect on pneumothorax.

2.3.1. Studies that treated < 50% of mothers with antenatal steroids

Four studies treated < 50% of mothers with antenatal steroids (Bevilacqua 1996; Dunn 1991; Egberts 1993; Walti 1995).

The meta-analysis of these four studies suggests that prophylactic surfactant administration may result in a reduction in the risk of pneumothorax in comparison with selective administration of surfactant in this population of infants (RR 0.62, 95% CI 0.35 to 1.10; $I^2 = 0\%$; RD -0.03, 95% CI -0.06 to 0.01; $I^2 = 0\%$; 4 trials, 788 infants; low-certainty evidence; Analysis 2.3). No important heterogeneity was noted. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

2.3.2. Studies that treated $\geq 50\%$ of mothers with antenatal steroids

Only Dunn 2011 reported that $\geq 50\%$ of mothers were treated with antenatal steroids.

The results of this study on pneumothorax are uncertain and do not allow us to determine the effects of prophylactic surfactant on the risk of pneumothorax compared with selective surfactant

administration in this population of infants whose mothers mostly received antenatal steroids (RR 0.89, 95% CI 0.39 to 2.01; RD -0.01, 95% CI -0.05 to 0.04; I^2 not applicable; 1 trial, 431 infants; very low-certainty evidence; Analysis 2.3). We downgraded the certainty of the evidence to very low due to risk of bias and very serious imprecision in the effect estimates.

2.4. Chronic lung disease or death (use of supplemental oxygen or death at 36 weeks' postmenstrual age)

Only four studies providing information on neonatal steroid use reported on the combined outcome of CLD or death (Dunn 1991; Dunn 2011; SUPPORT 2010; Walti 1995).

Subgroups based on antenatal steroid treatment are reported below in 2.4.1 and 2.4.2. The test for subgroup differences indicates that there is high heterogeneity ($I^2 = 52.0\%$, $P = 0.15$; Analysis 2.4), suggesting that treatment with antenatal steroids modifies the prophylactic surfactant effect on CLD or death.

2.4.1. Chronic lung disease or death in studies that treated < 50% of mothers with antenatal steroids

Two studies treated < 50% of mothers with antenatal steroids (Dunn 1991; Walti 1995).

The meta-analysis of the two studies in this subgroup suggest that prophylactic surfactant administration may result in little to no difference in the risk of CLD or death in comparison with selective administration of surfactant in this population of infants with respiratory failure (RR 0.87, 95% CI 0.63 to 1.21; $I^2 = 47\%$; RD -0.04, 95% CI -0.13 to 0.05; $I^2 = 55\%$; 2 trials, 378 infants; low-certainty evidence; Analysis 2.4). There was moderate heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

2.4.2. Chronic lung disease or death in studies that treated $\geq 50\%$ of mothers with antenatal steroids

Both studies with more than $\geq 50\%$ of mothers treated with antenatal steroids reported a concerning trend toward increased risk of CLD or death in the prophylactic surfactant administration group (Dunn 2011; SUPPORT 2010).

The meta-analysis of these two studies demonstrated that prophylactic surfactant administration likely results in little to no difference in the risk of CLD or death when compared to selective use of surfactant in infants with respiratory failure whose mothers mostly received antenatal steroids (RR 1.12, 95% CI 1.02 to 1.24; $I^2 = 0\%$; RD 0.06, 95% CI 0.01 to 0.10; $I^2 = 0\%$; 2 trials, 1744 infants; moderate-certainty evidence; Analysis 2.4). No heterogeneity was noted. We downgraded the certainty of evidence to moderate due to the risk of bias in the included studies.

2.5. Bronchopulmonary dysplasia (oxygen requirement at 28 to 30 days of age)

Six studies on antenatal steroid use reported on bronchopulmonary dysplasia at 28 to 30 days of age (Bevilacqua 1996; Bevilacqua 1997; Dunn 1991; Dunn 2011; Egberts 1993; Walti 1995).

Subgroups based on maternal treatment with antenatal steroids are reported below in 2.5.1 and 2.5.2. The test for subgroup differences indicates that there is no subgroup effect ($I^2 = 0\%$,

$P = 0.45$; [Analysis 2.5](#)), suggesting that treatment with antenatal steroids may not modify the prophylactic surfactant effect on BPD.

2.5.1. Studies in which < 50% of mothers received antenatal steroids

Five studies treated < 50% of mothers with antenatal steroids ([Bevilacqua 1996](#); [Bevilacqua 1997](#); [Dunn 1991](#); [Egberts 1993](#); [Walti 1995](#)).

The meta-analysis of the five studies in this subgroup suggest that prophylactic surfactant administration may result in little to no difference in the risk of bronchopulmonary dysplasia in comparison with the selective administration of surfactant in this population of infants (RR 1.04, 95% CI 0.83 to 1.31; $I^2 = 70\%$; RD 0.01 95% CI -0.05 to 0.07; $I^2 = 74\%$; 5 trials, 798 infants; low-certainty evidence; [Analysis 2.5](#)). There was moderate heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

2.5.2. Studies in which $\geq 50\%$ of mothers received antenatal steroids

[Dunn 2011](#) was the only study that treated $\geq 50\%$ of mothers with antenatal steroids and reported on this outcome.

The results of this study suggest that prophylactic surfactant administration may result in little to no difference in the risk of bronchopulmonary dysplasia in comparison with the selective surfactant administration in infants whose mothers mostly received antenatal steroids (RR 1.16, 95% CI 0.99 to 1.36; RD 0.09 95% CI -0.01 to 0.18; I^2 not applicable; 1 trial, 402 infants; low-certainty evidence; [Analysis 2.5](#)). We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

2.6 Bronchopulmonary dysplasia or death at 28 to 30 days of age

Seven studies on the use of antenatal steroids reported on the combined outcome of bronchopulmonary dysplasia or death at 28 to 30 days ([Bevilacqua 1996](#); [Bevilacqua 1997](#); [Dunn 1991](#); [Dunn 2011](#); [Egberts 1993](#); [Walti 1995](#)). Subgroups based on antenatal steroid treatment are reported below in 2.6.1 and 2.6.2. The test for subgroup differences indicates that there is high heterogeneity ($I^2 = 83.8\%$, $P = 0.001$; [Analysis 2.6](#)), suggesting that treatment with antenatal steroids modifies the prophylactic surfactant effect on the combined outcome of bronchopulmonary dysplasia or death.

2.6.1. Bronchopulmonary dysplasia or death at 28 to 30 days of age in studies that treated < 50% of mothers with antenatal steroids

Five studies treated < 50% of mothers with antenatal steroids ([Bevilacqua 1996](#); [Bevilacqua 1997](#); [Dunn 1991](#); [Egberts 1993](#); [Walti 1995](#)).

The results of the meta-analysis of the five studies are uncertain and do not allow us to determine the effects of prophylactic surfactant administration in the risk of bronchopulmonary dysplasia or death at 28 to 30 days in comparison with selective administration of surfactant in this population of infants with respiratory failure (RR 0.85, 95% CI 0.73 to 0.99; $I^2 = 78\%$; RD -0.07, 95% CI -0.13 to -0.00; NNTB 14, 95% CI 8 to >1000; $I^2 = 79\%$; 5 trials, 886 infants; low-certainty evidence; [Analysis 2.6](#)). There was substantial heterogeneity. We downgraded the certainty of the evidence to very low due to the risk of bias in the included studies, inconsistency of results, and serious imprecision in the effect estimates.

2.6.2. Bronchopulmonary dysplasia or death at 28 to 30 days of age in studies that treated $\geq 50\%$ of mothers with antenatal steroids

Only [Dunn 2011](#) treated more than $\geq 50\%$ of mothers with antenatal steroids. The results of this study suggest that prophylactic surfactant administration may result in little to no difference in the risk of bronchopulmonary dysplasia or death at 28 to 30 days in comparison with the selective administration in infants with respiratory failure whose mothers mostly received antenatal steroids (RR 1.12, 95% CI 0.96 to 1.31; RD 0.07, 95% CI -0.02 to 0.16; I^2 not applicable; 1 trial, 420 infants; low-certainty evidence; [Analysis 2.6](#)). We downgraded the certainty of the evidence to low due to the risk of bias in the included study and serious imprecision in the effect estimates.

Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed on the threshold used to apply selective treatment)

Under this subgroup heading, we discuss the results of 10 studies that reported on the criteria used to determine the decision to use selective treatment ([Bevilacqua 1996](#); [Bevilacqua 1997](#); [Dunn 1991](#); [Dunn 2011](#); [Egberts 1993](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Merritt 1991](#); [SUPPORT 2010](#); [Walti 1995](#)). The subgroups are based on the requirement for oxygen supplementation (FiO_2 required) to maintain adequate blood oxygen saturation. This specific subgroup analysis was performed only for the main outcomes.

Critical and important outcomes

3.1 Chronic lung disease (use of supplemental oxygen at 36 weeks' postmenstrual age)

Six studies reported on chronic lung disease ([Bevilacqua 1997](#); [Bevilacqua 1996](#); [Dunn 1991](#); [Dunn 2011](#); [Egberts 1993](#); [SUPPORT 2010](#)). Subgroups based on the criteria used to apply selective treatment are reported below in 3.1.1 to 3.1.5. The test for subgroup differences indicates that there is no subgroup effect ($I^2 = 0\%$, $P = 0.61$; [Analysis 3.1](#)), suggesting that the FiO_2 threshold used to apply selective treatment may not modify the surfactant effect on chronic lung disease (CLD).

3.1.1 Studies applying selective treatment at $FiO_2 > 0.3$

None of the included studies within this subgroup reported on CLD.

3.1.2 Studies applying selective treatment at $FiO_2 > 0.4$

Two studies treated infants in the selective group at $FiO_2 > 0.4$ and reported on CLD ([Bevilacqua 1997](#); [Dunn 2011](#)). Meta-analysis of these two studies suggests that prophylactic surfactant administration may result in little to no difference in the risk of CLD when compared to selective use of surfactant in infants with respiratory failure requiring a $FiO_2 > 0.4$ (RR 1.19, 95% CI 0.91 to 1.55; $I^2 = 0\%$; RD 0.05, 95% CI -0.03 to 0.13; $I^2 = 1\%$; 2 trials, 497 infants; low-certainty evidence). No heterogeneity was noted. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

3.1.3 Studies applying selective treatment at $FiO_2 > 0.5$

[SUPPORT 2010](#) treated infants in the selective group at $FiO_2 > 0.5$. The results of this study suggest that prophylactic surfactant administration may result in little to no difference in the risk of CLD in comparison with the selective treatment in infants with

respiratory failure requiring a $\text{FiO}_2 > 0.5$ (RR 1.10, 95% CI 0.96 to 1.26; RD 0.04, 95% CI -0.02 to 0.10; I^2 not applicable; 1 trial, 1108 infants; low-certainty evidence). We downgraded the certainty of the evidence to low due to the risk of bias in the included study and serious imprecision in the effect estimates.

3.1.4 Studies applying selective treatment at $\text{FiO}_2 > 0.6$

One study treated infants in the selective group at $\text{FiO}_2 > 0.6$ (Egberts 1993). The results of this study reveal that the evidence is very uncertain about the effect of prophylactic surfactant administration on the risk of CLD compared to selective treatment in infants with respiratory failure requiring a $\text{FiO}_2 > 0.6$ (RR 1.34, 95% CI 0.45 to 4.04; RD 0.02 95% CI -0.06 to 0.11; I^2 not applicable; 1 trial, 147 infants; very low-certainty evidence). We downgraded the certainty of the evidence to very low due to the risk of bias in the included study and very serious imprecision in the effect estimates.

3.1.5 Studies applying selective treatment based on other indications

Two studies treated infants in the selective group based on other indications (Bevilacqua 1996; Dunn 1991). The meta-analysis of these studies suggests that prophylactic surfactant administration may result in little to no difference in the risk of CLD when compared to selective surfactant administration in infants with respiratory failure based on other indications than FiO_2 (RR 0.88 95% CI 0.49 to 1.58; $I^2 = 41\%$; RD -0.01 95% CI -0.07 to 0.05; $I^2 = 46\%$; 2 trials, 390 infants; low-certainty evidence). There was low heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

3.2 Mortality at 36 weeks' PMA or prior to hospital discharge

Six studies reported on mortality at 36 weeks' PMA or prior to hospital discharge (Dunn 2011; Egberts 1993; Kendig 1991; Merritt 1991; SUPPORT 2010; Walti 1995). Subgroups based on the threshold used to apply selective treatment are reported below in 3.2.1 to 3.2.5. The test for subgroup differences indicates that there is a subgroup effect ($I^2 = 71.7\%$, $P = 0.01$; Analysis 3.2), suggesting that the threshold used to apply selective treatment modifies the prophylactic surfactant effect on mortality.

3.2.1 Studies applying selective treatment at $\text{FiO}_2 > 0.3$

There were no studies reporting on mortality at 36 weeks' PMA in this subgroup.

3.2.2 Studies applying selective treatment at $\text{FiO}_2 > 0.4$

Two studies treated infants in the selective group at $\text{FiO}_2 > 0.4$ (Dunn 2011; Kendig 1991). The meta-analysis of the two studies in this subgroup shows very uncertain results that do not allow us to determine the effects of prophylactic surfactant administration on mortality at 36 weeks or prior to discharge in comparison with selective administration of surfactant in infants with respiratory failure requiring a $\text{FiO}_2 > 0.4$ (RR 0.79, 95% CI 0.55 to 1.14; $I^2 = 81\%$; RD -0.03, 95% CI -0.07 to 0.01; $I^2 = 89\%$; 2 trials, 911 infants; low-certainty evidence). There was substantial heterogeneity. We downgraded the certainty of the evidence to very low due to the risk of bias in the included studies, important inconsistency, and serious imprecision in the effect estimates.

3.2.3 Studies applying selective treatment at $\text{FiO}_2 > 0.5$

Two studies treated infants in the selective group using a threshold of $\text{FiO}_2 > 0.5$ (Merritt 1991; SUPPORT 2010). Meta-analysis of these two studies suggests that prophylactic surfactant administration may result in little to no difference in the risk of mortality at 36 weeks or prior to discharge when compared to selective use of surfactant in infants with respiratory failure requiring a $\text{FiO}_2 > 0.5$ (RR 1.22, 95% CI 0.98 to 1.52; $I^2 = 0\%$; RD 0.04, 95% CI -0.00 to 0.07; $I^2 = 0\%$; 2 trials, 1464 infants; low-certainty evidence). No heterogeneity was noted. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

3.2.4 Studies applying selective treatment at $\text{FiO}_2 > 0.6$

Only Egberts 1993 treated infants in the selective group at $\text{FiO}_2 > 0.6$. The results of this study are very uncertain and do not allow us to determine the effects of prophylactic surfactant on the risk of mortality at 36 weeks or prior to discharge compared with selective surfactant administration in infants requiring a $\text{FiO}_2 > 0.6$ (RR 0.64, 95% CI 0.31 to 1.33; RD -0.08, 95% CI -0.20 to 0.05; I^2 not applicable; 1 trial, 147 infants; low-certainty evidence). We downgraded the certainty of the evidence to very low due to the risk of bias in the included study and very serious imprecision in the effect estimates.

3.2.5 Studies applying selective treatment based on other indications

Walti 1995 treated infants in the selective group based on other indications. The results of this study suggested that prophylactic surfactant administration may reduce the risk of mortality at 36 weeks or prior to discharge compared with selective surfactant administration in infants with respiratory failure based on other indications than FiO_2 (RR 0.57, 95% CI 0.33 to 0.96; RD -0.10 95% CI -0.20 to -0.01; I^2 not applicable; 1 trial, 256 infants; low-certainty evidence). We downgraded the certainty of the evidence to low due to the risk of bias in the included study and imprecision in the effect estimates.

3.3. Pneumothorax

Of the eight studies reporting on pneumothorax (Bevilacqua 1996; Dunn 1991; Dunn 2011; Egberts 1993; Kattwinkel 1993; Kendig 1991; Merritt 1991; Walti 1995), only Kattwinkel 1993 applied the selective surfactant treatment at $\text{FiO}_2 > 0.3$; Merritt 1991 applied selective treatment at $\text{FiO}_2 > 0.5$; Egberts 1993 applied $\text{FiO}_2 > 0.6$; and three studies treated infants in the selective group based on other indications (Bevilacqua 1996; Dunn 1991; Walti 1995). Subgroups based on the threshold used to apply selective treatment are reported below in 3.3.1 to 3.3.5. The test for subgroup differences indicates that there is a subgroup effect ($I^2 = 40.5\%$, $P = 0.15$; Analysis 3.3), suggesting that the threshold used to apply selective treatment may not modify the prophylactic surfactant effect on pneumothorax.

3.3.1 Pneumothorax in studies that treated infants in the selective group at $\text{FiO}_2 > 0.3$

Only Kattwinkel 1993 treated infants in the selective group at $\text{FiO}_2 > 0.3$. The evidence is very uncertain regarding the effect of the prophylactic surfactant administration on the risk of pneumothorax when compared to selective use of surfactant in infants requiring a $\text{FiO}_2 > 0.3$ (RR 0.72, 95% CI 0.29 to 1.78; RD -0.00, 95% CI -0.02 to 0.01; I^2 not applicable; 1 trial, 1248 infants; very low

certainty evidence). We downgraded the certainty of the evidence to very low due to the risk of bias in the included study and very serious imprecision in the effect estimates.

3.3.2 Pneumothorax in studies that treated infants in the selective group at $\text{FiO}_2 > 0.4$

Two studies treated infants in the selective group at $\text{FiO}_2 > 0.4$ (Dunn 2011; Kendig 1991). Meta-analysis of these two studies suggests that prophylactic surfactant administration may result in a reduction of pneumothorax compared to selective use of surfactant in infants requiring a $\text{FiO}_2 > 0.4$ (RR 0.66, 95% CI 0.41 to 1.06; $I^2 = 0\%$; RD -0.03, 95% CI -0.06 to 0.00; $I^2 = 46\%$; 2 trials, 910 infants; low-certainty evidence). We detected no heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

3.3.3 Pneumothorax in studies that treated infants in the selective group at $\text{FiO}_2 > 0.5$

Merritt 1991 study treated infants in the selective group at $\text{FiO}_2 > 0.5$. The results of this study are very uncertain regarding the effect of prophylactic surfactant administration on the risk of pneumothorax compared to selective use of surfactant in infants requiring a $\text{FiO}_2 > 0.5$ (RR 2.46, 95% CI 0.92 to 6.56; RD 0.10, 95% CI -0.00 to 0.20; I^2 not applicable; 1 trial, 148 infants; very low certainty evidence). We downgraded the certainty of the evidence to very low due to the risk of bias in the included study and very serious imprecision in the effect estimates.

3.3.4 Pneumothorax in studies that treated infants in the selective group at $\text{FiO}_2 > 0.6$

One study treated infants in the selective group at $\text{FiO}_2 > 0.6$ (Egberts 1993). The results of this study are uncertain and do not allow us to determine the effects of prophylactic surfactant on the risk of pneumothorax compared to selective use of surfactant in infants requiring a $\text{FiO}_2 > 0.6$ (RR 0.38, 95% CI 0.08 to 1.92; RD -0.04, 95% CI -0.11 to 0.03; I^2 not applicable; 1 trial, 147 infants; very low-certainty evidence). We downgraded the certainty of the evidence to very low due to the risk of bias in the included study and very serious imprecision in the effect estimates.

3.3.5 Studies applying selective treatment based on other indications

Three studies treated infants in the selective group based on other indications (Bevilacqua 1996; Dunn 1991; Walti 1995). Results of the meta-analysis of these three studies suggest that prophylactic surfactant administration may result in a slight reduction of pneumothorax compared to the selective use of surfactant in infants with respiratory failure based on other indications than FiO_2 (RR 0.67, 95% CI 0.36 to 1.24; $I^2 = 0\%$; RD -0.02, 95% CI -0.06 to 0.01; $I^2 = 0\%$; 3 trials, 641 infants; low-certainty evidence). No heterogeneity was noted. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

3.4. Bronchopulmonary dysplasia (BPD) (oxygen requirement at 28 to 30 days of age)

Nine studies reported on bronchopulmonary dysplasia (Bevilacqua 1996; Bevilacqua 1997; Dunn 1991; Dunn 2011; Egberts 1993; Kattwinkel 1993; Kendig 1991; Merritt 1991; Walti 1995). Subgroups based on the threshold used to apply selective treatment are

reported below in 3.4.1 to 3.4.5. The test for subgroup differences indicates that there is no subgroup effect ($I^2 = 20.5\%$, $P = 0.28$; Analysis 3.4), suggesting that treatment of established respiratory distress analyzed by the threshold used to apply selective treatment may not modify the prophylactic surfactant effect on pneumothorax.

3.4.1 Studies applying selective treatment at $\text{FiO}_2 > 0.3$

Kattwinkel 1993 reported on this outcome. The results of this study suggest that prophylactic surfactant administration may result in a reduction in bronchopulmonary dysplasia compared to selective surfactant administration in infants requiring a $\text{FiO}_2 > 0.3$ (RR 0.65, 95% CI 0.41 to 1.03; RD -0.02, 95% CI -0.05 to 0.00; I^2 not applicable; 1 trial, 1248 infants; low-certainty evidence). We downgraded the certainty of the evidence to low due to the risk of bias in the included study and serious imprecision in the effect estimates.

3.4.2 Studies applying selective treatment at $\text{FiO}_2 > 0.4$

Three studies treated infants in the selective group at $\text{FiO}_2 > 0.4$ (Bevilacqua 1997; Dunn 1991; Kendig 1991). The meta-analysis of the three trials in this subgroup suggest that prophylactic surfactant administration may result in little to no difference in the risk of bronchopulmonary dysplasia in comparison with selective administration of surfactant in infants with respiratory failure requiring a $\text{FiO}_2 > 0.4$ (RR 1.09, 95% CI 0.95 to 1.25; $I^2 = 0\%$; RD 0.04, 95% CI -0.02 to 0.10; $I^2 = 1\%$; 3 trials, 974 infants; low-certainty evidence). No heterogeneity was noted. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

3.4.3 Studies applying selective treatment at $\text{FiO}_2 > 0.5$

Only Merritt 1991 treated infants in the selective group at $\text{FiO}_2 > 0.5$. The results of this study are highly uncertain, and do not allow us to determine the effects that prophylactic surfactant administration may have on the risk of bronchopulmonary dysplasia when compared with selective surfactant administration in infants requiring a $\text{FiO}_2 > 0.5$ (RR 0.95, 95% CI 0.38 to 2.39; RD -0.01, 95% CI -0.11 to 0.09; I^2 not applicable; 1 trial, 148 infants; very low-certainty evidence). We downgraded the certainty of the evidence to very low due to the risk of bias in the included study and very serious imprecision in the effect estimates.

3.4.4 Studies applying selective treatment at $\text{FiO}_2 > 0.6$

Egberts 1993 reported on this outcome.

The results of this study are highly uncertain, making it difficult to draw conclusions about the effects of prophylactic surfactant administration on the risk of bronchopulmonary dysplasia when compared with selective surfactant administration in infants requiring a $\text{FiO}_2 > 0.6$ (RR 1.24, 95% CI 0.72 to 2.14; RD 0.06, 95% CI -0.09 to 0.20; I^2 not applicable; 1 trial, 147 infants; very low certainty evidence). We downgraded the certainty of the evidence to very low due to the risk of bias in the included study and very serious imprecision in the effect estimates.

3.4.5 Studies applying selective treatment based on other indications

Three studies treated infants in the selective group based on other indications (Bevilacqua 1996; Dunn 1991; Walti 1995).

The results of the meta-analysis of trials in this subgroup are uncertain and do not allow us to determine the effects of prophylactic surfactant administration on the risk of bronchopulmonary dysplasia in comparison with the use of selective administration of surfactant in infants with respiratory failure based on other indications than FiO_2 (RR 1.00, 95% CI 0.77 to 1.30; $I^2 = 78\%$; RD -0.00, 95% CI -0.06 to 0.06; $I^2 = 80\%$; 3 trials, 646 infants; very low certainty evidence). There was substantial heterogeneity. We downgraded the evidence to very low due to the risk of bias in the included studies, inconsistency, and serious imprecision in estimates of the effect.

3.5 Bronchopulmonary dysplasia or death at 28 to 30 days of age

Eight studies reported on bronchopulmonary dysplasia or death at 28 to 30 days of age (Bevilacqua 1996; Bevilacqua 1997; Dunn 1991; Dunn 2011; Egberts 1993; Kattwinkel 1993; Kendig 1991; Walti 1995). Subgroups based on the threshold used to apply selective treatment are reported below in 3.5.1 to 3.5.5. The test for subgroup differences indicates that there is a subgroup effect ($I^2 = 61.1\%$, $P = 0.05$; Analysis 3.5), suggesting that the threshold used to apply selective treatment may modify the prophylactic surfactant effect on bronchopulmonary dysplasia or death.

3.5.1 Studies applying selective treatment at $\text{FiO}_2 > 0.3$

Kattwinkel 1993 treated infants in the selective group at $\text{FiO}_2 > 0.3$. The results of this study suggest that prophylactic surfactant administration probably reduces the risk of bronchopulmonary dysplasia or death compared with selective surfactant administration in infants requiring a $\text{FiO}_2 > 0.3$ (RR 0.59, 95% CI 0.38 to 0.90; RD -0.04, 95% CI -0.06 to -0.01; NNTB 25 95% 17 to 100; I^2 not applicable; 1 trial, 1248 infants; low-certainty evidence). We downgraded the certainty of the evidence to low due to the risk of bias in the included study and imprecision in estimates of the effect.

3.5.2 Studies applying selective treatment at $\text{FiO}_2 > 0.4$

Three studies treated infants in the selective group at $\text{FiO}_2 > 0.4$ (Bevilacqua 1997; Dunn 2011; Kendig 1991). Meta-analysis of these three studies suggests that there may be little to no difference between groups of surfactant administration in the risk of bronchopulmonary dysplasia or death at 28 to 30 days of age when selective treatment is applied based on $\text{FiO}_2 > 0.4$ (RR 1.00, 95% CI 0.89 to 1.12; $I^2 = 56\%$; RD -0.00, 95% CI -0.06 to 0.06; $I^2 = 56\%$; 3 trials, 992 infants; low-certainty evidence). There was moderate heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and inconsistency.

3.5.3 Studies applying selective treatment at $\text{FiO}_2 > 0.5$

There were no studies reporting on bronchopulmonary dysplasia or death that used $\text{FiO}_2 > 0.5$ as the criteria to apply selective treatment.

3.5.4 Studies applying selective treatment at $\text{FiO}_2 > 0.6$

Egberts 1993 treated infants in the selective group at $\text{FiO}_2 > 0.6$. The results of this study are highly uncertain, making it difficult to draw conclusions about the effects of prophylactic surfactant administration on the risk of bronchopulmonary dysplasia or death when compared to the selective administration of surfactant in infants requiring $\text{FiO}_2 > 0.6$ (RR 0.93, 95% CI 0.63 to 1.36; RD -0.03,

95% CI -0.19 to 0.13; I^2 not applicable; 1 trial, 147 infants; very low-certainty evidence). We downgraded the certainty of the evidence to very low due to the risk of bias in the included study and very serious imprecision in the effect estimates.

3.5.5 Studies applying selective treatment based on other indications

Three studies treated infants in the selective group based on other indications (Bevilacqua 1996; Dunn 1991; Walti 1995). The results of meta-analysis of these three studies suggests that prophylactic surfactant administration may result in little to no difference in the risk of bronchopulmonary dysplasia or death compared with the use of selective surfactant in infants with respiratory failure based on other indications than FiO_2 (RR 0.82, 95% CI 0.69 to 0.98; $I^2 = 88\%$; RD -0.09, 95% CI -0.16 to -0.01; NNTB 11, 95% CI 7 to 100; $I^2 = 89\%$; 3 trials, 646 infants; low-certainty evidence). There was substantial heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious inconsistency in the effect estimates.

3.6 Chronic lung disease or death (at 36 weeks' PMA)

Four studies reported on chronic lung disease or death (Dunn 1991; Dunn 2011; SUPPORT 2010; Walti 1995). Subgroups based on the threshold used to apply selective treatment are reported below in 3.6.1 to 3.6.5. The test for subgroup differences indicates that there is no subgroup effect ($I^2 = 13.6\%$, $P = 0.31$; Analysis 3.6), suggesting that the threshold used to apply selective treatment may not modify the prophylactic surfactant effect on CLD or death.

3.6.1 Studies applying selective treatment at $\text{FiO}_2 > 0.3$

No study reporting on CLD or death used $\text{FiO}_2 > 0.3$ as the criteria to apply selective treatment.

3.6.2 Studies applying selective treatment at $\text{FiO}_2 > 0.4$

One study applying selective treatment at $\text{FiO}_2 > 0.4$ reported on this outcome (Dunn 2011). The results of this study suggest that prophylactic surfactant administration may result in little to no difference in the risk of CLD or death when compared to the selective administration of surfactant in infants requiring $\text{FiO}_2 > 0.4$ (RR 1.20, 95% CI 0.92 to 1.57; RD 0.06 95% CI -0.03 to 0.15; I^2 not applicable; 1 trial, 428 infants; low-certainty evidence). We downgraded the certainty of the evidence to low due to the risk of bias in the included study and serious imprecision in the effect estimates.

3.6.3 Studies applying selective treatment at $\text{FiO}_2 > 0.5$

SUPPORT 2010 treated infants in the selective group at $\text{FiO}_2 > 0.5$. The results of this study suggested that prophylactic surfactant administration may result in little to no difference in the risk of CLD or death compared to the selective administration of surfactant in infants requiring $\text{FiO}_2 > 0.5$ (RR 1.11, 95% CI 1.00 to 1.23; RD 0.05, 95% CI -0.00 to 0.11; I^2 not applicable; 1 trial, 1316 infants; moderate certainty evidence). We downgraded the certainty of the evidence to moderate due to the risk of bias in the included study. Using thresholds of $\text{RR} < 0.75$ or > 1.25 to define a clinically important effect, we did not downgrade the certainty of the evidence for imprecision.

3.6.4 Studies applying selective treatment at $\text{FiO}_2 > 0.6$

No study reporting on CLD or death used $\text{FiO}_2 > 0.6$ as the criteria to apply selective treatment reported on CLD or death.

3.6.5 Studies applying selective treatment based on other indications

Two studies treated infants in the selective group based on other indications (Dunn 1991; Walti 1995).

Meta-analysis of these two studies showed that prophylactic surfactant administration may result in little to no difference in the risk of bronchopulmonary dysplasia or death when compared to selective surfactant administration in infants with respiratory failure, based on other indications than FiO_2 (RR 0.87, 95% CI 0.63 to 1.21; $I^2 = 47\%$; RD -0.04, 95% CI -0.13 to 0.05; $I^2 = 55\%$; 2 trials, 378 infants; low-certainty evidence). There was moderate heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by the method of surfactant administration)

Eight studies treated infants with intubation and ongoing ventilation (Dunn 1991; Dunn 2011; Egberts 1993; Kattwinkel 1993; Kendig 1991; Merritt 1991; SUPPORT 2010; Walti 1995). Two studies treated infants with intubation, surfactant administration, and rapid extubation (Bevilacqua 1996; Bevilacqua 1997). No studies treated infants with thin catheter surfactant administration, laryngeal mask airway administration, or hypopharyngeal surfactant administration.

Critical and important outcomes

4.1 Chronic lung disease (use of supplemental oxygen at 36 weeks' postmenstrual age)

Five studies reported on this outcome (Bevilacqua 1997; Dunn 1991; Dunn 2011; Egberts 1993; SUPPORT 2010).

Subgroups based on the method used in these studies for surfactant administration are reported below in 4.1.1 to 4.1.2. The test for subgroup differences indicates that there is no subgroup effect ($I^2 = 0\%$, $P = 0.53$; Analysis 4.1), suggesting that the method of surfactant administration may not modify the prophylactic surfactant effect on CLD.

4.1.1 Studies that treated infants with intubation and ongoing ventilation

Four studies used intubation and ongoing ventilation for surfactant administration and reported on CLD (Dunn 1991; Dunn 2011; Egberts 1993; SUPPORT 2010). The meta-analysis of these four studies suggests that prophylactic surfactant administration may result in little to no difference in the risk of CLD in comparison with selective administration of surfactant in infants with respiratory failure treated with intubation and ongoing ventilation (RR 1.13, 95% CI 1.00 to 1.28; $I^2 = 0\%$; RD 0.04, 95% CI 0.00 to 0.09; $I^2 = 0\%$; NNTH 25, 95% CI 12 to > 1000 ; 4 trials, 1781 infants; low-certainty evidence; Analysis 4.1. We detected no heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included study and serious imprecision in the effect estimates.

4.1.2 Studies that treated infants with intubation surfactant administration and rapid extubation

One study reported CLD in this subgroup (Bevilacqua 1997). The results of this study are very uncertain about the effect of prophylactic surfactant administration on chronic lung disease compared to selective treatment in infants treated with intubation and rapid extubation (RR 2.70, 95% CI 0.11 to 64.61; RD 0.02, 95% CI -0.04 to 0.08; I^2 not applicable; 1 trial, 93 infants; very low certainty evidence; Analysis 4.1). We downgraded the certainty of the evidence to very low due to the risk of bias in the included studies and very serious imprecision in the effect estimates.

4.2 Mortality at 36 weeks' PMA or prior to hospital discharge

Six studies reported on this outcome (Dunn 2011; Egberts 1993; Kendig 1991; Merritt 1991; SUPPORT 2010; Walti 1995). Only Kendig 1991 and Walti 1995 found differences. Subgroups based on the threshold used to apply selective treatment are reported below in 4.2.1 to 4.2.2. The test for subgroup differences was not applicable because all studies treated infants with intubation and ongoing ventilation (Analysis 4.2).

4.2.1 Studies that treated infants with intubation and ongoing ventilation

Six studies reported on this outcome (Dunn 2011; Egberts 1993; Kendig 1991; Merritt 1991; SUPPORT 2010; Walti 1995). Meta-analysis of the six studies suggests that the prophylactic surfactant administration may result in a little to no difference in mortality at 36 weeks' PMA or prior to hospital discharge in comparison with the selective administration in infants with respiratory failure treated with intubation and ongoing ventilation (RR 0.98, 95% CI 0.83 to 1.16; $I^2 = 69\%$; RD -0.00, 95% CI -0.03 to 0.02; $I^2 = 70\%$; 6 trials, 2778 infants; low-certainty evidence; Analysis 4.2). There was moderate heterogeneity. We downgraded the certainty of the evidence to low for risk of bias in the included studies and serious imprecision in the effect estimates.

4.2.2 Studies that treated infants with intubation surfactant administration and rapid extubation

None of the studies applied intubation, surfactant administration, and rapid extubation and reported this outcome.

4.3. Pneumothorax

Eight studies reported on pneumothorax (Bevilacqua 1996; Dunn 1991; Dunn 2011; Egberts 1993; Kattwinkel 1993; Kendig 1991; Merritt 1991; Walti 1995). Subgroups based on the method used in these studies for surfactant administration are reported below in 4.3.1 to 4.3.2. The test for subgroup differences indicates that there is no subgroup effect ($I^2 = 0\%$, $P = 0.89$; Analysis 4.3), suggesting that the method of surfactant administration may not modify the prophylactic surfactant effect on pneumothorax.

4.3.1 Studies that treated infants with intubation and ongoing ventilation

Seven studies treated infants with intubation and ongoing ventilation (Dunn 1991; Dunn 2011; Egberts 1993; Kattwinkel 1993; Kendig 1991; Merritt 1991; Walti 1995).

The meta-analysis of trials in this subgroup suggested that prophylactic surfactant administration may result in a slight reduction in pneumothorax in comparison with selective administration of surfactant in infants with respiratory failure

treated with intubation and ongoing ventilation (RR 0.76, 95% CI 0.54 to 1.06; $I^2 = 24\%$; RD -0.01, 95% CI -0.03 to 0.00; $I^2 = 36\%$; 7 trials, 2826 infants; low-certainty evidence; [Analysis 4.3](#)). There was low heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

4.3.2 Studies that treated infants with intubation surfactant administration and rapid extubation

One study treated infants with intubation, surfactant administration, and rapid extubation ([Bevilacqua 1996](#)).

This study showed very uncertain results regarding the effect of prophylactic surfactant on the risk of pneumothorax in comparison with selective administration of surfactant in infants with respiratory failure treated with intubation and rapid extubation (RR 0.81, 95% CI 0.36 to 1.81; RD -0.02 95% CI -0.08 to 0.05; I^2 not applicable; 1 trial, 268 infants; very low-certainty evidence; [Analysis 4.3](#)). We downgraded the certainty of the evidence to very low due to risk of bias in the included studies and very serious imprecision in the effect estimates.

4.4 Bronchopulmonary dysplasia (oxygen requirement at 28 to 30 days of age)

Nine studies reported on bronchopulmonary dysplasia ([Bevilacqua 1996](#); [Bevilacqua 1997](#); [Dunn 1991](#); [Dunn 2011](#); [Egberts 1993](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Merritt 1991](#); [Walti 1995](#)). Subgroups based on the method used in these studies for surfactant administration are reported below in 4.4.1 to 4.4.2. The test for subgroup differences indicates that there is no subgroup effect ($I^2 = 0\%$, $P = 0.44$; [Analysis 4.4](#)), suggesting that the method of surfactant administration may not modify the prophylactic surfactant effect on BPD.

4.4.1 Studies that treated infants with intubation and ongoing ventilation

Seven studies treated infants with intubation and ongoing ventilation ([Dunn 1991](#); [Dunn 2011](#); [Egberts 1993](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Merritt 1991](#); [Walti 1995](#)).

The meta-analysis of studies in this subgroup suggested that prophylactic surfactant administration may result in little to no difference in bronchopulmonary dysplasia in comparison with selective administration of surfactant in infants with respiratory failure with intubation and ongoing ventilation (RR 1.03, 95% CI 0.92 to 1.16; $I^2 = 57\%$; RD 0.01, 95% CI -0.02 to 0.04; $I^2 = 67\%$; 7 trials, 2802 infants; low-certainty evidence; [Analysis 4.4](#)). There was moderate heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

4.4.2 Studies that treated infants with intubation surfactant administration and rapid extubation

Two studies treated infants with intubation, surfactant administration, and rapid extubation ([Bevilacqua 1996](#); [Bevilacqua 1997](#)).

The meta-analysis of studies in this subgroup suggests that prophylactic surfactant administration may result in little to no difference in the risk of bronchopulmonary dysplasia in comparison with selective administration of surfactant in infants with respiratory failure treated with intubation and rapid extubation (RR

0.82, 95% CI 0.46 to 1.46; $I^2 = 0\%$; RD -0.02, 95% CI -0.09 to 0.04; $I^2 = 0\%$; 2 trials, 316 infants; low-certainty evidence; [Analysis 4.4](#)). No heterogeneity was noted. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

4.5 Bronchopulmonary dysplasia or death at 28 to 30 days of age

Eight studies reported on this outcome ([Bevilacqua 1996](#); [Bevilacqua 1997](#); [Dunn 1991](#); [Dunn 2011](#); [Egberts 1993](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Walti 1995](#)). Subgroups based on the method used in these studies for surfactant administration are reported below in 4.5.1 to 4.5.2. The test for subgroup differences indicates that there is no statistically significant subgroup effect ($I^2 = 73.9\%$, $P = 0.05$; [Analysis 4.5](#)), suggesting that the method of surfactant administration may not modify the prophylactic surfactant effect on bronchopulmonary dysplasia or death at 28 to 30 days of age.

4.5.1 Studies that treated infants with intubation and ongoing ventilation

Six studies reported on this outcome ([Dunn 1991](#); [Dunn 2011](#); [Egberts 1993](#); [Kattwinkel 1993](#); [Kendig 1991](#); [Walti 1995](#)).

The meta-analysis including all six studies reveals that the prophylactic surfactant administration may reduce the risk of bronchopulmonary dysplasia or death at 28 to 30 days of age slightly when compared to selective surfactant administration in infants with respiratory failure with intubation and ongoing ventilation (RR 0.93, 95% CI 0.84 to 1.03; $I^2 = 78\%$; RD -0.02, 95% CI -0.05 to 0.01; $I^2 = 74\%$; 6 trials, 2672 infants; low-certainty evidence; [Analysis 4.5](#)). There was substantial heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and serious imprecision in the effect estimates.

4.5.2 Studies that treated infants with intubation surfactant administration and rapid extubation

Two studies reported on this outcome ([Bevilacqua 1996](#); [Bevilacqua 1997](#)).

The meta-analysis of these two studies suggests that prophylactic surfactant administration may reduce the risk of bronchopulmonary dysplasia or death at 28 to 30 days of age in infants with respiratory failure treated with intubation and rapid extubation (RR 0.69, 95% CI 0.53 to 0.91; $I^2 = 55\%$; RD -0.13, 95% CI -0.23 to -0.04; $I^2 = 65\%$; NNTB 8, 95% CI 5 to 25; 2 trials, 361 infants; low-certainty evidence; [Analysis 4.5](#)). There was moderate heterogeneity. We downgraded the certainty of the evidence to low due to the risk of bias in the included studies and inconsistency of results.

4.6 Chronic lung disease or death (at 36 weeks' PMA)

Four studies reported on this outcome ([Dunn 1991](#); [Dunn 2011](#); [SUPPORT 2010](#); [Walti 1995](#)). All four studies treated infants with intubation and ongoing ventilation. The test for subgroup differences was not applicable because all studies treated infants with intubation and ongoing ventilation ([Analysis 4.6](#)).

4.6.1 Studies that treated infants with intubation and ongoing ventilation

The meta-analysis including all four studies reveals that the prophylactic surfactant administration probably increases the risk of chronic lung disease or death at 36 weeks' PMA slightly compared

with selective surfactant administration in infants treated with intubation and ongoing ventilation (RR 1.09, 95% CI 0.99 to 1.20; $I^2 = 31\%$; RD 0.04 95% CI -0.00 to 0.08; $I^2 = 41\%$; 4 trials, 2122 infants; moderate-certainty evidence; Analysis 4.6) (Dunn 1991; Dunn 2011; SUPPORT 2010; Walti 1995). There was low heterogeneity. We downgraded the certainty of the evidence to moderate due to the risk of bias in the included studies and serious imprecision in the effect estimates.

DISCUSSION

Changes in neonatology practice and implications for our review findings

This is the update of a review that has had a great impact on treatment decisions for preterm neonates with respiratory failure. Its initial version (Soll 2001), based on five randomized controlled trials, showed that prophylactic surfactant administration significantly decreased the risks of pneumothorax, pulmonary interstitial emphysema, mortality, and bronchopulmonary dysplasia (BPD) or death, in comparison with surfactant treatment of established RDS. Ten years later, the updated version (Rojas-Reyes 2012), including six additional studies, two of which reflected the most current clinical practice (Dunn 2011; SUPPORT 2010), revealed that the beneficial effect associated with the use of prophylactic surfactant was not that clear when infants were stabilized on CPAP early or their mothers had received a complete course of steroids during the antenatal period. Our 2026 updated review with subgroup analysis based on the routine application of CPAP showed a reduction in the combined outcome "chronic lung disease (CLD) or death" and a slight reduction in the risk of pneumothorax in the selective surfactant administration arm compared with the prophylactic arm when infants had early stabilization on CPAP. For other important clinical outcomes such as neonatal mortality, bronchopulmonary dysplasia (BDP), CLD, or sepsis, the evidence suggested little to no difference in risk associated with the use of prophylactic surfactant.

The growing appreciation of the role of non-invasive support using CPAP has led us to rethink the real benefit of using prophylactic surfactant in high-risk preterm infants. Morley 2008 conducted a large randomized controlled trial to investigate whether CPAP, rather than intubation and ventilation shortly after birth, would reduce the rates of death or BPD in spontaneously breathing preterm infants. The study demonstrated that in infants born at 25 to 28 weeks' postmenstrual age, early nasal CPAP (NCPAP) was associated with lower need of oxygen supplementation at 28 days and fewer days of mechanical ventilation. Evidence from randomized controlled trials (Rojas 2009; Sandri 2010) suggested that, for patients treated with CPAP, an "early strategy" of surfactant treatment (administration of surfactant during a brief period of intubation soon after the first symptoms of RDS appear) may, in fact, be a better approach than immediate intubation and prophylactic surfactant administration.

Other changes in practice may have influenced the results observed in the previous version of this review (Rojas-Reyes 2012). There is strong evidence that supports the effectiveness of the use of a single course of antenatal corticosteroids to accelerate fetal lung maturation in women at risk of preterm birth (McGoldrick 2020). The use of antenatal steroids has almost tripled in the last two decades. The trials included in the original review in general had a low rate of steroid exposure; in fact, antenatal steroid use was

not even reported in two studies (Kattwinkel 1993; Merritt 1991). In those studies that commented on the use of antenatal steroids, antenatal steroids were administered to mothers prior to delivery in as few as 14% (Walti 1995) to as many as 50% of pregnancies (Dunn 1991), while in the most recent trials included in the previously updated review, at least 50% of the mothers had received a complete course of steroids (Dunn 2011; SUPPORT 2010). As these trials dominated our subgroup analysis regarding antenatal steroid exposure, the results were consistent with the findings regarding routine stabilization on CPAP. Therefore, we speculated that these two interventions may act synergistically to reduce the risk of lung damage.

For this 2026 update, we did not find any new eligible studies, but we identified a new follow-up study (Vaucher 2012), which reported on long-term outcomes in infants enrolled in the SUPPORT 2010 trial. We performed new subgroup analyses that sought to answer some questions that have arisen since the last update. These related to the effect of prophylactic surfactant in neonates less than 28 weeks of gestation stabilized on CPAP, the FiO_2 threshold used to apply selective treatment, and the method of ventilatory support used to treat patients when surfactant was applied. We returned to the original studies and re-extracted the data to be able to do so.

In the subgroup analysis based on prenatal corticosteroids, the findings reinforced our premises. Specifically, the use of prophylactic surfactant in infants whose mothers mostly received steroids and who were stabilized early on CPAP does not provide any benefit compared to selective treatment. In fact, it may result in an increased risk of bronchopulmonary dysplasia and death.

Due to the small number of studies reporting what FiO_2 threshold they used to guide selective treatment, evidence for differential effects of interventions according to this potential effect modifier is of low or very low certainty for most outcomes, primarily due to imprecision. As a result, no conclusions can be drawn regarding effect modification.

Likewise, the limited number of studies reporting outcomes stratified by gestational age precluded subgroup analyses based on gestational age at birth (< 28 weeks versus ≥ 28 weeks). We judged such analyses to be inappropriate with so few studies, as they would be highly imprecise, resulting in very low-certainty evidence.

Recent studies have proposed a variety of new methods for surfactant administration, including thin catheter surfactant administration, laryngeal mask airway administration, and hypopharyngeal surfactant administration (Abdel-Latif 2011a; Abdel-Latif 2011b; Abdel-Latif 2012; Abdel-Latif 2021; Isayama 2016), which may impact the surfactant administration effect, but this was not possible to assess with the available studies. Most studies included in this review administered surfactant to intubated infants and then continued positive pressure ventilation. Only two studies used the technique of intubation, surfactant administration, and rapid extubation (InSurE). A few differences are noted in subgroup analyses of these two administration techniques.

Even though subgroup analysis does not allow us to draw definitive conclusions, it illustrates the underlying differences in the populations studied, particularly regarding the evolution of neonatal respiratory care practices over time. While gestational age across studies is similar, management strategies differ

significantly, especially in the use of antenatal steroids and the routine application of CPAP. These clinical management differences represent important contextual shifts that affect how the interventions (prophylactic versus selective surfactant) are implemented and interpreted.

Our intention in maintaining these subgroup distinctions is not to infer causality from the statistical differences observed, but rather to allow clinicians to understand how the effects of the same intervention strategy may vary across these clinically relevant populations. Therefore, they should be considered exploratory and hypothesis-generating, and the results should be interpreted with caution to avoid overstating the implications of the subgroup findings.

Finally, based on the evidence provided by the included studies, the true benefit of prophylactic surfactant administration in infants treated under current practices is no longer clear. It appears that infants who have received antenatal steroids and can be stabilized early after birth on CPAP may not require prophylactic surfactant treatment. Instead, they could potentially achieve better outcomes if treated with an early selective surfactant.

Summary of main results

We identified 10 studies (3151 infants) that met the inclusion criteria: eight in which continuous positive airway pressure (CPAP) was not routinely applied in the selective treatment group, and two in which CPAP was routinely applied in the selective treatment group.

In the overall analysis, which includes both studies that used CPAP and those that did not, prophylactic surfactant does not demonstrate greater benefits than selective surfactant administered after the onset of respiratory distress syndrome (RDS). Our analyses suggest a slight reduction of pneumothorax but little to no difference in chronic lung disease (CLD), or moderate-to-severe neurodevelopmental impairment.

When compared with the selective use of surfactant in infants not stabilized on CPAP, the use of prophylactic surfactant may result in a slight reduction of pneumothorax and little to no difference in CLD at 36 weeks' postmenstrual age ([Summary of findings 1](#)); however, high-certainty evidence suggests that prophylactic surfactant reduces neonatal mortality in these infants.

Meta-analyses of more recent studies comparing prophylactic surfactant with the selective use of surfactant in infants stabilized on CPAP suggest that prophylactic surfactant probably slightly increases neonatal mortality and makes little to no difference in the combined outcome of chronic lung disease or death.

In our view, these findings favor the management of neonates with respiratory distress based on stabilization with CPAP and the selective use of surfactant.

Overall completeness and applicability of evidence

Overall, based on 10 RCTs, the completeness of the evidence and its applicability to the study question were limited by study heterogeneity, possibly unreported cointerventions, and incomplete reporting of most outcomes of interest.

The lack of information on baseline confounders and care practices, particularly in studies conducted in the 1990s, combined with the small number of included studies, limited our ability to conduct subgroup analyses. Additionally, variations in clinical practices over the periods when the studies were conducted may act as a confounding factor when estimating the relative effects of prophylactic versus selective surfactant.

We identified and excluded four RCTs published between 2008 and 2010 in which surfactant was administered early after birth (within the first 30 minutes of life) to infants with evidence of RDS. This approach, referred to as “early surfactant” rather than “prophylactic,” may better reflect current practice in NICUs caring for preterm neonates in resource-limited settings. We also excluded trials in which the comparison group received “early surfactant” instead of selective surfactant administration.

Quality of the evidence

Our judgment, according to the GRADE approach, of the overall certainty of the evidence for the reported critical outcomes was mostly low to moderate, though our judgments spanned the whole range from high to very low. For the main analysis, we downgraded the certainty of the evidence for all outcomes because of study limitations (mainly due to lack of blinded participants and personnel, i.e. performance bias was at either high or unclear risk of bias) and imprecision of the estimates (wide confidence intervals).

Potential biases in the review process

We are confident that this review is a comprehensive summary of all currently available trials comparing the selective versus prophylactic use of surfactant in preterm neonates. We did not apply any date restrictions to the searches. We have updated searches each year for the last six years until we identified new potentially eligible studies, which prompted us to conduct this update. At least two review authors independently assessed studies for inclusion, extracted the studies' data, and assessed the risk of bias in selected studies. The GRADE certainty assessment for each outcome was conducted by two authors; one of them, a senior author with expertise in applying GRADE, also drew the conclusions.

In this 2026 update, we adopted pragmatic rules for interpreting the findings with regard to the magnitude of a clinically important effect. However, in cases close to the boundaries of these rules, we erred on the side of caution when the results could suggest only a small effect of the intervention. For example, for the pooled estimate of the outcome pneumothorax, the effect estimate was 0.76 and only just crossed the line of no effect.

We focused our interpretation of the magnitude of the effect on the point estimate, while using the 95% confidence interval to assess the precision of the estimates according to the same rules, and consequently, our degree of certainty regarding the magnitude of the effect.

We believe the current subgroup structure meaningfully reflects underlying differences in the populations and interventions studied—particularly with respect to the evolution of neonatal respiratory care practices over time. As explained earlier in this Discussion section, while gestational age across studies is similar, management strategies differ significantly, particularly regarding the use of antenatal steroids and routine application of CPAP. These

clinical management differences define important contextual shifts that affect how the interventions (prophylactic versus selective surfactant) are implemented and interpreted.

As noted earlier in the Discussion, our intention in maintaining the subgroup distinctions is not to infer causality from the observed statistical differences, but rather to facilitate understanding of how the effects of the same intervention strategy may differ across clinically relevant populations. Given the limited number of studies included in most subgroups, these analyses are exploratory and hypothesis-generating in nature and should therefore be interpreted with appropriate caution.

Agreements and disagreements with other studies or reviews

In updating the review, we address a fundamental question regarding the applicability of older studies to current practice. The original Cochrane review of prophylactic versus selective surfactant, which included studies conducted exclusively in the 1990s, reported a clinically significant reduction in pneumothorax and mortality (Soll 2001). The 2012 update included trials conducted in populations that had received antenatal steroids and were stabilized on nasal continuous positive airway pressure (Rojas-Reyes 2012). Incorporating newer studies that reflected contemporary practice, the 2012 update revealed important differences in the relative effects of prophylactic versus selective surfactant use. Our 2026 update further explored additional hypotheses, including the impact of the administration method and the FiO_2 treatment threshold on the selective surfactant administration strategy.

AUTHORS' CONCLUSIONS

Implications for practice

Studies conducted in the 1990s, before the widespread use of antenatal corticosteroids and early continuous positive airway pressure (CPAP), suggested that prophylactic surfactant administration in infants at risk of developing respiratory distress syndrome (RDS) improved clinical outcomes more than selective surfactant use in infants with established RDS. In these populations, prophylactic surfactant was associated with a reduced risk of pneumothorax and mortality.

In contrast, more recent larger trials that reflect current neonatal care practices, including the use of antenatal corticosteroids and routine early stabilization on CPAP, no longer appear to demonstrate these benefits. Instead, meta-analyses of these studies show that, compared with selective surfactant administration in infants stabilized on CPAP, prophylactic surfactant may result in little to no difference in the risk of chronic lung disease and probably slightly increases mortality.

Implications for research

Further trials are needed to determine the role of early CPAP combined with selective surfactant administration in very preterm infants (< 28 weeks' gestation), for whom the evidence is scarce and the effects of the intervention strategies explored here remain

uncertain. Research should also evaluate whether improved assessment of lung maturity, either early after birth or before delivery, can refine patient selection for prophylactic surfactant. It is clear that any strategy that reduces the need for surfactant administration has economic implications and the potential to improve the cost-effectiveness of interventions. Finally, further research should explore the effectiveness and safety of less invasive surfactant administration techniques in early treatment strategies.

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The following people conducted the editorial process for this 2026 version of the review.

- Sign-off Editor (final editorial decision): Professor Mohamed E Abdel-Latif, Australian National University, College of Health and Medicine, Discipline of Neonatology, School of Medicine and Psychology, Canberra, AUSTRALIA; Toby Lasserson, Acting Editor In Chief
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Ben Ridley, Cochrane Central Editorial Service
- Editorial Assistant (conducted editorial policy checks, collated peer-reviewer comments and supported editorial team): Leticia Rodrigues, Cochrane Central Editorial Service
- Copy Editor (copy editing and production): Laura MacDonald, Cochrane Central Production Service
- Peer reviewers (provided comments and recommended an editorial decision): Clare Miles, Evidence Production and Methods Directorate (methods); Yuan Chi, Beijing Yealth Technology Co Ltd., McMaster University (search); Hsiu-Lin Chen, Kaohsiung Medical University Hospital, Kaohsiung, Taiwan (clinical); Anna Lavizzari, Fondazione IRCCS Cà Granda Ospedale Maggiore Policlinico, NICU, Milan, Italy (consumer). One additional person provided clinical peer-review but chose not to be publicly acknowledged.

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* Indicates the major publication for the study

CHARACTERISTICS OF STUDIES

Characteristics of included studies [ordered by study ID]

Bevilacqua 1996

Study characteristics

Methods	Design: RCT Country(ies): Italy Single site/multiple sites: multiple sites (18) Period of study: not stated
Participants	Inclusion criteria: all preterm babies born at a gestational age of 24 to 30 weeks without antenatal evidence of major defects were eligible for the study. Exclusion criteria 1. Babies presenting at birth with major malformations possibly interfering with cardiopulmonary function, neurodevelopment, or survival 2. Babies who subsequently showed evidence of congenital heart defects or congenital infection Number enrolled 1. Targeted sample size was 140 infants in each group. 2. A total of 287 infants were randomized in 18 Italian centers. 3. 10 infants in the prophylactic group and 9 infants in the control group were excluded.
Interventions	Intervention (prophylactic treatment): curosurf (200 mg/kg) within 10 minutes of birth Control (selective treatment): curosurf treatment (200 mg/kg) for intubated infants with RDS within 24 hours of age A second dose of curosurf (200 mg/kg) was permitted in the prophylactic group if an infant presented with RDS requiring assisted ventilation within 24 hours of age.
Outcomes	Primary outcome 1. Reduction in RDS based on radiological chest examination Secondary outcomes 1. Patent ductus arteriosus 2. BPD 3. IVH 4. Pneumonia 5. Retinopathy of prematurity
Notes	Funding source: not stated Conflicts of interest: not stated

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized multicenter trial; stratification based on gestational age
Allocation concealment (selection bias)	Low risk	Blinding of randomization: yes (sealed envelopes)

Bevilacqua 1996 (Continued)

Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding has been described; it is unlikely that personnel were blinded.
Blinding of outcome assessment (detection bias) Hospital outcomes	Unclear risk	Radiological findings were graded by a pediatric radiologist not aware of the treatment group assignment of individual patients; not clear for other outcomes (not related to radiology).
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition rate 7%
Selective reporting (reporting bias)	Unclear risk	No protocol or register available
Other bias	Low risk	There are no other potential risks.

Bevilacqua 1997

Study characteristics

Methods	Design: RCT Country(ies): Italy, Bulgaria and Romania Single site/multiple sites: multiple sites (4) (2 NICUs in Italy, 1 NICU in Bulgaria and 1 NICU in Romania) Period of study: not stated
Participants	Inclusion criteria 1. Gestational age 24 to 30 weeks Exclusion criteria 1. Prolonged rupture of membranes ≥ 3 weeks 2. Major congenital malformations evident at birth or diagnosed prenatally 3. Uncertain gestational age Number enrolled 1. The targeted sample size was 174. 2. Study stopped at enrollment of 93 (61 in Italy and 32 in Bulgaria). The Romanian center did not start the study because it was impossible in the scheduled times to equip it for MV of the newborn infants. 3. No exclusions were reported.
Interventions	Intervention (prophylactic treatment): curosurf (200 mg/kg) within 10 minutes of birth Control (selective treatment): curosurf treatment (200 mg/kg) for intubated infants with respiratory distress syndrome within 24 hours of age A second dose of curosurf (200 mg/kg) was allowed in the prophylactic group if infant presented with RDS requiring assisted ventilation within 24 hours of age.
Outcomes	Primary outcome 1. Reduction in RDS based on radiological chest examination

Bevilacqua 1997 (Continued)

Secondary outcomes

1. Patent ductus arteriosus
2. IVH
3. Oxygen dependence
4. Pulmonary hemorrhage
5. Retinopathy of prematurity

Notes

Funding source: not stated

Conflict of interest: not stated

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized multicenter trial; stratification based on gestational age
Allocation concealment (selection bias)	Low risk	Blinding of randomization: yes (sealed envelopes)
Blinding of participants and personnel (performance bias) All outcomes	High risk	It is unlikely that blinding was performed.
Blinding of outcome assessment (detection bias) Hospital outcomes	Unclear risk	Radiologic findings were graded by a pediatric radiologist who was not aware of treatment; not clear for other outcomes (not related to radiology).
Incomplete outcome data (attrition bias) All outcomes	Low risk	Intention-to-treat analysis performed
Selective reporting (reporting bias)	Unclear risk	No protocol or register available
Other bias	Low risk	There are no other sources of bias.

Dunn 1991

Study characteristics

Methods

Design: RCT

Country(ies): Canada

Single site/multiple sites: single-center trial

Period of study: between July 1986 and July 1989

Participants

Inclusion criteria

1. Estimated gestational age < 30 weeks

Exclusion criteria

Dunn 1991 (Continued)

1. Congenital abnormalities
2. ROM $\geq$ 2 weeks
3. Mature L/S ratio

Numbers enrolled

1. A sample size of 60 infants per group was calculated.
2. Preterm infants: (n = 182) 62 were randomly assigned to the prophylactic treatment group, 60 to the late treatment group, and 60 to the control group.
3. No exclusions were reported.

Interventions	Intervention (prophylactic surfactant): BLSE (75 to 100 mg) prior to first breath Intervention (selective treatment): BLSE treatment (75 to 100 mg) for intubated infants with respiratory insufficiency less than 6 hours of age Control (no treatment/instillation of air) Up to 3 additional treatments allowed in either surfactant treatment group.	
Outcomes	Primary outcome 1. Improvement in a/A ratio Secondary outcomes 1. Ventilatory requirements 2. Duration of ventilation 3. Complications of prematurity	
Notes	Funding source: (quote:) "This work was supported by grants from the Hospital for Sick Children Foundation, Toronto, and the Medical Research Council of Canada" Conflict of interest: not stated	
Risk of bias		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized single center; stratification based on gestational age and exposure to antenatal steroids
Allocation concealment (selection bias)	Low risk	Blinding of randomization: yes (sealed envelopes)
Blinding of participants and personnel (performance bias) All outcomes	High risk	There was no attempt to blind the administration.
Blinding of outcome assessment (detection bias) Hospital outcomes	Unclear risk	It has not been described whether outcome assessors were blinded.
Incomplete outcome data (attrition bias) All outcomes	Low risk	All enrolled participants were randomized and measured during follow-up.

Dunn 1991 (Continued)

Selective reporting (reporting bias)	Unclear risk	No protocol or register available
Other bias	Low risk	There are no other sources of potential bias.

Dunn 2011
Study characteristics

Methods	Design: RCT Country(ies): USA Single site/multiple sites: multicenter (27 North American NICUs participating in the Vermont Oxford Network) Period of study: between September 2003 and March 2009.
Participants	Inclusion criteria 1. Infants between 26 and 29 weeks' gestation at birth 2. No congenital malformations at birth Exclusion criteria 1. Women who were carrying a fetus with a potentially life-threatening anomaly or condition Number enrolled 1. Planned sample size was 876 (292 in each arm). 2. 648 infants enrolled from 27 Vermont Oxford Network centers. 3. Enrollment stopped prior to achieving the desired sample size (n = 876).
Interventions	Intervention group 1: prophylactic surfactant within 5 to 15 minutes of birth followed by a period of assisted ventilation for a minimum of 6 hours; Intervention group 2: Intubation-surfactant treatment within 5 to 15 minutes after birth and rapid (15 to 30 minutes after surfactant) extubation to nasal CPAP Control group: early stabilization in nasal CPAP within 15 minutes of birth.
Outcomes	Primary outcome 1. Incidence of death or moderate to severe BPD at 36 weeks' postmenstrual age Secondary outcomes 1. Number of infants who received surfactant 2. Number of surfactant doses 3. Use of postnatal steroids 4. Days on assisted ventilation 5. Days on NCPAP 6. Days on supplemental oxygen Other outcomes included the incidence of common complications of prematurity and mortality.
Notes	Funding: Vermont Oxford Network

Dunn 2011 (Continued)

Conflict of interest: "The authors have indicated they have no financial relationships relevant to this article to disclose."

Roger Soll (review author) is a co-author of this study.

Trial registered: (quote:) "this trial has been registered at www.clinicaltrials.gov (identifier NCT00244101)."

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Stratification and block randomization was according to center and gestational age.
Allocation concealment (selection bias)	Low risk	Blinding of randomization: yes (sealed envelopes)
Blinding of participants and personnel (performance bias) All outcomes	High risk	No blinding could be performed.
Blinding of outcome assessment (detection bias) Hospital outcomes	High risk	No blinding was performed.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition rate 1.2%
Selective reporting (reporting bias)	Low risk	Study is in line with clinical trials register.
Other bias	Unclear risk	Study stopped prematurely

Egberts 1993

Study characteristics

Methods	Design: RCT Country(ies): the Netherlands and Sweden Single site/multiple sites: multiple sites The trial was conducted in the perinatal and neonatal units of the University Hospitals of Leiden (Netherlands) and Uppsala (Sweden), the Karolinska Hospital in Stockholm (Sweden), and the St. Joseph Hospital in Veldhoven (Netherlands). Period of study: between 1989 and 1991
Participants	Inclusion criteria 1. Preterm infants 6 to 24 hours of age 2. Gestational age 26 to 30 weeks Exclusion criteria

Egberts 1993 (Continued)

1. Major congenital malformations, diagnosed prenatally or at birth
2. ROM < 3 weeks

Number enrolled

1. No targeted sample size was provided.
2. 82 neonates were enrolled in the Netherlands. 67 neonates were enrolled in Sweden.
3. 2 neonates were excluded from the statistical analysis for violation of entry criteria: 1 neonate in the rescue-eligible group had tetralogy of Fallot and 1 neonate in the prophylaxis group was delivered after 30 days of ROM.

Interventions	Intervention: prophylactic curosurf (200 mg/kg) within 10 minutes after birth Control (rescue-eligible group): curosurf treatment (200 mg/kg) for intubated infants with moderate to severe RDS (supplemental oxygen equal to or greater than 60%) All neonates, both in the prophylaxis and the rescue-eligible groups, were treated with surfactant at the age of 6 to 24 hours if they needed assisted ventilation with FiO₂ 0.6 (retreatment and rescue treatment). A second dose of curosurf (200 mg/kg) was allowed in the prophylactic group if the infant required assisted ventilation and supplemental oxygen 60% or greater 6 hours after the first dose.
Outcomes	Primary outcome 1. Reduction in RDS according to Giedion 1973 Secondary outcomes 1. Apgar scores 2. Heart rate 3. Frequency of ventilation 4. FIO₂ 5. tcPO₂ 6. tcPO₂/FIO₂ ratio 7. Peak inspiratory pressure 8. MAP 9. Duration of mechanical ventilation 10. Time in > 21% oxygen 11. Pulmonary interstitial emphysema 12. Pneumothorax 13. Intracranial hemorrhage 14. Periventricular leukomalacia 15. Patent ductus arteriosus 16. BPD 17. Mortality
Notes	Funding sources: this study was supported in part by the Swedish Medical Research Council (project 3351 and grant 19M-7544), Praeventie-fonds (grant 28-1559), GiSela Thierfonds from the Netherlands, and Oscar U's Jubileumsfond and Axel Tielman's Minnesfond, both from Sweden. Conflicts of interest: not stated
Risk of bias	
Bias	Authors' judgement Support for judgement

Egberts 1993 (Continued)

Random sequence generation (selection bias)	Low risk	Randomized multicenter trial; randomization cards for each center were prepared in blocks ranging between 4 and 8.
Allocation concealment (selection bias)	Low risk	Blinding of randomization: yes (sealed envelopes)
Blinding of participants and personnel (performance bias) All outcomes	High risk	Personnel not blinded
Blinding of outcome assessment (detection bias) Hospital outcomes	Unclear risk	Whether assessors were blinded was not described.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition rate 1.3%
Selective reporting (reporting bias)	Unclear risk	No protocol or register available
Other bias	Low risk	There are no other sources of potential bias.

Kattwinkel 1993
Study characteristics

Methods	Design: RCT Country(ies): USA Single site/multiple sites: multicenter (Nine neonatal intensive care units originally agreed to participate in this trial. One center was found to have a high incidence of protocol deviations after submitted data on their first 15 patients were retrieved. Because of these inaccuracies and since this center was not willing to modify its behavior, the data management and advisory committee advised dropping all collected and subsequent data from this center from inclusion in the study.) Period of study: between August 1989 and January 1992
Participants	Inclusion criteria 1. Preterm infants 2. Gestational age 29 to 32 weeks Exclusion criteria 1. Congenital malformation 2. Congenital sepsis 3. Perinatal asphyxia 4. Judged too mature 5. Severe perinatal asphyxia 6. Lost data 7. Outborn baby Number enrolled

Kattwinkel 1993 (Continued)

1. A sample size of 1320 neonates was targeted.
2. 1398 infants (from 8 centers) were enrolled.
3. After exclusions, for malformations and other factors, 1248 were analyzed.

Interventions	Intervention: prophylactic CLSE (150 mg) within 5 minutes of birth Control: CLSE treatment (150 mg) in infants with mild RDS (compatible radiograph, greater than 30% supplemental oxygen). Re-treatment of either group was allowed if the infant reached criteria for severe RDS (MAP 10 cmH ₂ O or greater, supplemental oxygen 60% or greater).	
Outcomes	Primary outcome 1. Development of moderate RDS Secondary outcomes 1. Need for mechanical ventilation during the first 4 days 2. Need for supplemental oxygen during the first 4 days 3. Requirement for supplemental oxygen at 28 days 4. Mortality 5. 1-minute Apgar score 6. 5-minute Apgar score 7. Endotracheal intubation 8. CLSE 9. Pulmonary interstitial emphysema 10.Pneumothorax 11.Pulmonary hemorrhage 12.Postextubation stridor 13.Sent home dependent on oxygen 14.Patent ductus arteriosus 15.Neonatal sepsis 16.ROP 17.Necrotizing enterocolitis 18.Persistent pulmonary hypertension 19.Intracranial hemorrhage 20.Periventricular leukomalacia	
Notes	Funding source: funds for this project were provided by ONY, Inc, Amherst, NY, who provided some financial support for the data management and analysis, and from a Children’s Mirade Network Telethon grant from University of Kansas, School of Medicine, Wichita. Conflicts of interest: none stated	
Risk of bias		
Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized multicenter trial; no methods of allocation sequence generation described
Allocation concealment (selection bias)	Low risk	Blinding of randomization: yes (sealed envelopes)

Kattwinkel 1993 (Continued)

Blinding of participants and personnel (performance bias) All outcomes	High risk	Open-label protocol
Blinding of outcome assessment (detection bias) Hospital outcomes	Unclear risk	No blinding of outcome assessors described
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	A number of infants were not included because one center was dropped due to a high incidence of protocol deviations. Attrition rate 10.7%
Selective reporting (reporting bias)	Unclear risk	No protocol or register available
Other bias	Low risk	There are no other sources of potential bias.

Kendig 1991

Study characteristics

Methods	Design: RCT Country(ies): USA Single site/multiple sites: multicenter (3: perinatal and neonatal units in New York) Period of study: between 5 December 1985 and 31 August 1988
Participants	Inclusion criteria 1. Preterm infants 2. Gestational age < 30 weeks Exclusion criteria 1. Congenital abnormalities such as anencephaly and renal agenesis Number enrolled 1. Calculated they needed a sample size of 450 infants 2. 479 infants enrolled a. Surfactant as prophylaxis: n = 235 b. Rescue therapy: n = 244 3. No infants were excluded from the primary analysis.
Interventions	Intervention: prophylactic CLSE (90 mg) prior to first breath Control: CLSE treatment (90 mg) in intubated infants with RDS requiring greater than 40% supplemental oxygen or MAP greater than 7 cm H ₂ O Retreatment of either group was allowed at 12 hours if the infant remained on assisted ventilation, requiring greater than 40% supplemental oxygen or MAP greater than 7 cm H ₂ O.
Outcomes	Primary outcome 1. Mortality

Kendig 1991 (Continued)

Secondary outcomes

1. Pneumothorax
2. Severity of RDS
3. Persistent need for supplemental oxygen at 28 days
4. Chronic lung disease
5. Intraventricular hemorrhage
6. Patent ductus arteriosus
7. ROP

The article by Sinkin 1998 published pulmonary and neurodevelopmental outcomes in children at 4.5 to 8 years of age.

Notes

Funding: supported by a Specialized Center of Research (SCOR) grant (HL-36453) and in part by a General Research Centers grant (RR000-44), both from the National Institutes of Health

Declarations of interest: not stated

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomized multicenter trial; randomization prepared in blocks of 8
Allocation concealment (selection bias)	Low risk	Blinding of randomization: yes (sealed envelopes)
Blinding of participants and personnel (performance bias) All outcomes	High risk	Blinding of personnel was not attempted.
Blinding of outcome assessment (detection bias) Hospital outcomes	Unclear risk	Blinding of outcome assessors not described
Blinding of outcome assessment (detection bias) Follow-up outcomes	Unclear risk	Blinding of outcome assessors not described
Incomplete outcome data (attrition bias) All outcomes	Low risk	All infants were included in the intention-to-treat analysis.
Selective reporting (reporting bias)	Unclear risk	No protocol or register available
Other bias	Low risk	There are no other sources of potential bias.

Merritt 1991

Study characteristics

Methods

Design: placebo-controlled RCT

Merritt 1991 (Continued)

Country(ies): Finland and USA

Single site/multiple sites: multicenter (4: Helsinki, Finland; San Diego (2), California (1) USA)

Period of study: not stated

Participants	Inclusion criteria 1. Preterm infants 2. Gestational age 24 to 29 weeks Exclusion criteria 1. Congenital malformation 2. No mature L/S ratio 3. ROM < 3 weeks Number enrolled 1. Gravidas of 243 singletons and 29 multiple gestations were admitted to the participating institutions. 2. Of 282 potentially eligible fetuses: a. 10 sets of parents declined participation. i. 246 infants received treatment at birth. ii. 200 infants had RDS. 3. Of the 246 infants treated, 46 did not have RDS leaving 200 infants who received the full course of surfactant therapy.
Interventions	Singleton fetuses were randomly assigned to receive (1) placebo (air), (2) prophylactic surfactant treatment, or (3) rescue surfactant treatment. Infants of multiple births received either (1) prophylactic or (2) rescue treatment. 1. Prophylactic human amniotic fluid extract (70 mg/kg) immediately post intubation/minimal ventilation 2. Treatment with human amniotic fluid extract (70 mg/kg) in intubated infants with RDS requiring greater than 50% supplemental oxygen and MAP of 7 cm H₂O or greater prior to 12 hours of age 3. Control treatment
Outcomes	Primary outcomes 1. Mortality at 28 days 2. BPD at 28 days Secondary outcomes 1. BPD at 38 weeks 2. Pulmonary air leaks 3. Severity of RDS 4. Pneumothorax 5. Intraventricular hemorrhage 6. Necrotizing enterocolitis
Notes	Funding sources and conflicts of interest: supported by grants HL/HD 35036 (Dr Merritt) from the National Institutes of Health, via General Clinical Research Center grant (RR-00827) to the University of California, San Diego ~ Office of Orphan Drug Products, U.S. Food and Drug Administration 000112 (Dr. Merritt), the Finnish Academy, the Sigrid Juselius Foundation (Dr. Hallman), and a Children's Hospital and Health Center Development grant (Dr. Wozniak and Dr. Merritt).

Merritt 1991 (Continued)

A follow-up study published by Vaucher 1993 including 143 infants from the trial, reported growth, neurosensory, and neurologic outcomes.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Low risk	Randomized multicenter trial; variation of the biased-coin method; stratification based on gestational age; separate randomization schedules for multiple births
Allocation concealment (selection bias)	Low risk	Blinding of randomization: yes (sealed envelopes)
Blinding of participants and personnel (performance bias) All outcomes	Low risk	Health care providers were unaware of actual surfactant administration.
Blinding of outcome assessment (detection bias) Hospital outcomes	Low risk	Pediatric radiologists and pathologists were unaware of treatment allocation.
Blinding of outcome assessment (detection bias) Follow-up outcomes	Low risk	Staff were unaware of the treatment.
Incomplete outcome data (attrition bias) All outcomes	Low risk	Attrition rate 18.6%
Selective reporting (reporting bias)	Unclear risk	No protocol or register available
Other bias	Low risk	There are no other sources of potential bias.

SUPPORT 2010
Study characteristics

Methods	Design: RCT Country(ies): USA Single site/multiple sites: multicenters in the NICHD Neonatal Research Network Period of study: from February 2005 through February 2009
Participants	Inclusion criteria 1. Gestational age 24 to 27.6 weeks 2. A decision had been made to provide full resuscitation for them Exclusion criteria 1. Congenital malformation 2. No written informed consent obtained from a parent or guardian

SUPPORT 2010 (Continued)

Number enrolled

1. Target sample 1310 preterm infants
2. A total of 1316 infants were enrolled in the study.
 - a. 565 were in the lower gestational-age stratum (24 weeks 0 days to 25 weeks 6 days).
 - b. And 751 were in the higher stratum (26 weeks 0 days to 27 weeks 6 days).

Interventions

Intervention: prophylactic surfactant given within the first 60 minutes of birth followed by conventional ventilation in NICU

Control: early intratracheal administration of surfactant (within 1 hour of birth) followed by a conventional ventilation strategy

All infants received CPAP at birth.

Outcomes
Primary outcomes

1. Death or BPD (physiological definition)

Secondary outcomes

1. BDP as defined by the receipt of any supplemental oxygen at 36 weeks
2. Need for mechanical ventilation
3. Air leak
4. Pneumothorax
5. Necrotizing enterocolitis
6. Patent ductus arteriosus
7. Severe intraventricular hemorrhage
8. Severe ROP

Notes

Funding source: supported by grants (U10 HD21364, U10 HD21373, U10 HD21385, U10 HD21397, U10 HD27851, U10 HD27853, U10 HD27856, U10 HD27880, U10 HD27871, U10 HD27904, U10 HD34216, U10 HD36790, U10 HD40461, U10 HD40492, U10 HD40498, U10 HD40521, U10 HD40689, U10 HD53089, U10 HD53109, U10 HD53119, U10 HD53124) from the Eunice Kennedy Shriver National Institute of Child Health and Human Development, cofunding from the National Heart, Lung, and Blood Institute, and grants (M01 RR30, M01 RR32, M01 RR39, M01 RR44, M01 RR54, M01 RR59, M01 RR64, M01 RR70, M01 RR80, M01 RR125, M01 RR633, M01 RR750, M01 RR997, M01 RR6022, M01 RR7122, M01 RR8084, M01 RR16587, UL1 RR25008, UL1 RR24139, UL1 RR24979, UL1 RR25744) from the National Institutes of Health

Conflicts of interest: Namasivayam Ambalavanan's institution received grant funding from NIH. The other authors declared they had no conflicts of interest.

Registration details: ClinicalTrials.gov number, NCT00233324

Vaucher 2012 assessed the neurodevelopmental impairment of infants enrolled in the SUPPORT 2010 trial at 18 to 22 months of corrected age.

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized multicenter trial; randomization stratified by age (24 to 25 + 6 weeks; 26 to 27 + 6 weeks) and center
Allocation concealment (selection bias)	Low risk	Double-sealed envelopes

SUPPORT 2010 (Continued)

Blinding of participants and personnel (performance bias) All outcomes	High risk	It is unlikely that blinding was performed (not described).
Blinding of outcome assessment (detection bias) Hospital outcomes	Unclear risk	It has not been described whether outcome assessors were blinded.
Blinding of outcome assessment (detection bias) Follow-up outcomes	Low risk	Neurodevelopmental testers were unaware of the treatment assignment.
Incomplete outcome data (attrition bias) All outcomes	Low risk	All randomized infants were analyzed in intention-to-treat analysis.
Selective reporting (reporting bias)	Low risk	Study in line with clinicaltrials.gov register
Other bias	Low risk	There are no other sources of potential bias.

Walti 1995

Study characteristics

Methods	Design: RCT Country(ies): France Single site/multiple sites: 12 perinatal and NICUs Period of study: between 19 September 1990 and 15 June 1991
Participants	Inclusion criteria 1. Preterm infants 3 to 18 hours of age 2. Gestational age 25 to 31 weeks 3. Written consents obtained before delivery Exclusion criteria (before birth) 1. Suspected congenital anomaly 2. Suspected disease requiring special care Exclusion criteria (after enrollment) 1. Stillborn 2. Had an unanticipated congenital malformation or disease requiring special care outside the expected gestational age range of 25 to 31 weeks at birth 3. Delivered after more than 3 weeks of ROM 4. Parents declined their baby's participation Number enrolled 1. Calculated they needed a sample size of 250 infants 2. 288 random envelopes were opened. 32 of these infants were excluded after enrollment.

Walti 1995 (Continued)

3. Exclusions
 - a. 14 babies were excluded from the prophylaxis group.
 - b. 18 babies were excluded from the rescue group.
4. The final analysis was conducted on only 256 infants.
 - a. 134 in the prophylactic group
 - b. 122 in the rescue group

Interventions	Intervention: (prophylactic) curosurf (100 mg/kg) within 15 minutes after birth Control: (rescue administration) curosurf treatment (100 mg/kg) in intubated infants with moderate respiratory distress syndrome ($\text{PaO}_2/\text{FiO}_2 < 150$ mm Hg on assisted ventilation with $\text{MAP} > 8$ cm H_2O) Retreatment of either group was allowed with up to 3 doses of curosurf (100 mg/kg) within 48 hours if the above criteria were met.
Outcomes	Primary outcome 1. Survival without BPD at 28 days after birth Secondary outcomes 1. Radiographic evidence of RDS 2. Immediate respiratory course 3. Complications of prematurity
Notes	Funding source: "Chiesi Farmaceutici, Parma, Italy (liaison, M Podesta) funded the trial and provided Curosurf. However, the trial was designed, conducted and analyzed independently of the company." Conflicts of interest: not stated

Risk of bias

Bias	Authors' judgement	Support for judgement
Random sequence generation (selection bias)	Unclear risk	Randomized multicenter trial; no information provided on allocation sequence generation
Allocation concealment (selection bias)	Low risk	Blinding of randomization: yes (sealed envelopes)
Blinding of participants and personnel (performance bias) All outcomes	High risk	Treatment was open, and no blinded procedure was used.
Blinding of outcome assessment (detection bias) Hospital outcomes	Unclear risk	Not described whether assessors were blinded
Incomplete outcome data (attrition bias) All outcomes	Unclear risk	Attrition rate 11%
Selective reporting (reporting bias)	Unclear risk	No register or protocol available
Other bias	Low risk	There are no other sources of potential bias.

CLSE: calf lung surfactant extract; MAP: mean airway pressure; MV: mechanical ventilation; NICHD: National Institute of Child Health and Human Development; NICU: neonatal intensive care unit; Outborn baby: a neonate born outside the hospital or facility where they are currently receiving neonatal intensive care, having been transferred there shortly after birth; RCT: randomized controlled trial; RDS: respiratory distress syndrome; ROP: retinopathy of prematurity

Characteristics of excluded studies *[ordered by study ID]*

Study	Reason for exclusion
Iarŭkova 1999	Design: retrospective study comparing the use of prophylactic curosurf (200 mg/kg) vs rescue curosurf treatment (200 mg/kg) if indicated. This study had, mistakenly, been included in the previous version of the review.
Lefort 2003	Excluded from this review because the treatment group received surfactant as "very early surfactant" once the first RDS signs appeared, instead of prophylactic administration of surfactant.
Morley 2008	Excluded from this review because the trial did not give prophylactic surfactant treatment to all infants in a high risk group, and the research team selected those infants who neither needed acute resuscitation or had no signs of respiratory distress. In the remaining infants, surfactant treatment was not considered as mandatory and followed local protocols.
Rojas 2009	Excluded from this review because the treatment group received surfactant as "very early surfactant" once the first RDS signs appeared, instead of prophylactic administration of surfactant.
Sandri 2010	Excluded because of the comparison: the control group received "early surfactant" soon after treatment failure on CPAP instead of selective administration of surfactant, according to the definitions of interventions used in this review.

CPAP: continuous positive airway pressure; RDS: respiratory distress syndrome; vs: versus

Characteristics of studies awaiting classification *[ordered by study ID]*

[Murphy 2024](#)

Methods	Design: RCT Country(ies): 6 European countries (Ireland, Norway, Czech Republic, Belgium, Sweden, Portugal) Single site/multiple sites: 9 NICUs (Ireland: 2; Norway: 2; Czech Republic: 2; Belgium: 1; Sweden: 1 and Portugal: 1)
Participants	Inclusion criteria 1. Born at a participating center before 29 weeks# gestational age 2. There is a plan to offer intensive care. Exclusion criteria 1. Major congenital anomalies Target sample size: 125 infants per arm (planned target sample: 250) Final sample size: not stated
Interventions	1. Infants are randomized at birth to treatment with oropharyngeal surfactant (120 mg vial < 26 weeks gestational age stratum; 240 mg vial 26 to 28 ⁺⁶ weeks' GA stratum) in addition to CPAP 2. CPAP alone
Outcomes	Primary outcome

Murphy 2024 (Continued)

1. Incidence of endotracheal intubation for respiratory failure within 120 hours of birth

Secondary outcomes include

1. Incidence of mechanical ventilation
2. Endotracheal surfactant use
3. Chronic lung disease
4. Death before hospital discharge

Notes

Funding sources (from Erratum): "Chiesi Farmaceutici, manufacturers of poractant alfa (Curo-surf), supply the study drug free of charge; they had no role in study design; have no role in study conduct; and will have no role in data collection, analysis or interpretation. Further, they will have no role in the decision to present, publish or otherwise report results."

Paediatric Clinical Research Infrastructure Network (PedCRIN), a Project of the European Union Horizon 2020 programme designed to support the extension of clinical drug trials in children throughout Europe, also support the study.

Declarations of interest (from Erratum): Madeleine Murphy is the recipient of a Clinical Research Fellowship by the National Children's Research Centre, Dublin (R17637).

Classified as 'awaiting' classification': because only published in 2024

CPAP: continuous positive airway pressure; GA: gestational age; NICU: neonatal intensive care unit; RCT: randomised controlled trial

Characteristics of ongoing studies [ordered by study ID]

DRKS00028086

Study name	A randomized controlled trial in preterm infants comparing prophylactic with selective "Less Invasive Surfactant Administration"
Methods	Design: RCT Country(ies): Germany Single site/multiple sites: multicenter
Participants	Inclusion criteria 1. Preterm infants with a gestational age between 25 weeks + 0 days and 28 weeks + 6 days 2. Age < 60 minutes 3. Spontaneous breathing on CPAP with supplemental oxygen < 30% Exclusion criteria 1. Lethal malformations Target sample size: 698 Final sample size: not stated
Interventions	Group 1: prophylactic administration of surfactant (100 mg/kg body weight) within the first 60 minutes of life using Less Invasive Surfactant Administration (LISA) Group 2: surfactant only provided if a preterm infant's condition is deteriorating (e.g. oxygen demand rises above 30% for more than 30 minutes or breathing is labored). Surfactant will also be administered using LISA.
Outcomes	Primary outcome

DRKS00028086 (Continued)

1. Forced expiratory volume within the first second is determined by spirometry at the age of 5 years

Secondary outcomes

1. Other spirometry outcomes
2. Cerebral palsy defined as Gross Motor Function Classification System values > 1
3. Values in Movement Assessment Battery for Children's score
4. Intelligence Quotient as determined by the WPPSI IV
5. Medicated obstructive bronchitis

Starting date	Planned study start: 14 February 2022 Actual study starts: 5 March 2022 Planned study completion date: 2024
Contact information	Principal investigator: Prof Dr Wolfgang Göpel, Universitätsklinikum Schleswig-Holstein, Campus Lübeck, Ratzeburger Allee 160, 23538 Lübeck, Deutschland
Notes	Funding source: Federal Ministry of Education and Research Headquarters, Berlin, Germany Conflicts of interest: not stated Upon examining this trial registration record on 28 July 2025, we found a different, updated registration record, and we listed the reference : https://trialsearch.who.int/Trial2.aspx?TrialID=C-TIS2024-516504-42-00 . This record indicated that the trial, as of 7 July 2025, was still recruiting.

NCT04715373

Study name	Less Invasive Administration of Surfactant in the Delivery Room for extremely preterm infants: a randomized control trial
Methods	"single center, unblinded, randomized control, feasibility trial"
Participants	Preterm infants at 22 to 25 weeks GA Exclusion criteria • Major congenital anomalies • Infants receiving only comfort care • Mothers with impaired decision-making capacity
Interventions	Intervention: LISA in delivery room Control: resuscitation per guidelines
Outcomes	Primary - need for intubation and/any mechanical ventilation Secondary - need for mechanical ventilation within 7 days of life; mechanical ventilation; high-frequency ventilation; mechanical ventilator days; bronchopulmonary dysplasia; severe intraventricular hemorrhage; pneumothorax; desaturation and bradycardia; proportion of infants requiring > 1 LISA attempt; death or bronchopulmonary dysplasia
Starting date	22 June 2021
Contact information	Principal investigator: Venkatakrisna Kakkilaya, University of Texas Southwestern Medical Center

NCT04715373 (Continued)

Notes

NCT06557551

Study name	Prophylactic surfactant by thin endotracheal catheter for preterm infants at birth: the ProTeCt trial
Methods	Design: randomized open-label, parallel group trial Country: Ireland Single/multiple sites: single center - National Maternity Hospital Dublin, Ireland Target sample size: 164 Final sample size: not stated
Participants	Inclusion criteria 1. Infants of either sex born before 28 weeks of gestation 2. Clinicians plan to provide intensive care. 3. Consent for their participation provided by parent(s)/guardian(s) before their birth Exclusion criteria 1. Major congenital anomalies (e.g. upper airway anomaly, diaphragmatic hernia, gastroschisis, ex-omphalos major, intestinal atresias, open neural tube defect, aneuploidy, cardiac lesions other than ASD/PFO/VSD/PDA).
Interventions	Group 1: at or after 5 minutes of life, participants will have a thin endotracheal catheter inserted under video laryngoscopy, through which they will receive surfactant. Group 2: no intervention/standard care, participants will continue on face mask breathing support.
Outcomes	Primary outcome 1. Number of participants intubated in the first 72 hours of life. Secondary outcomes 1. Success in the first thin catheter insertion attempt 2. Bradycardia in the delivery room 3. Mask positive pressure ventilation in the delivery room 4. Endotracheal intubation in the delivery room 5. Chest compressions in the delivery room 6. Adrenaline administration in the delivery room 7. Volume expansion in the delivery room 8. Intubation at any time during hospitalization 9. Intratracheal surfactant post intervention 10. Pneumothorax drained with needs aspiration or chest drain insertion 11. Duration of mechanical ventilation 12. Duration of any respiratory support 13. Duration of oxygen therapy 14. Bronchopulmonary dysplasia at 36 weeks 15. Treatment with systemic postnatal corticosteroids 16. Severe intraventricular hemorrhage on cranial ultrasound

NCT06557551 (Continued)

- 17.Cystic periventricular leukomalacia on cranial ultrasound
- 18.Patent ductus arteriosus treated medically or with transcatheter closure or surgical ligation
- 19.Necrotizing enterocolitis
- 20.Spontaneous intestinal perforation
- 21.Retinopathy of prematurity stage 4 or greater
- 22.Death or BPD at 36 weeks
- 23.Duration of hospitalization
- 24.Discharged home receiving supplemental oxygen
- 25.Death during first hospitalization

Starting date	Actual study start: 19 August 2024 Planned study completion: 28 February 2027
Contact information	Colm PF O'Donnell, MB PhD Email: codonnell@nmh.ie
Notes	

ASD: atrial septal defect; BPD: bronchopulmonary dysplasia; CPAP: continuous positive airway pressure; GA: gestational age; LISA: less invasive surfactant administration; PDA: patent ductus arteriosus; PFO: patent foramen ovale; RCT: randomized controlled trial; VSD: ventricular septal defect; WPPSI IV: Wechsler Preschool and Primary Scale of Intelligence

DATA AND ANALYSES

Comparison 1. Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP)

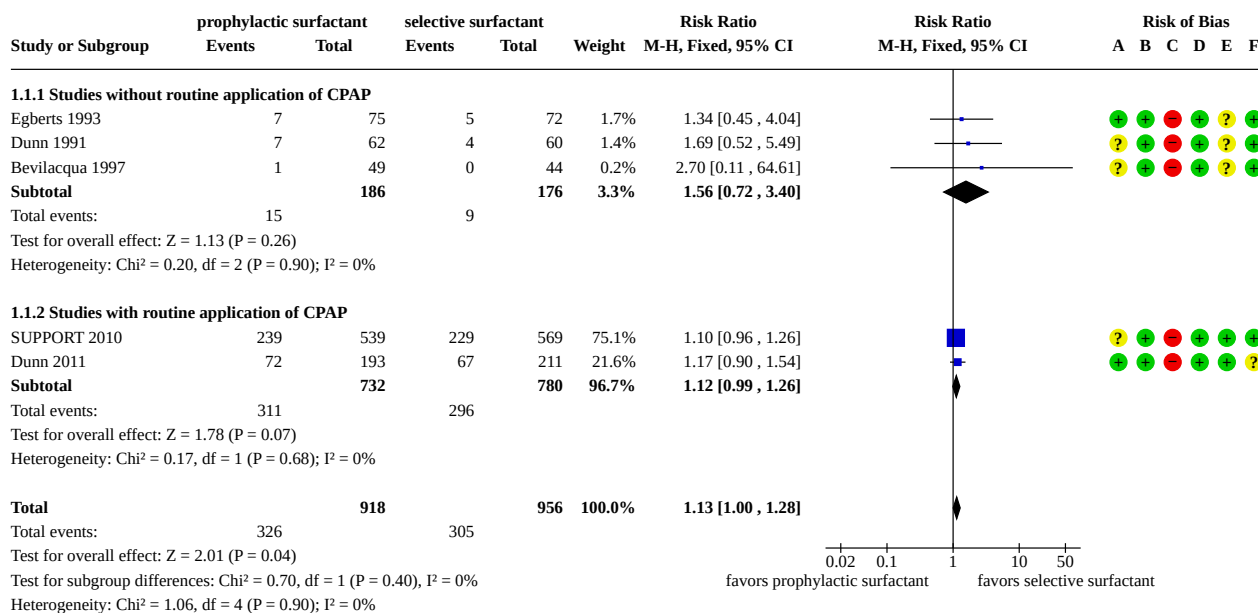
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.1 Chronic lung disease (supplemental oxygen at 36 weeks' PMA)	5	1874	Risk Ratio (M-H, Fixed, 95% CI)	1.13 [1.00, 1.28]
1.1.1 Studies without routine application of CPAP	3	362	Risk Ratio (M-H, Fixed, 95% CI)	1.56 [0.72, 3.40]
1.1.2 Studies with routine application of CPAP	2	1512	Risk Ratio (M-H, Fixed, 95% CI)	1.12 [0.99, 1.26]
1.2 Mortality at 36 weeks' PMA or prior to hospital discharge	6		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
1.2.1 Studies without routine application of CPAP	4	1030	Risk Ratio (M-H, Fixed, 95% CI)	0.72 [0.56, 0.93]
1.2.2 Studies with routine application of CPAP	2	1748	Risk Ratio (M-H, Fixed, 95% CI)	1.28 [1.01, 1.62]
1.3 Moderate-severe neurodevelopmental impairment (two years corrected age)	1		Risk Ratio (M-H, Random, 95% CI)	Totals not selected

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.4 Pneumothorax	8	3094	Risk Ratio (M-H, Fixed, 95% CI)	0.76 [0.56, 1.04]
1.4.1 Studies without routine application of CPAP	7	2663	Risk Ratio (M-H, Fixed, 95% CI)	0.75 [0.53, 1.05]
1.4.2 Studies with routine application of CPAP	1	431	Risk Ratio (M-H, Fixed, 95% CI)	0.89 [0.39, 2.01]
1.5 Pulmonary interstitial emphysema	6	2185	Risk Ratio (M-H, Fixed, 95% CI)	0.78 [0.56, 1.08]
1.5.1 Studies without routine application of CPAP	6	2185	Risk Ratio (M-H, Fixed, 95% CI)	0.78 [0.56, 1.08]
1.6 Any air leak syndromes (including pulmonary interstitial emphysema, pneumothorax, pneumomediastinum)	2	1409	Risk Ratio (M-H, Random, 95% CI)	1.13 [0.78, 1.63]
1.6.1 Studies without routine application of CPAP	1	93	Risk Ratio (M-H, Random, 95% CI)	1.57 [0.49, 5.01]
1.6.2 Studies with routine application of CPAP	1	1316	Risk Ratio (M-H, Random, 95% CI)	1.08 [0.73, 1.60]
1.7 Any pulmonary hemorrhage	4	2023	Risk Ratio (M-H, Fixed, 95% CI)	1.05 [0.49, 2.22]
1.7.1 Studies without routine application of CPAP	3	1592	Risk Ratio (M-H, Fixed, 95% CI)	0.73 [0.28, 1.87]
1.7.2 Studies with routine application of CPAP	1	431	Risk Ratio (M-H, Fixed, 95% CI)	2.12 [0.54, 8.39]
1.8 Bronchopulmonary dysplasia (oxygen requirement at 28 to 30 days of age)	9	3163	Risk Ratio (M-H, Fixed, 95% CI)	1.02 [0.91, 1.14]
1.8.1 Studies without routine application of CPAP	8	2761	Risk Ratio (M-H, Fixed, 95% CI)	0.95 [0.82, 1.11]
1.8.2 Studies with routine application of CPAP	1	402	Risk Ratio (M-H, Fixed, 95% CI)	1.16 [0.99, 1.36]
1.9 Bronchopulmonary dysplasia or death at 28 to 30 days of age	8	3033	Risk Ratio (M-H, Fixed, 95% CI)	0.89 [0.81, 0.98]
1.9.1 Studies without routine application of CPAP	7	2613	Risk Ratio (M-H, Fixed, 95% CI)	0.82 [0.73, 0.92]
1.9.2 Studies with routine application of CPAP	1	420	Risk Ratio (M-H, Fixed, 95% CI)	1.12 [0.96, 1.31]
1.10 Chronic lung disease or death (at 36 weeks' PMA)	4	2122	Risk Ratio (M-H, Fixed, 95% CI)	1.09 [0.99, 1.20]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.10.1 Studies without routine application of CPAP	2	378	Risk Ratio (M-H, Fixed, 95% CI)	0.87 [0.63, 1.21]
1.10.2 Studies with routine application of CPAP	2	1744	Risk Ratio (M-H, Fixed, 95% CI)	1.12 [1.02, 1.24]
1.11 Neonatal mortality	9	3191	Risk Ratio (M-H, Fixed, 95% CI)	0.71 [0.58, 0.87]
1.11.1 Studies without routine application of CPAP	8	2761	Risk Ratio (M-H, Fixed, 95% CI)	0.69 [0.56, 0.85]
1.11.2 Studies with routine application of CPAP	1	430	Risk Ratio (M-H, Fixed, 95% CI)	1.32 [0.53, 3.28]
1.12 Patent ductus arteriosus	8	3093	Risk Ratio (M-H, Fixed, 95% CI)	0.98 [0.89, 1.08]
1.12.1 Studies without routine application of CPAP	7	2663	Risk Ratio (M-H, Fixed, 95% CI)	0.98 [0.88, 1.10]
1.12.2 Studies with routine application of CPAP	1	430	Risk Ratio (M-H, Fixed, 95% CI)	0.97 [0.79, 1.20]
1.13 Culture-proven bacterial sepsis	5	2182	Risk Ratio (M-H, Fixed, 95% CI)	1.06 [0.80, 1.40]
1.13.1 Studies without routine application of CPAP	4	1757	Risk Ratio (M-H, Fixed, 95% CI)	0.91 [0.66, 1.25]
1.13.2 Studies with routine application of CPAP	1	425	Risk Ratio (M-H, Fixed, 95% CI)	1.70 [0.96, 3.03]
1.14 Culture-proven fungal sepsis	1		Risk Ratio (M-H, Fixed, 95% CI)	Totals not selected
1.15 Necrotizing enterocolitis	8	4237	Risk Ratio (M-H, Fixed, 95% CI)	0.90 [0.73, 1.10]
1.15.1 Studies without routine application of CPAP	6	2516	Risk Ratio (M-H, Fixed, 95% CI)	1.05 [0.76, 1.44]
1.15.2 Studies with routine application of CPAP	2	1721	Risk Ratio (M-H, Fixed, 95% CI)	0.79 [0.60, 1.04]
1.16 Intraventricular hemorrhage (in infants receiving cranial ultrasound)	9	3077	Risk Ratio (M-H, Fixed, 95% CI)	0.94 [0.85, 1.04]
1.16.1 Studies without routine application of CPAP	8	2656	Risk Ratio (M-H, Fixed, 95% CI)	0.93 [0.83, 1.03]
1.16.2 Studies with routine application of CPAP	1	421	Risk Ratio (M-H, Fixed, 95% CI)	1.05 [0.73, 1.50]

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
1.17 Severe intraventricular hemorrhage	10	4347	Risk Ratio (M-H, Fixed, 95% CI)	0.87 [0.74, 1.04]
1.17.1 Studies without routine application of CPAP	8	2656	Risk Ratio (M-H, Fixed, 95% CI)	0.87 [0.70, 1.08]
1.17.2 Studies with routine application of CPAP	2	1691	Risk Ratio (M-H, Fixed, 95% CI)	0.88 [0.67, 1.16]
1.18 Periventricular leukomalacia	5	2091	Risk Ratio (M-H, Fixed, 95% CI)	1.23 [0.88, 1.72]
1.18.1 Studies without routine application of CPAP	4	1695	Risk Ratio (M-H, Fixed, 95% CI)	1.26 [0.89, 1.77]
1.18.2 Studies with routine application of CPAP	1	396	Risk Ratio (M-H, Fixed, 95% CI)	0.72 [0.12, 4.28]
1.19 Retinopathy of prematurity (in infants screened)	8	3574	Risk Ratio (M-H, Fixed, 95% CI)	1.03 [0.89, 1.19]
1.19.1 Studies without routine application of CPAP	6	2215	Risk Ratio (M-H, Fixed, 95% CI)	1.17 [0.95, 1.45]
1.19.2 Studies with routine application of CPAP	2	1359	Risk Ratio (M-H, Fixed, 95% CI)	0.91 [0.75, 1.11]
1.20 Retinopathy of prematurity, Stage 3 or more	5	2738	Risk Ratio (M-H, Fixed, 95% CI)	0.97 [0.73, 1.28]
1.20.1 Studies without routine application of CPAP	3	1379	Risk Ratio (M-H, Fixed, 95% CI)	0.97 [0.46, 2.04]
1.20.2 Studies without routine application of CPAP	2	1359	Risk Ratio (M-H, Fixed, 95% CI)	0.97 [0.72, 1.30]

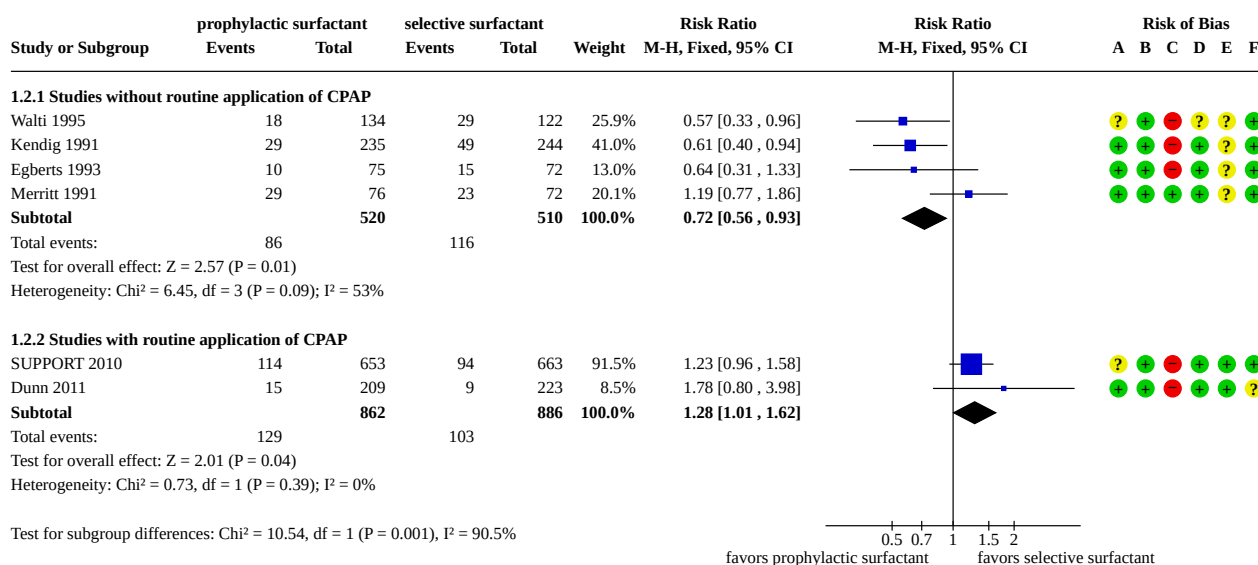
Analysis 1.1. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 1: Chronic lung disease (supplemental oxygen at 36 weeks' PMA)



Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Incomplete outcome data (attrition bias)
- (E) Selective reporting (reporting bias)
- (F) Other bias

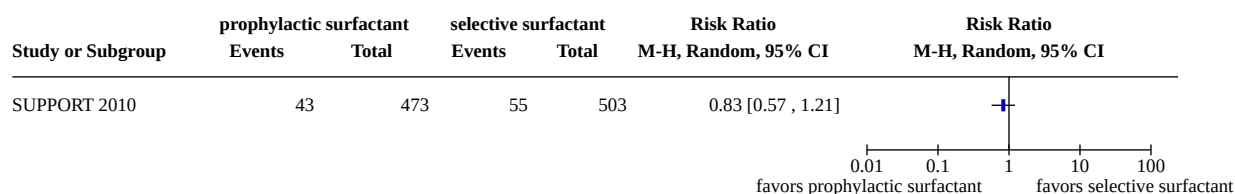
Analysis 1.2. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 2: Mortality at 36 weeks' PMA or prior to hospital discharge



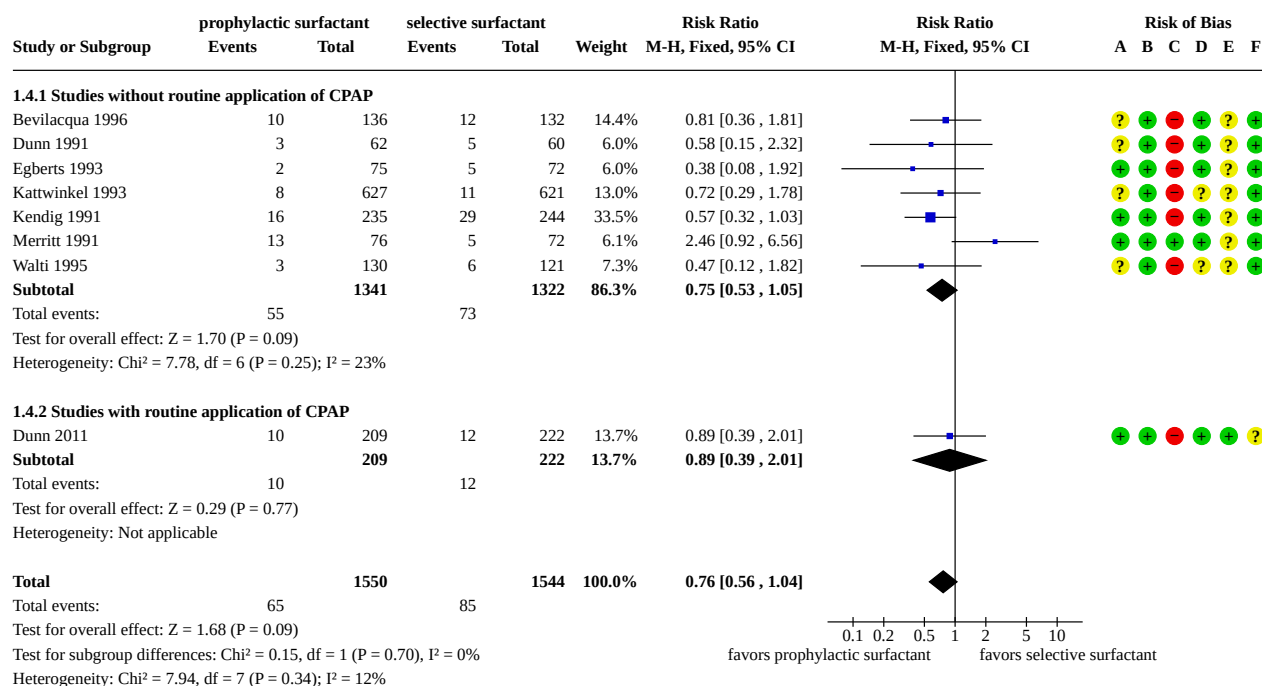
Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Incomplete outcome data (attrition bias)
- (E) Selective reporting (reporting bias)
- (F) Other bias

Analysis 1.3. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 3: Moderate-severe neurodevelopmental impairment (two years corrected age)

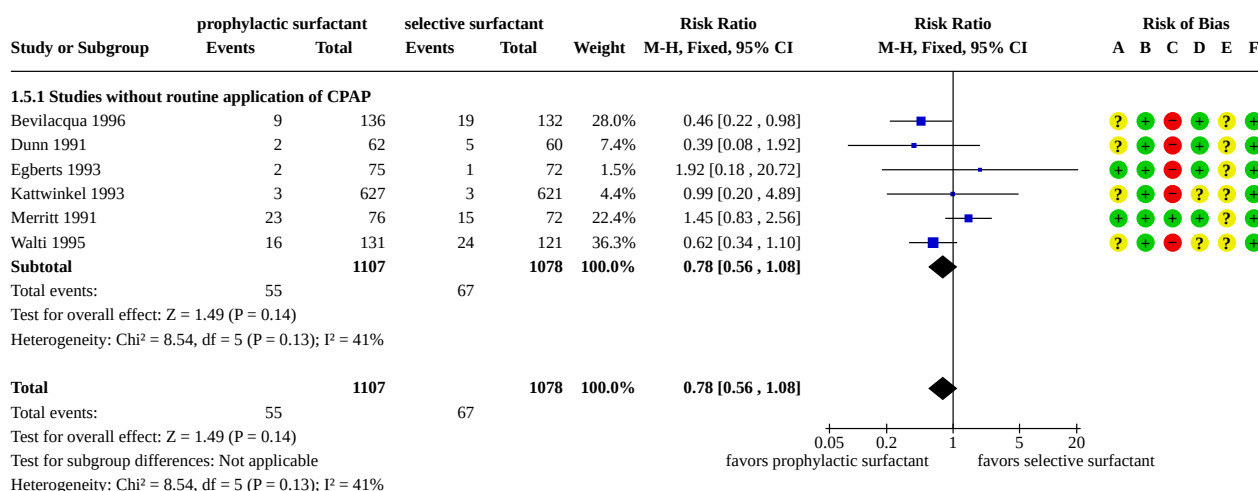


Analysis 1.4. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 4: Pneumothorax

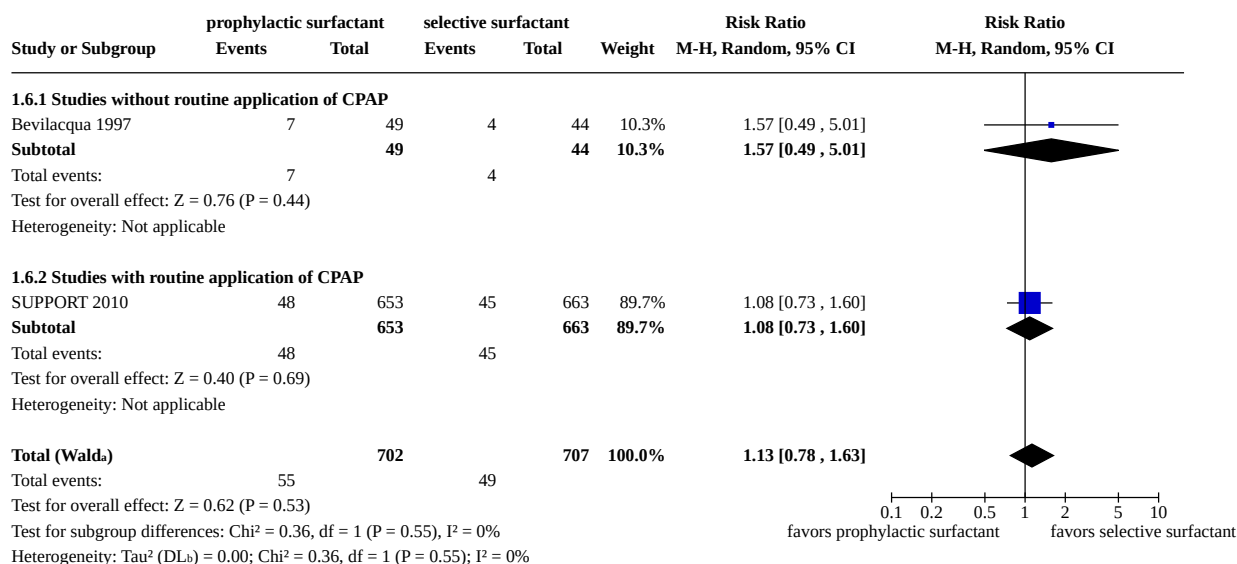


Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Incomplete outcome data (attrition bias)
- (E) Selective reporting (reporting bias)
- (F) Other bias

Analysis 1.5. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 5: Pulmonary interstitial emphysema**Risk of bias legend**

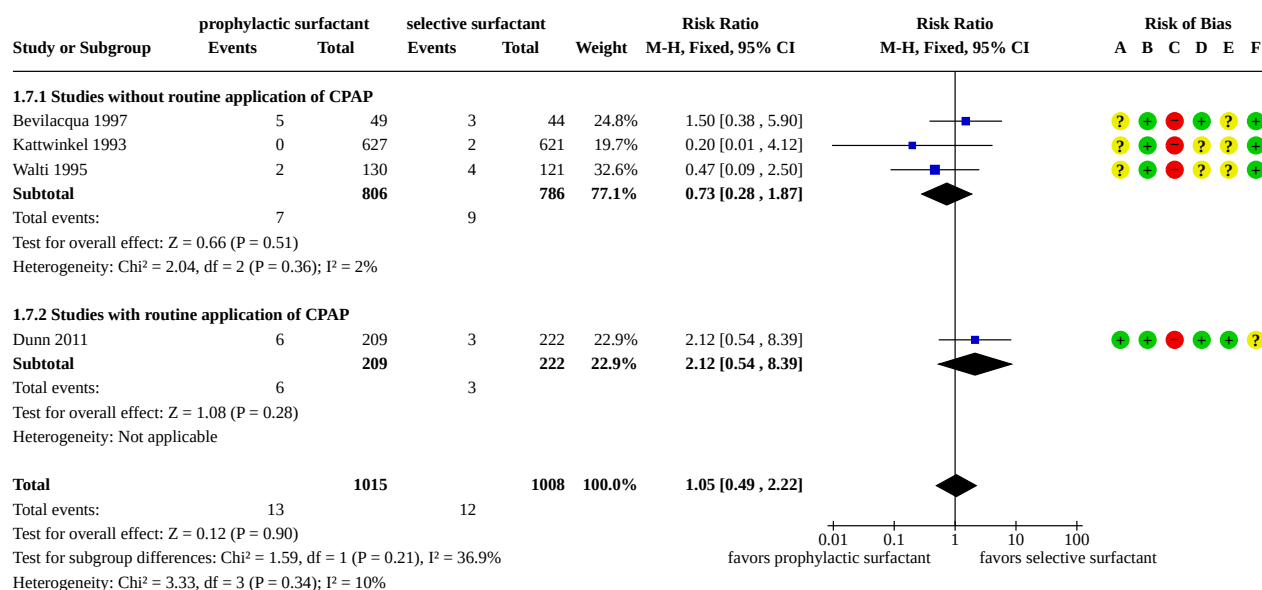
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- (C) Blinding of participants and personnel (performance bias)
- (D) Incomplete outcome data (attrition bias)
- (E) Selective reporting (reporting bias)
- (F) Other bias

Analysis 1.6. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 6: Any air leak syndromes (including pulmonary interstitial emphysema, pneumothorax, pneumomediastinum)**Footnotes**

^aCI calculated by Wald-type method.

^b Tau^2 calculated by DerSimonian and Laird method.

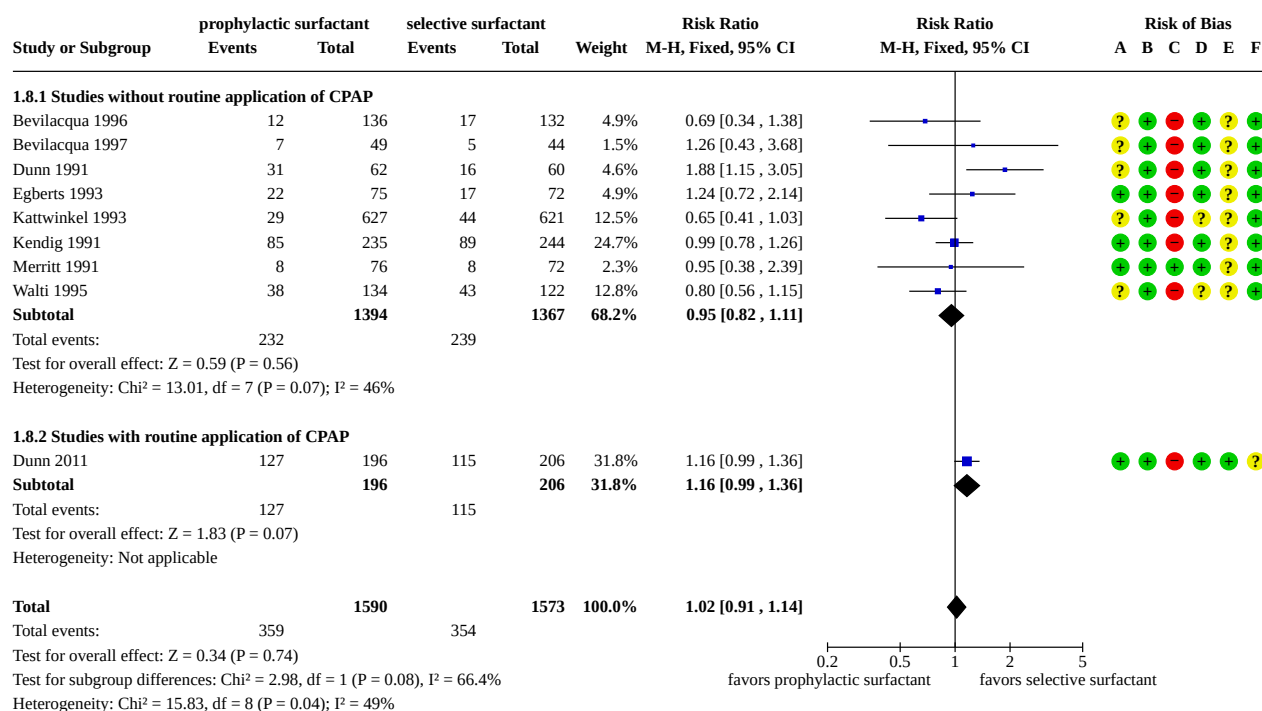
Analysis 1.7. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 7: Any pulmonary hemorrhage



Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Incomplete outcome data (attrition bias)
- (E) Selective reporting (reporting bias)
- (F) Other bias

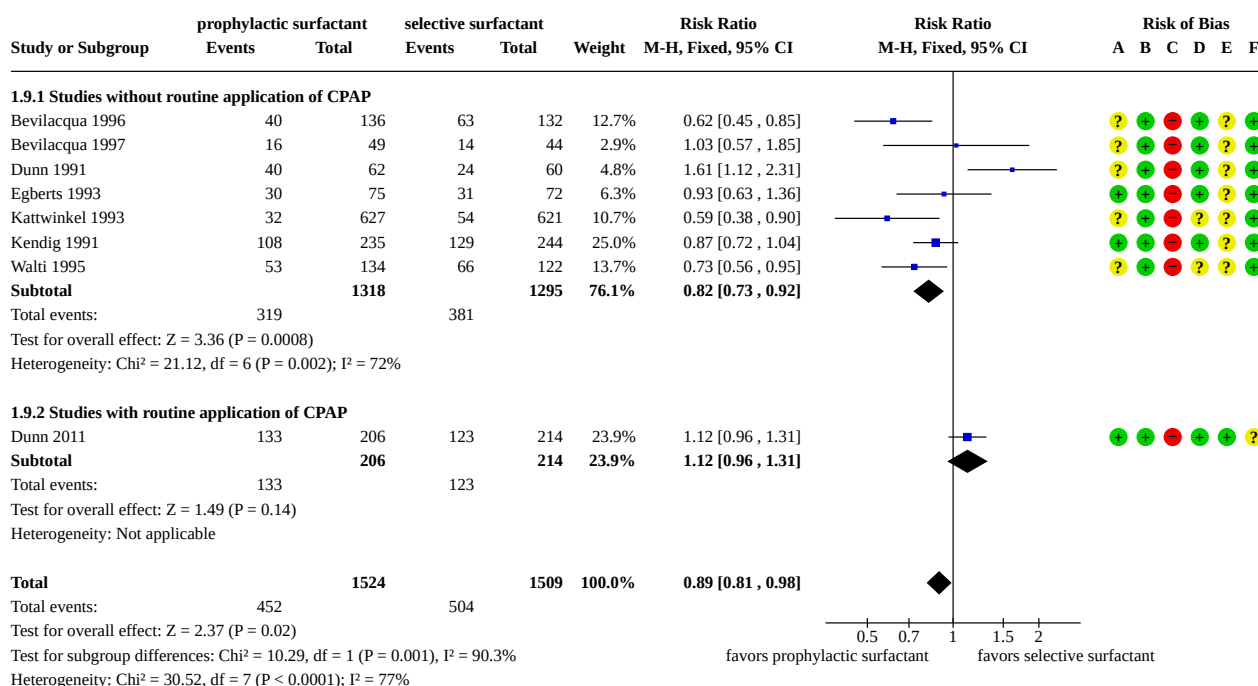
Analysis 1.8. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 8: Bronchopulmonary dysplasia (oxygen requirement at 28 to 30 days of age)



Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Incomplete outcome data (attrition bias)
- (E) Selective reporting (reporting bias)
- (F) Other bias

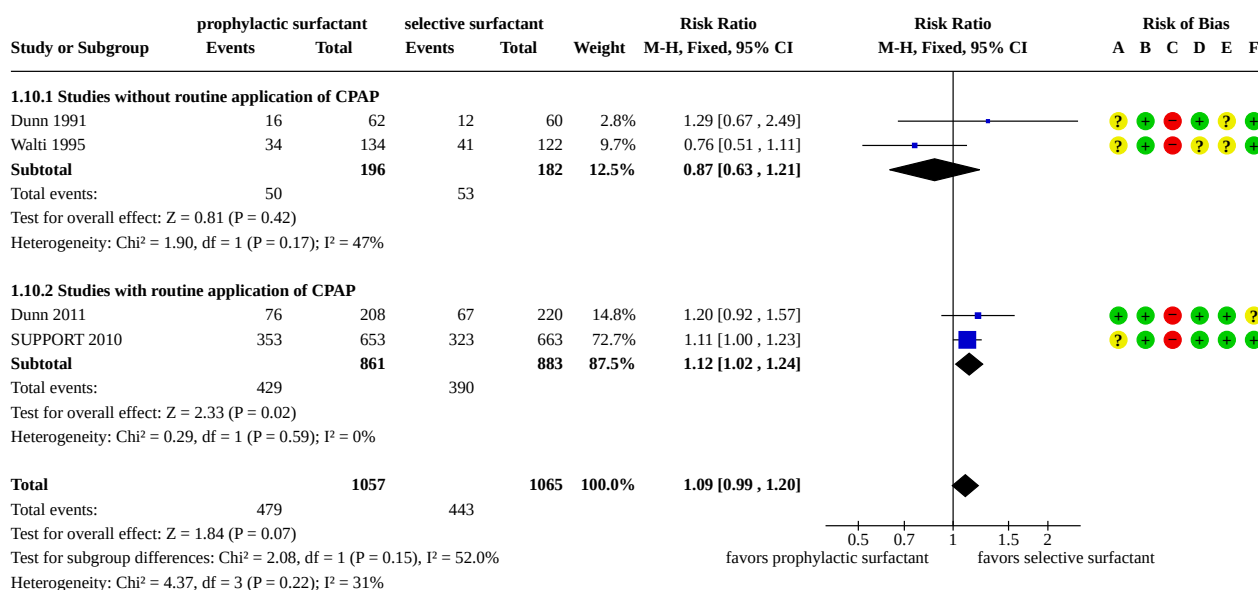
Analysis 1.9. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 9: Bronchopulmonary dysplasia or death at 28 to 30 days of age



Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Incomplete outcome data (attrition bias)
- (E) Selective reporting (reporting bias)
- (F) Other bias

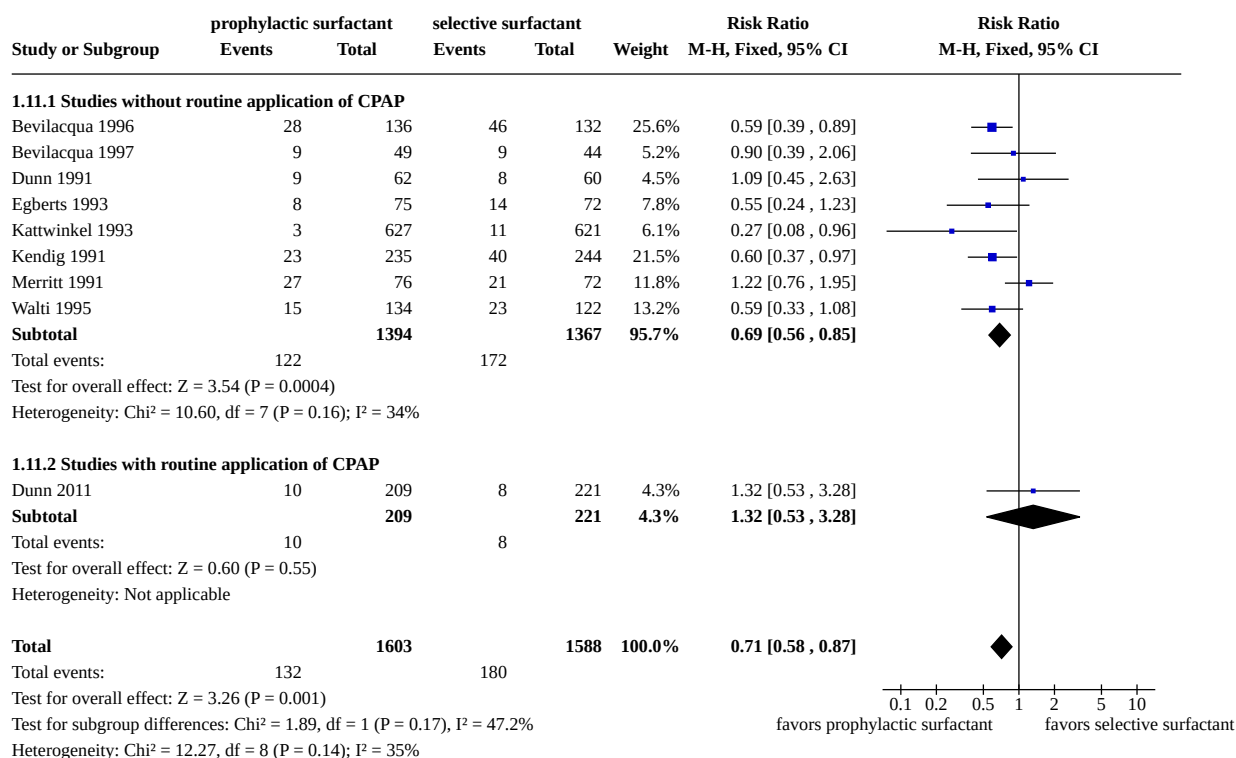
Analysis 1.10. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 10: Chronic lung disease or death (at 36 weeks' PMA)



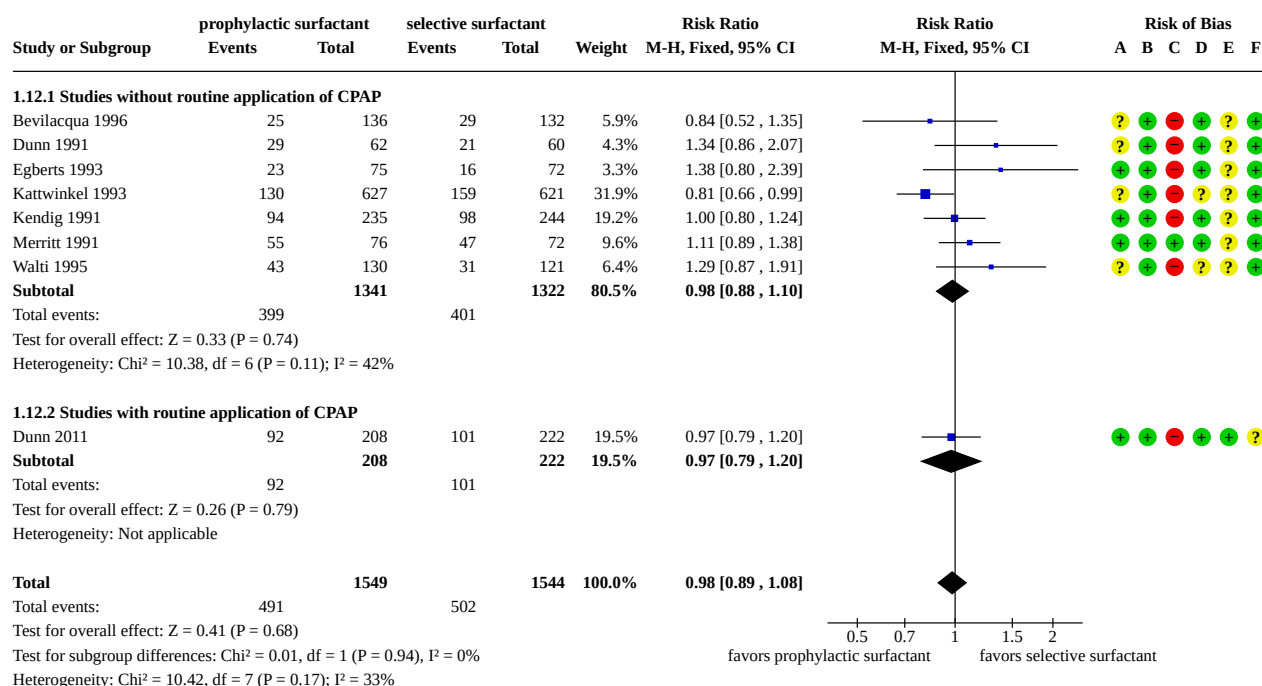
Risk of bias legend

- (A) Random sequence generation (selection bias)
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- (C) Blinding of participants and personnel (performance bias)
- (D) Incomplete outcome data (attrition bias)
- (E) Selective reporting (reporting bias)
- (F) Other bias

Analysis 1.11. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 11: Neonatal mortality



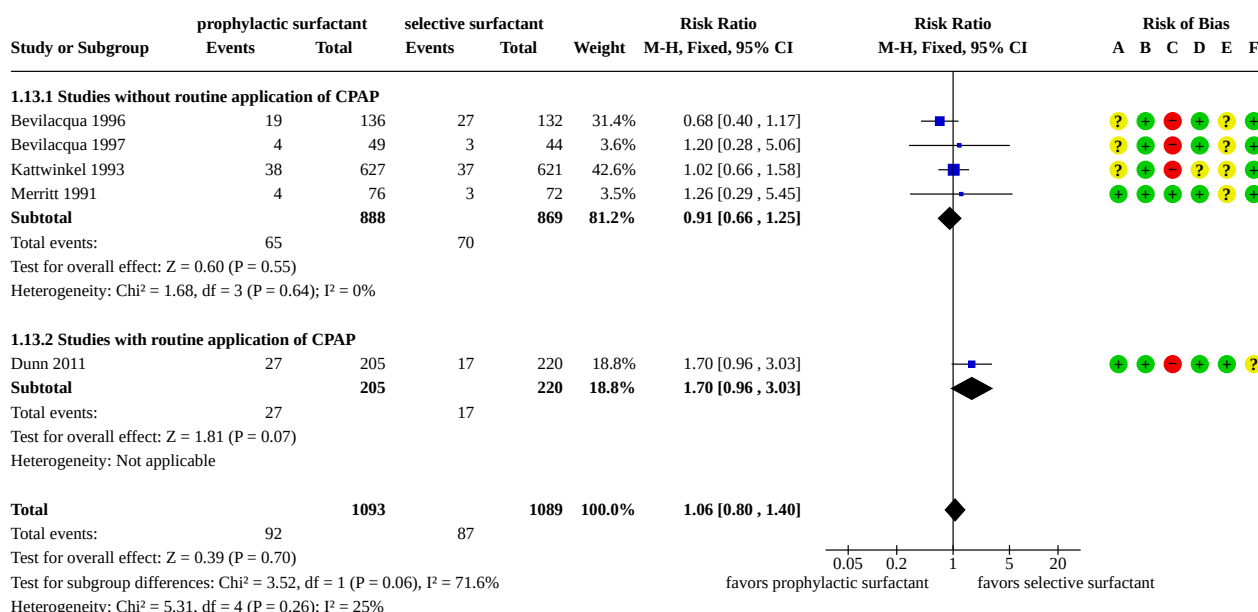
Analysis 1.12. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 12: Patent ductus arteriosus



Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Incomplete outcome data (attrition bias)
- (E) Selective reporting (reporting bias)
- (F) Other bias

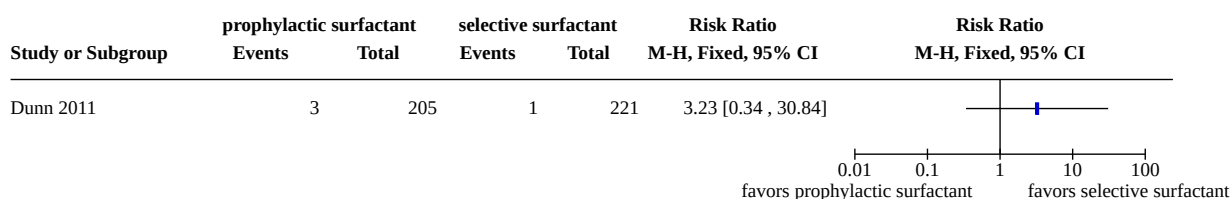
Analysis 1.13. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 13: Culture-proven bacterial sepsis



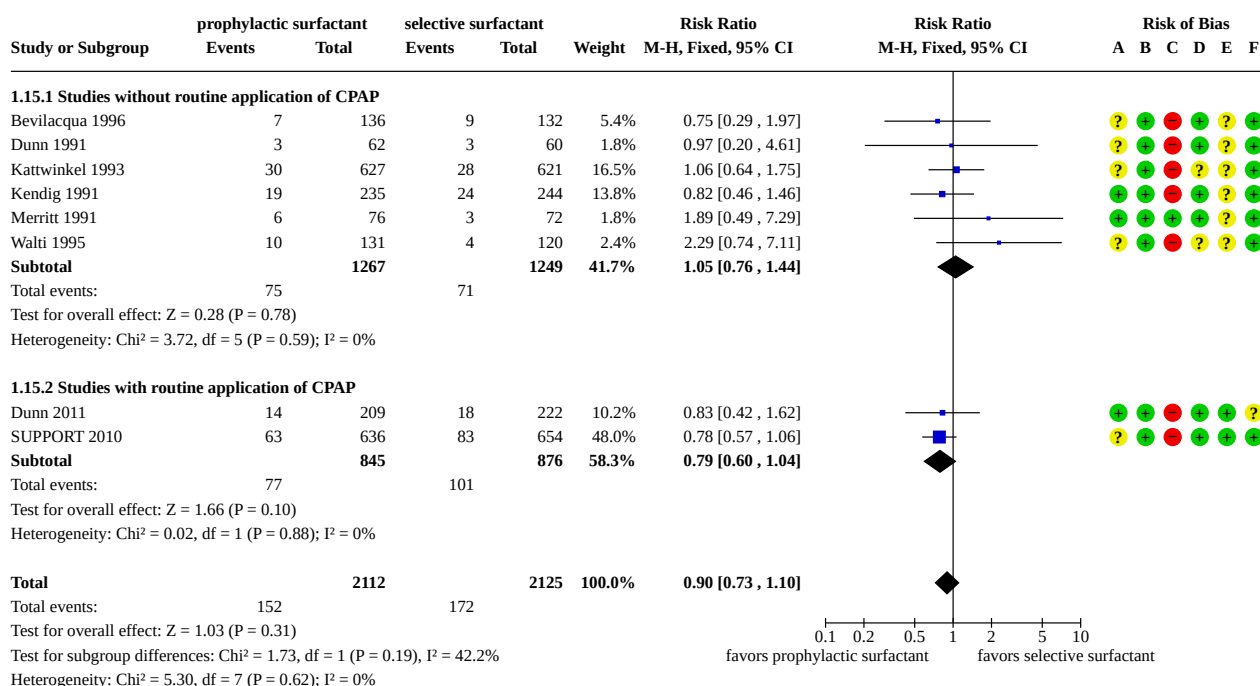
Risk of bias legend

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- (D) Incomplete outcome data (attrition bias)
- (E) Selective reporting (reporting bias)
- (F) Other bias

Analysis 1.14. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 14: Culture-proven fungal sepsis



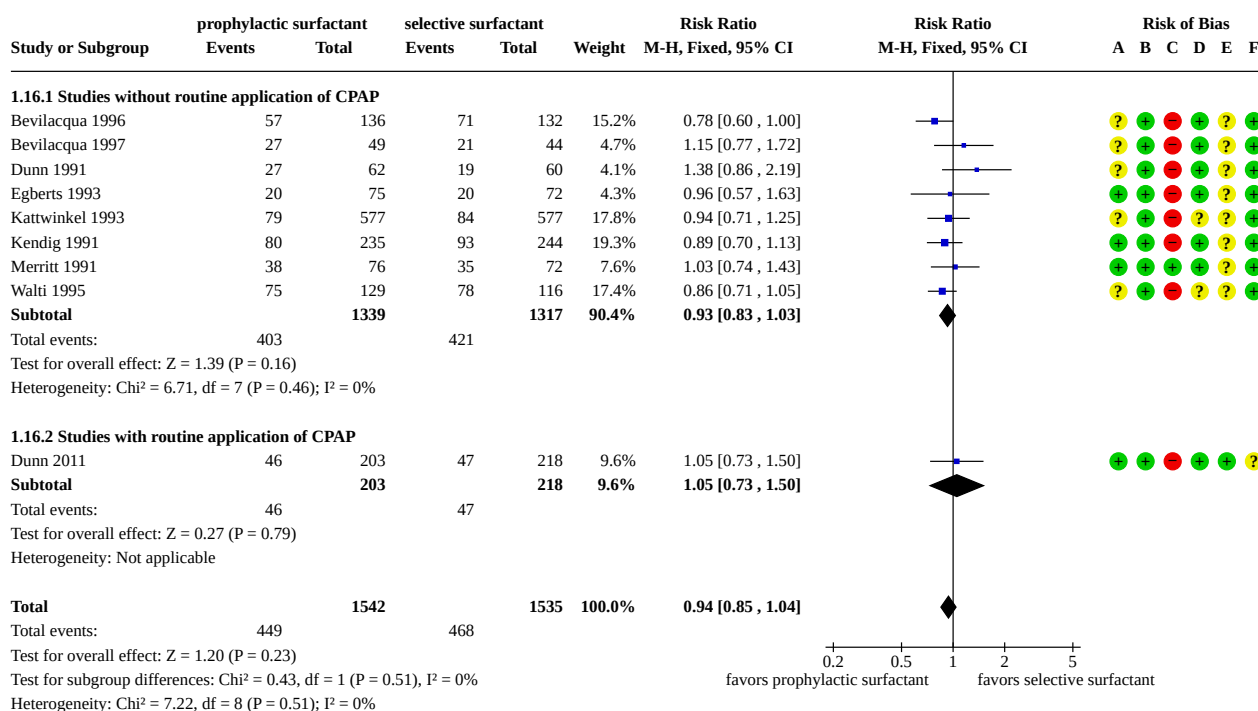
Analysis 1.15. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 15: Necrotizing enterocolitis



Risk of bias legend

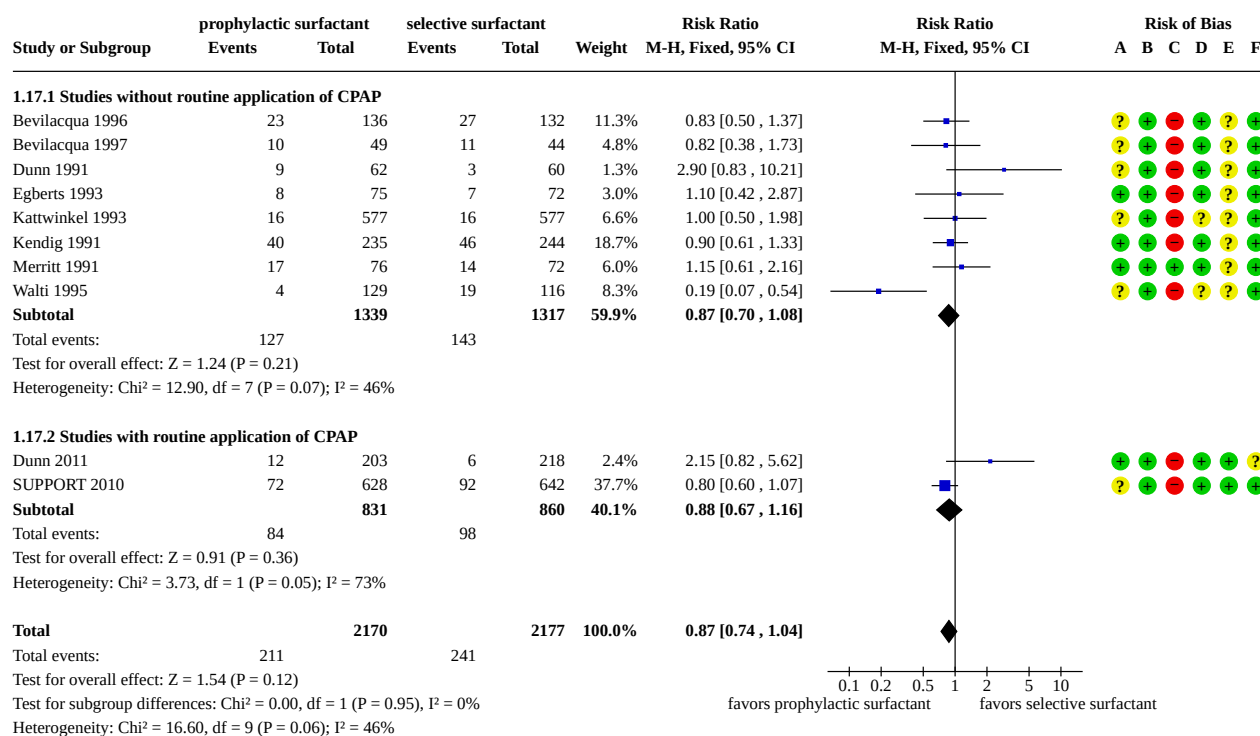
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- (D) Incomplete outcome data (attrition bias)
- (E) Selective reporting (reporting bias)
- (F) Other bias

Analysis 1.16. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 16: Intraventricular hemorrhage (in infants receiving cranial ultrasound)



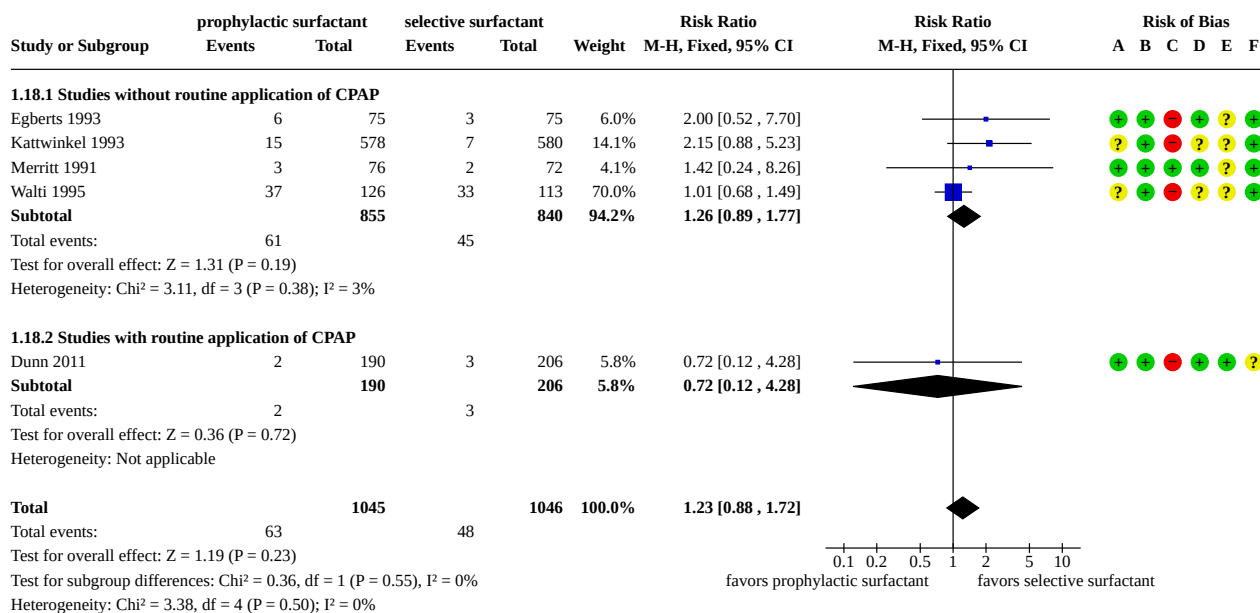
Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Incomplete outcome data (attrition bias)
- (E) Selective reporting (reporting bias)
- (F) Other bias

Analysis 1.17. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 17: Severe intraventricular hemorrhage**Risk of bias legend**

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Incomplete outcome data (attrition bias)
- (E) Selective reporting (reporting bias)
- (F) Other bias

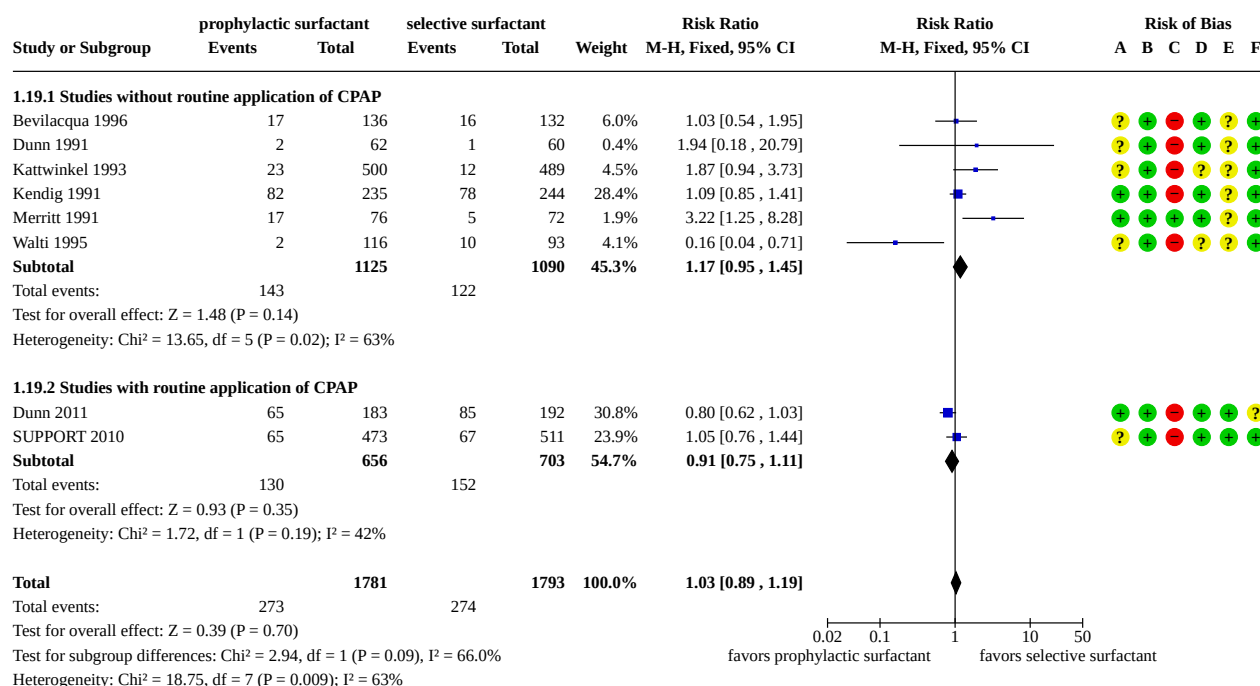
Analysis 1.18. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 18: Periventricular leukomalacia



Risk of bias legend

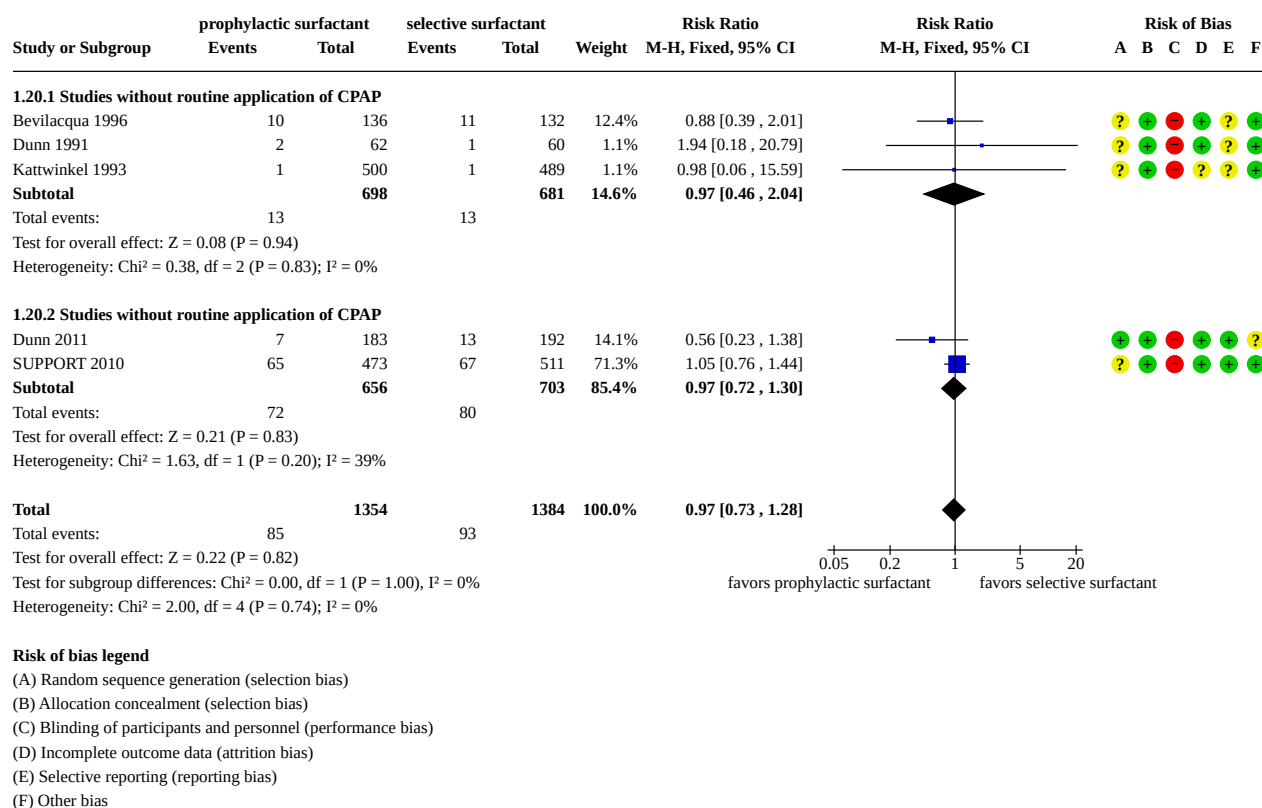
- (A) Random sequence generation (selection bias)
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- (C) Blinding of participants and personnel (performance bias)
- (D) Incomplete outcome data (attrition bias)
- (E) Selective reporting (reporting bias)
- (F) Other bias

Analysis 1.19. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 19: Retinopathy of prematurity (in infants screened)



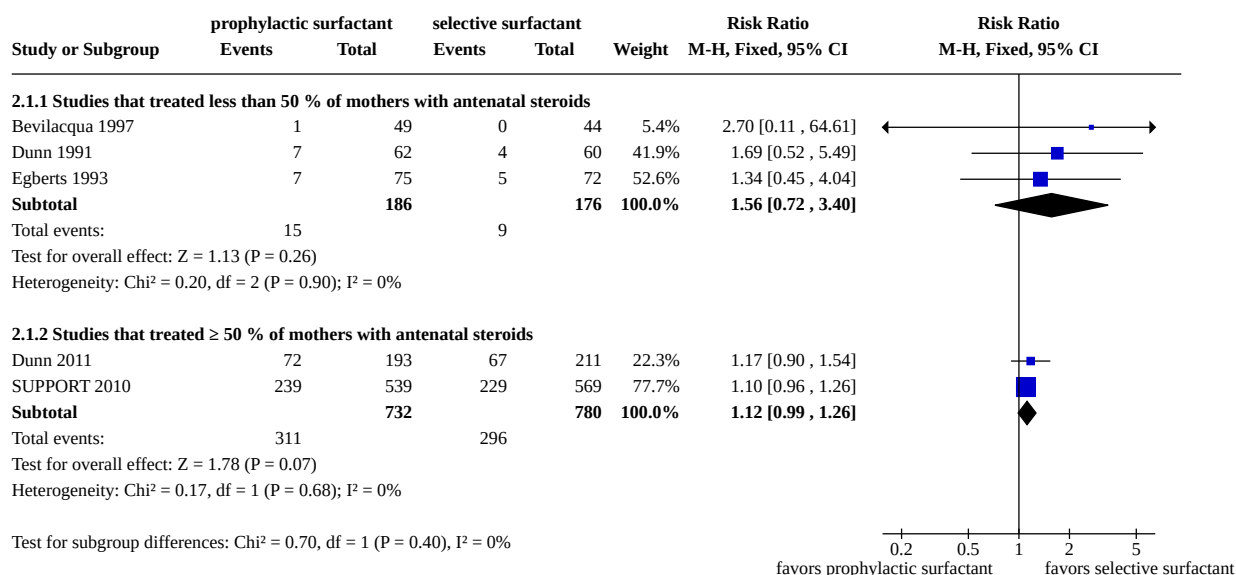
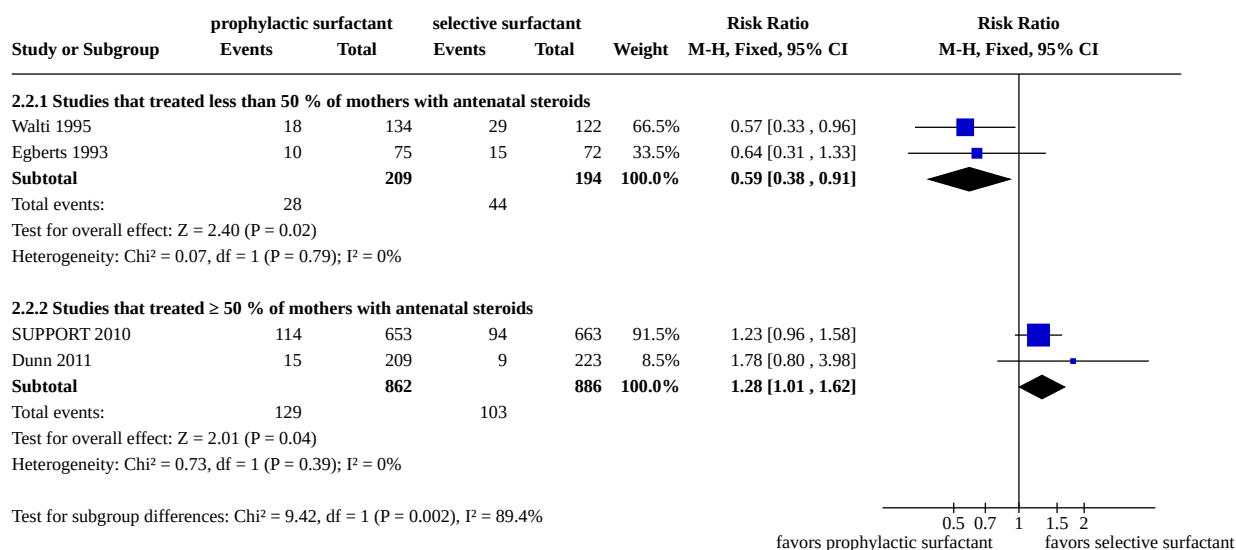
Risk of bias legend

- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Incomplete outcome data (attrition bias)
- (E) Selective reporting (reporting bias)
- (F) Other bias

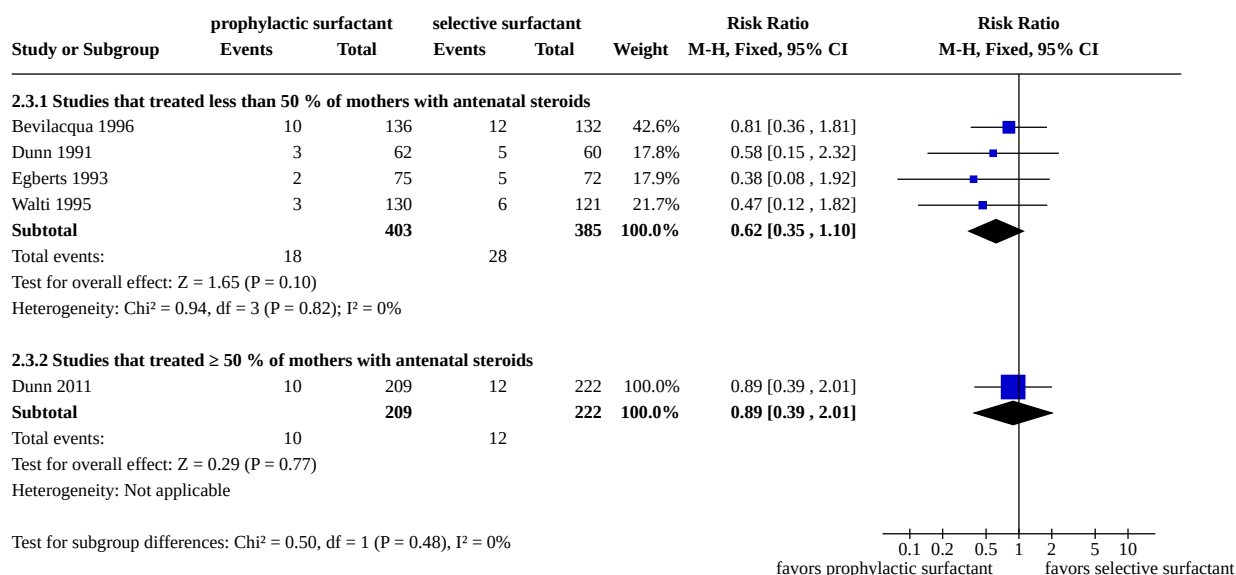
Analysis 1.20. Comparison 1: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (all infants analyzed by use of CPAP), Outcome 20: Retinopathy of prematurity, Stage 3 or more**Comparison 2. Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by receipt of antenatal steroids)**

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.1 Chronic lung disease (supplemental oxygen at 36 weeks' PMA)	5		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
2.1.1 Studies that treated less than 50 % of mothers with antenatal steroids	3	362	Risk Ratio (M-H, Fixed, 95% CI)	1.56 [0.72, 3.40]
2.1.2 Studies that treated $\geq 50\%$ of mothers with antenatal steroids	2	1512	Risk Ratio (M-H, Fixed, 95% CI)	1.12 [0.99, 1.26]
2.2 Mortality at 36 weeks' PMA or prior to hospital discharge	4		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
2.2.1 Studies that treated less than 50 % of mothers with antenatal steroids	2	403	Risk Ratio (M-H, Fixed, 95% CI)	0.59 [0.38, 0.91]
2.2.2 Studies that treated $\geq 50\%$ of mothers with antenatal steroids	2	1748	Risk Ratio (M-H, Fixed, 95% CI)	1.28 [1.01, 1.62]

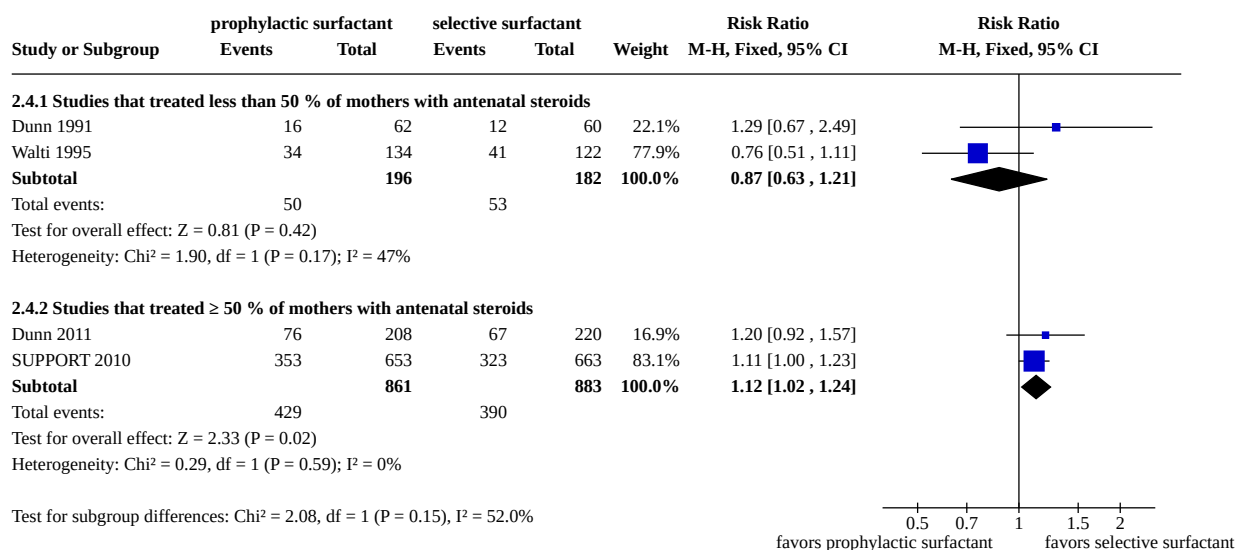
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
2.3 Pneumothorax	5		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
2.3.1 Studies that treated less than 50 % of mothers with antenatal steroids	4	788	Risk Ratio (M-H, Fixed, 95% CI)	0.62 [0.35, 1.10]
2.3.2 Studies that treated $\geq$ 50 % of mothers with antenatal steroids	1	431	Risk Ratio (M-H, Fixed, 95% CI)	0.89 [0.39, 2.01]
2.4 Chronic lung disease or death (at 36 weeks' PMA)	4		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
2.4.1 Studies that treated less than 50 % of mothers with antenatal steroids	2	378	Risk Ratio (M-H, Fixed, 95% CI)	0.87 [0.63, 1.21]
2.4.2 Studies that treated $\geq$ 50 % of mothers with antenatal steroids	2	1744	Risk Ratio (M-H, Fixed, 95% CI)	1.12 [1.02, 1.24]
2.5 Bronchopulmonary dysplasia (oxygen requirement at 28 to 30 days of age)	6		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
2.5.1 Studies that treated less than 50 % of mothers with antenatal steroids	5	798	Risk Ratio (M-H, Fixed, 95% CI)	1.04 [0.83, 1.31]
2.5.2 Studies that treated $\geq$ 50 % of mothers with antenatal steroids	1	402	Risk Ratio (M-H, Fixed, 95% CI)	1.16 [0.99, 1.36]
2.6 Bronchopulmonary dysplasia or death at 28 to 30 days of age	6		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
2.6.1 Studies that treated less than 50 % of mothers with antenatal steroids	5	886	Risk Ratio (M-H, Fixed, 95% CI)	0.85 [0.73, 0.99]
2.6.2 Studies that treated $\geq$ 50 % of mothers with antenatal steroids	1	420	Risk Ratio (M-H, Fixed, 95% CI)	1.12 [0.96, 1.31]

Analysis 2.1. Comparison 2: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by receipt of antenatal steroids), Outcome 1: Chronic lung disease (supplemental oxygen at 36 weeks' PMA)**Analysis 2.2. Comparison 2: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by receipt of antenatal steroids), Outcome 2: Mortality at 36 weeks' PMA or prior to hospital discharge**

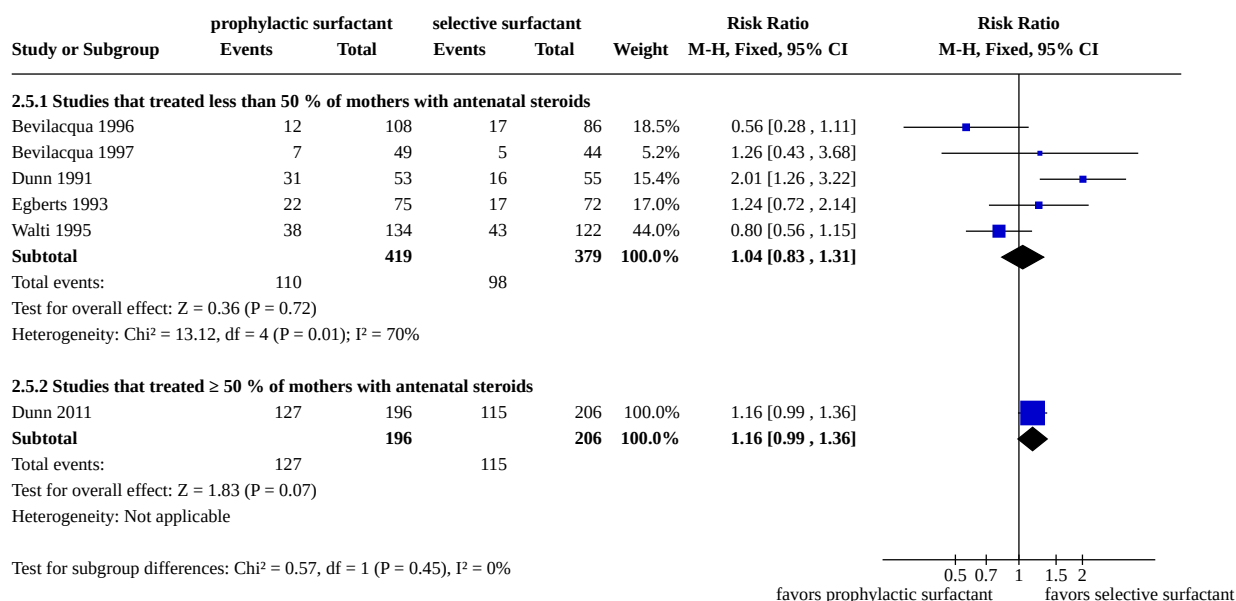
Analysis 2.3. Comparison 2: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by receipt of antenatal steroids), Outcome 3: Pneumothorax



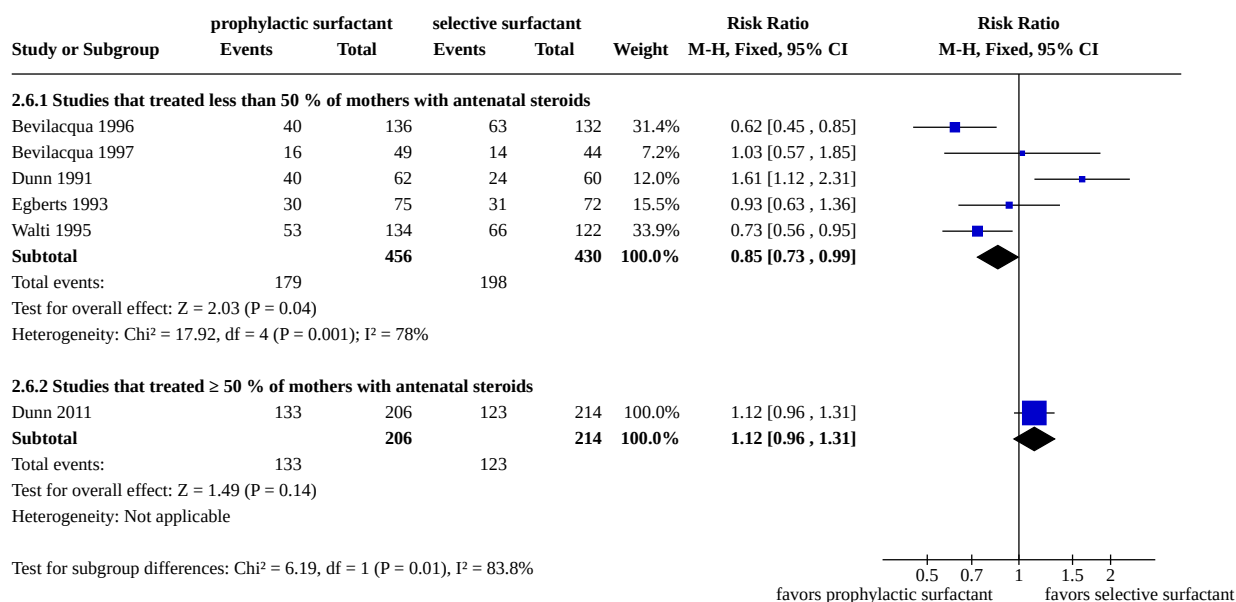
Analysis 2.4. Comparison 2: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by receipt of antenatal steroids), Outcome 4: Chronic lung disease or death (at 36 weeks' PMA)



Analysis 2.5. Comparison 2: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by receipt of antenatal steroids), Outcome 5: Bronchopulmonary dysplasia (oxygen requirement at 28 to 30 days of age)



Analysis 2.6. Comparison 2: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by receipt of antenatal steroids), Outcome 6: Bronchopulmonary dysplasia or death at 28 to 30 days of age



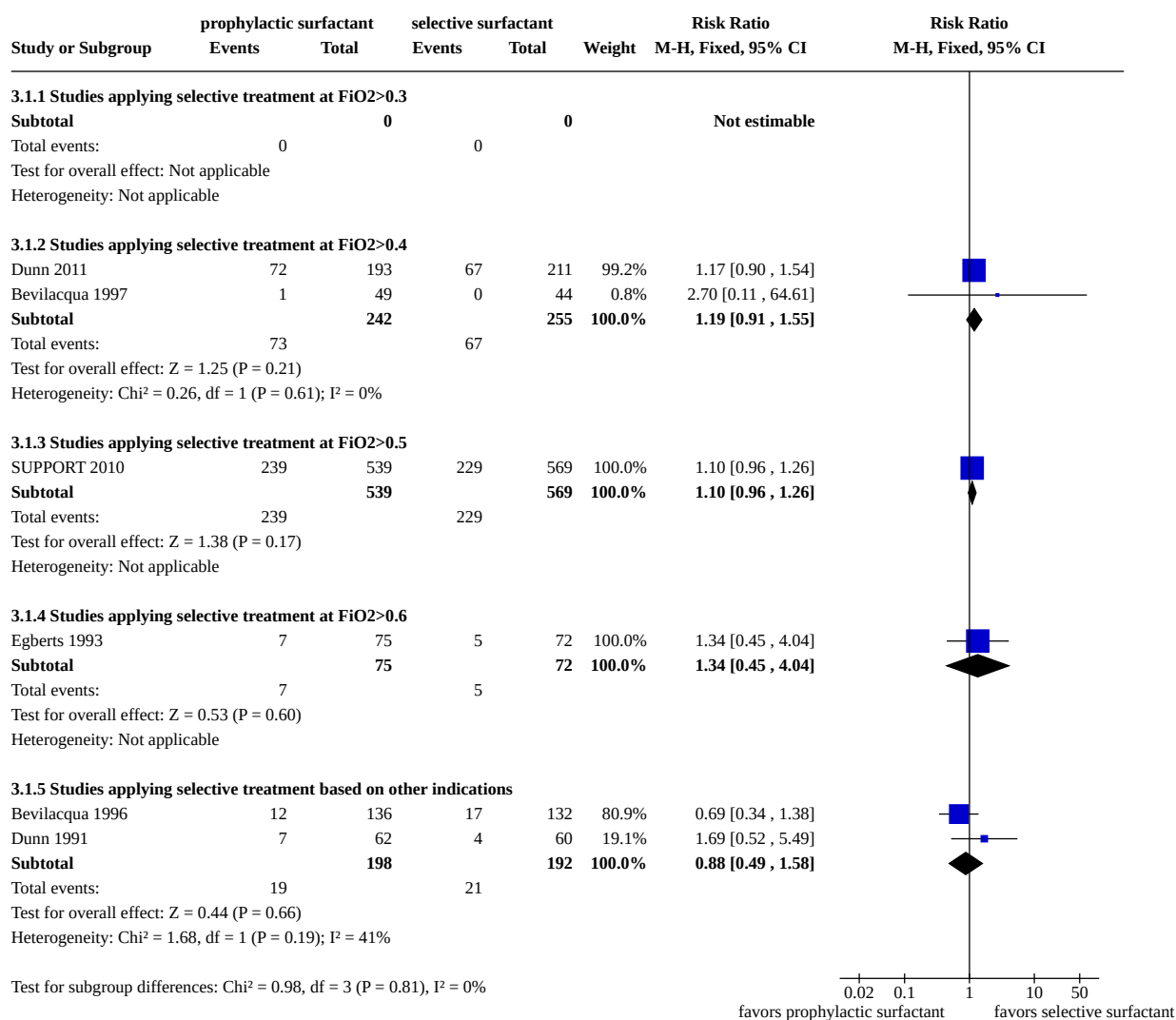
Comparison 3. Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed on the threshold used to apply selective treatment)

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
3.1 Chronic lung disease (supplemental oxygen at 36 weeks' PMA)	6		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
3.1.1 Studies applying selective treatment at $FiO_2 > 0.3$	0	0	Risk Ratio (M-H, Fixed, 95% CI)	Not estimable
3.1.2 Studies applying selective treatment at $FiO_2 > 0.4$	2	497	Risk Ratio (M-H, Fixed, 95% CI)	1.19 [0.91, 1.55]
3.1.3 Studies applying selective treatment at $FiO_2 > 0.5$	1	1108	Risk Ratio (M-H, Fixed, 95% CI)	1.10 [0.96, 1.26]
3.1.4 Studies applying selective treatment at $FiO_2 > 0.6$	1	147	Risk Ratio (M-H, Fixed, 95% CI)	1.34 [0.45, 4.04]
3.1.5 Studies applying selective treatment based on other indications	2	390	Risk Ratio (M-H, Fixed, 95% CI)	0.88 [0.49, 1.58]
3.2 Mortality at 36 weeks' PMA or prior to hospital discharge	6		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
3.2.1 Studies applying selective treatment at $FiO_2 > 0.3$	0	0	Risk Ratio (M-H, Fixed, 95% CI)	Not estimable
3.2.2 Studies applying selective treatment at $FiO_2 > 0.4$	2	911	Risk Ratio (M-H, Fixed, 95% CI)	0.79 [0.55, 1.14]
3.2.3 Studies applying selective treatment at $FiO_2 > 0.5$	2	1464	Risk Ratio (M-H, Fixed, 95% CI)	1.22 [0.98, 1.52]
3.2.4 Studies applying selective treatment at $FiO_2 > 0.6$	1	147	Risk Ratio (M-H, Fixed, 95% CI)	0.64 [0.31, 1.33]
3.2.5 Studies applying selective treatment based on other indications	1	256	Risk Ratio (M-H, Fixed, 95% CI)	0.57 [0.33, 0.96]
3.3 Pneumothorax	8		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
3.3.1 Studies applying selective treatment at $FiO_2 > 0.3$	1	1248	Risk Ratio (M-H, Fixed, 95% CI)	0.72 [0.29, 1.78]
3.3.2 Studies applying selective treatment at $FiO_2 > 0.4$	2	910	Risk Ratio (M-H, Fixed, 95% CI)	0.66 [0.41, 1.06]
3.3.3 Studies applying selective treatment at $FiO_2 > 0.5$	1	148	Risk Ratio (M-H, Fixed, 95% CI)	2.46 [0.92, 6.56]
3.3.4 Studies applying selective treatment at $FiO_2 > 0.6$	1	147	Risk Ratio (M-H, Fixed, 95% CI)	0.38 [0.08, 1.92]

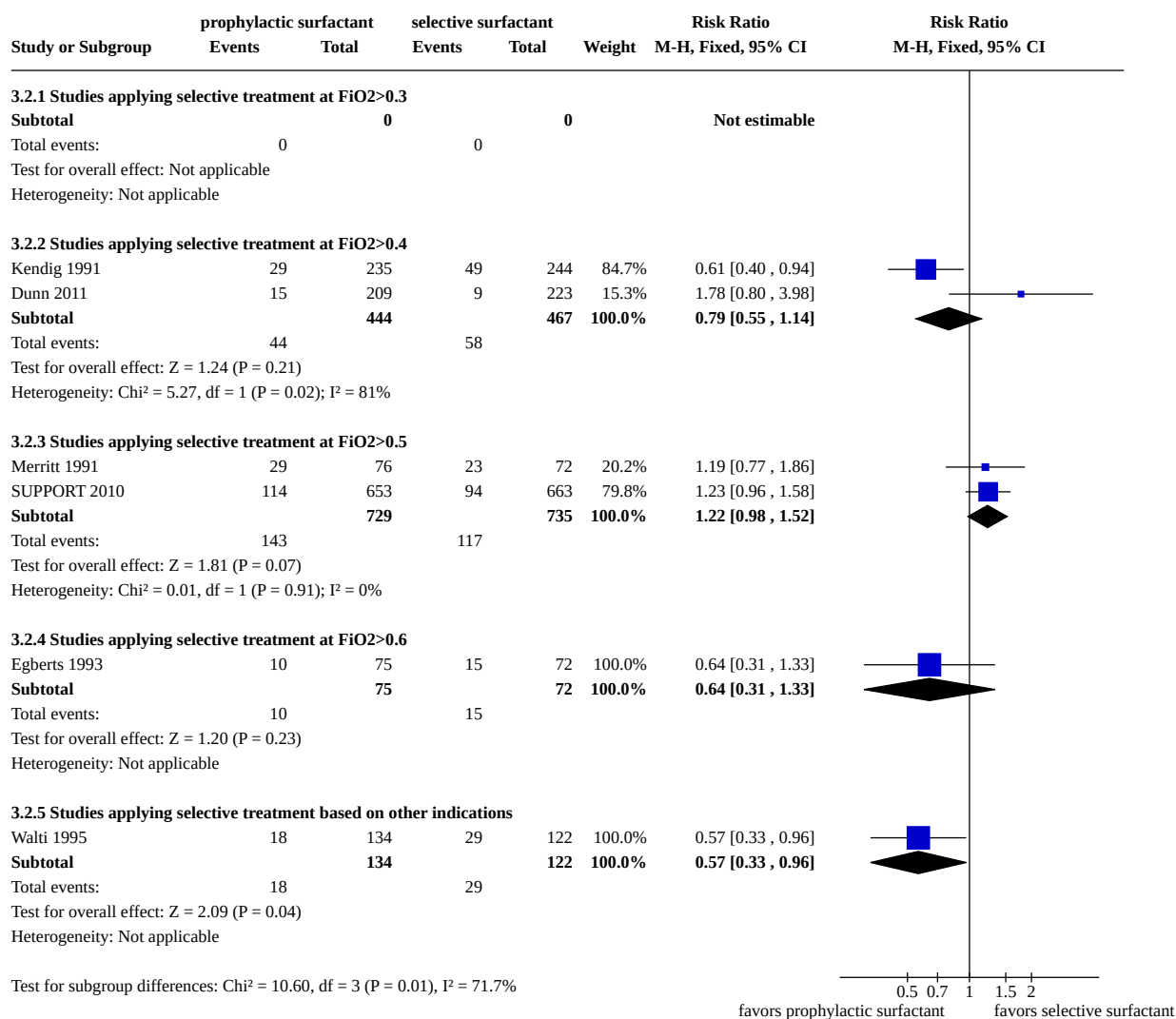
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
3.3.5 Studies applying selective treatment based on other indications	3	641	Risk Ratio (M-H, Fixed, 95% CI)	0.67 [0.36, 1.24]
3.4 Bronchopulmonary dysplasia (oxygen requirement at 28 to 30 days of age)	9		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
3.4.1 Studies applying selective treatment at FiO ₂ >0.3	1	1248	Risk Ratio (M-H, Fixed, 95% CI)	0.65 [0.41, 1.03]
3.4.2 Studies applying selective treatment at FiO ₂ >0.4	3	974	Risk Ratio (M-H, Fixed, 95% CI)	1.09 [0.95, 1.25]
3.4.3 Studies applying selective treatment at FiO ₂ >0.5	1	148	Risk Ratio (M-H, Fixed, 95% CI)	0.95 [0.38, 2.39]
3.4.4 Studies applying selective treatment at FiO ₂ >0.6	1	147	Risk Ratio (M-H, Fixed, 95% CI)	1.24 [0.72, 2.14]
3.4.5 Studies applying selective treatment based on other indications	3	646	Risk Ratio (M-H, Fixed, 95% CI)	1.00 [0.77, 1.30]
3.5 Bronchopulmonary dysplasia or death at 28 to 30 days of age	8		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
3.5.1 Studies applying selective treatment at FiO ₂ >0.3	1	1248	Risk Ratio (M-H, Fixed, 95% CI)	0.59 [0.38, 0.90]
3.5.2 Studies applying selective treatment at FiO ₂ >0.4	3	992	Risk Ratio (M-H, Fixed, 95% CI)	1.00 [0.89, 1.12]
3.5.3 Studies applying selective treatment at FiO ₂ >0.5	0	0	Risk Ratio (M-H, Fixed, 95% CI)	Not estimable
3.5.4 Studies applying selective treatment at FiO ₂ >0.6	1	147	Risk Ratio (M-H, Fixed, 95% CI)	0.93 [0.63, 1.36]
3.5.5 Studies applying selective treatment based on other indications	3	646	Risk Ratio (M-H, Fixed, 95% CI)	0.82 [0.69, 0.98]
3.6 Chronic lung disease or death (at 36 weeks' PMA)	4		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
3.6.1 Studies applying selective treatment at FiO ₂ >0.3	0	0	Risk Ratio (M-H, Fixed, 95% CI)	Not estimable
3.6.2 Studies applying selective treatment at FiO ₂ >0.4	1	428	Risk Ratio (M-H, Fixed, 95% CI)	1.20 [0.92, 1.57]
3.6.3 Studies applying selective treatment at FiO ₂ >0.5	1	1316	Risk Ratio (M-H, Fixed, 95% CI)	1.11 [1.00, 1.23]
3.6.4 Studies applying selective treatment at FiO ₂ >0.6	0	0	Risk Ratio (M-H, Fixed, 95% CI)	Not estimable

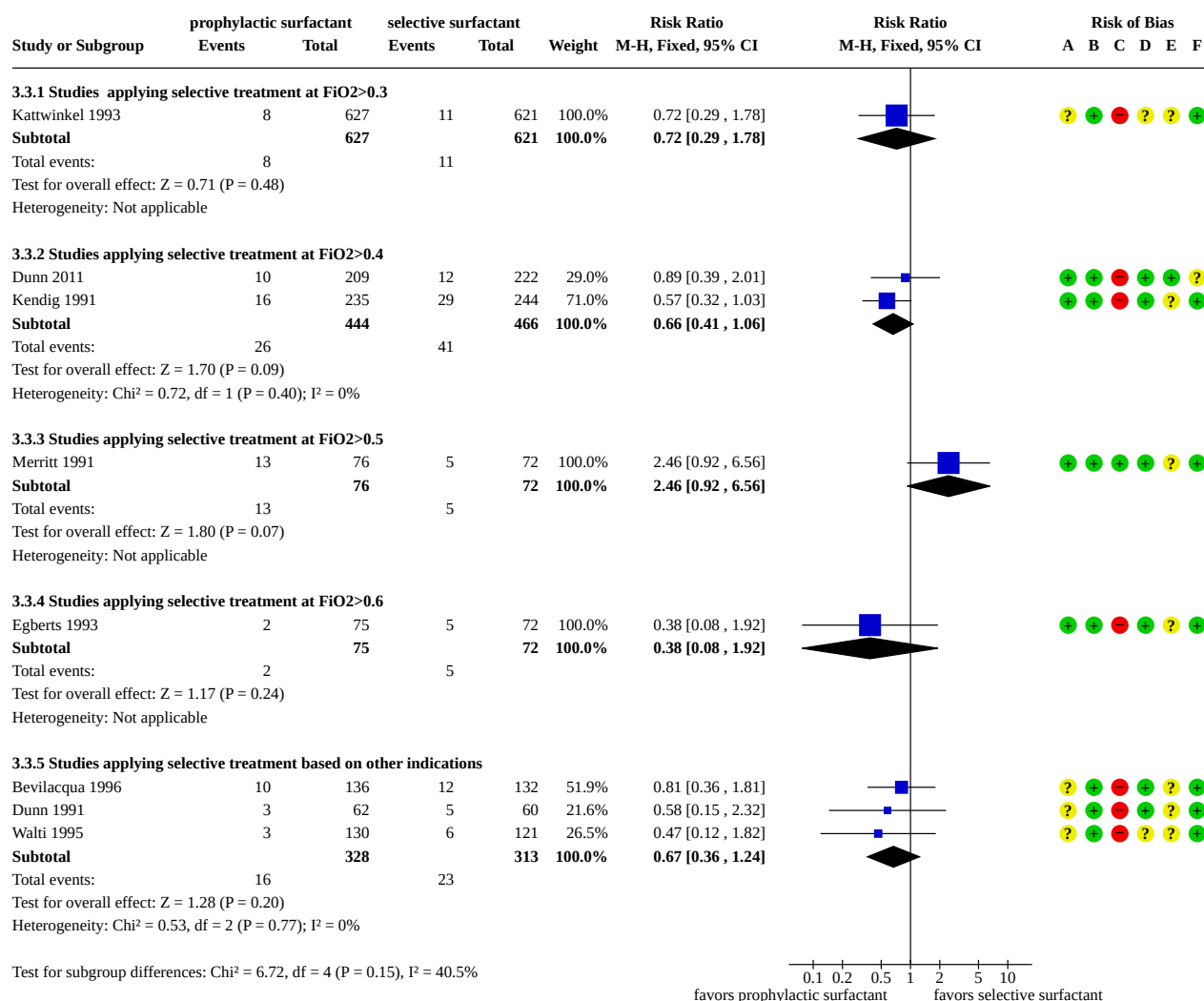
Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
3.6.5 Studies applying selective treatment based on other indications	2	378	Risk Ratio (M-H, Fixed, 95% CI)	0.87 [0.63, 1.21]

Analysis 3.1. Comparison 3: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed on the threshold used to apply selective treatment), Outcome 1: Chronic lung disease (supplemental oxygen at 36 weeks' PMA)



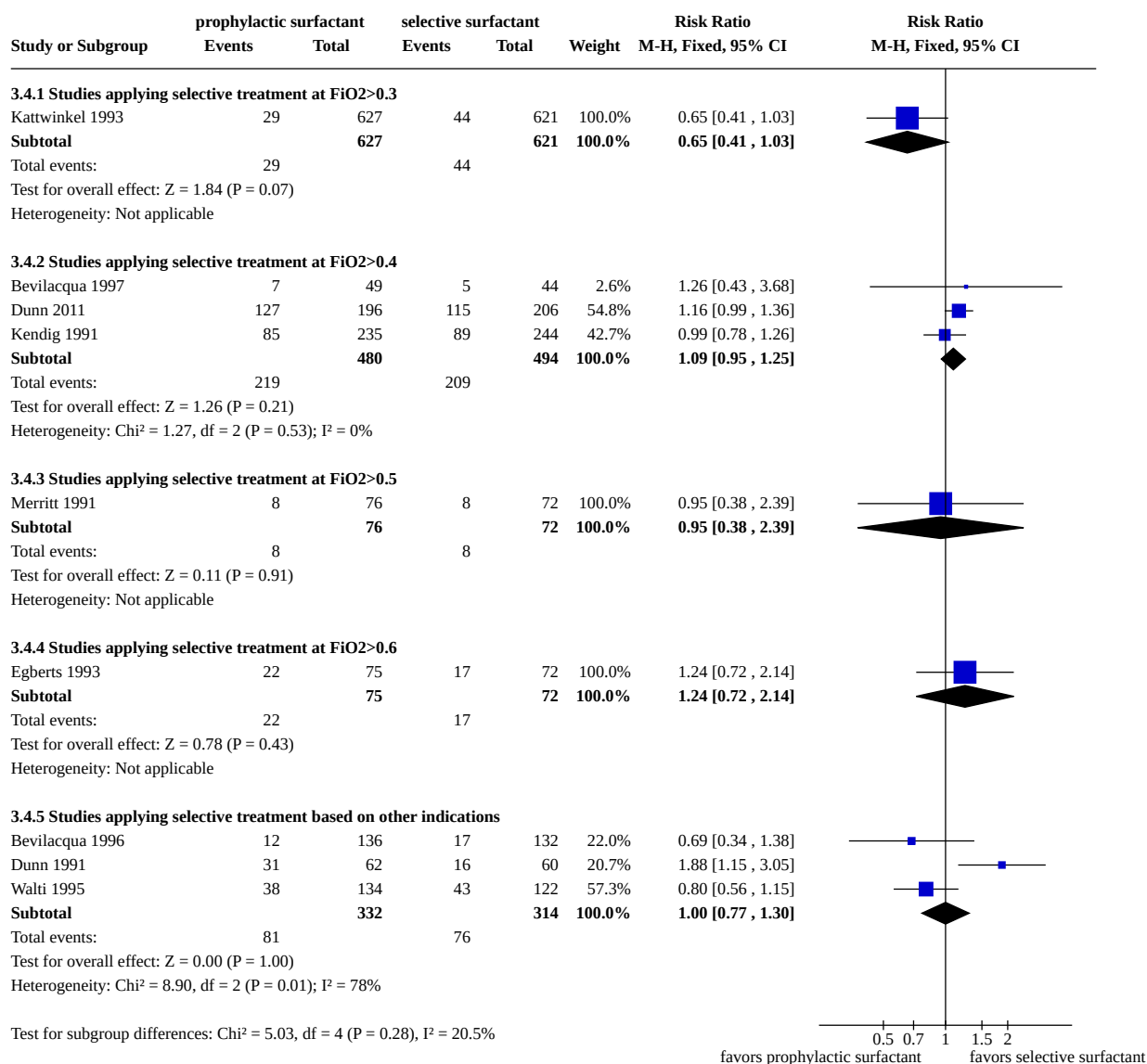
Analysis 3.2. Comparison 3: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed on the threshold used to apply selective treatment), Outcome 2: Mortality at 36 weeks' PMA or prior to hospital discharge



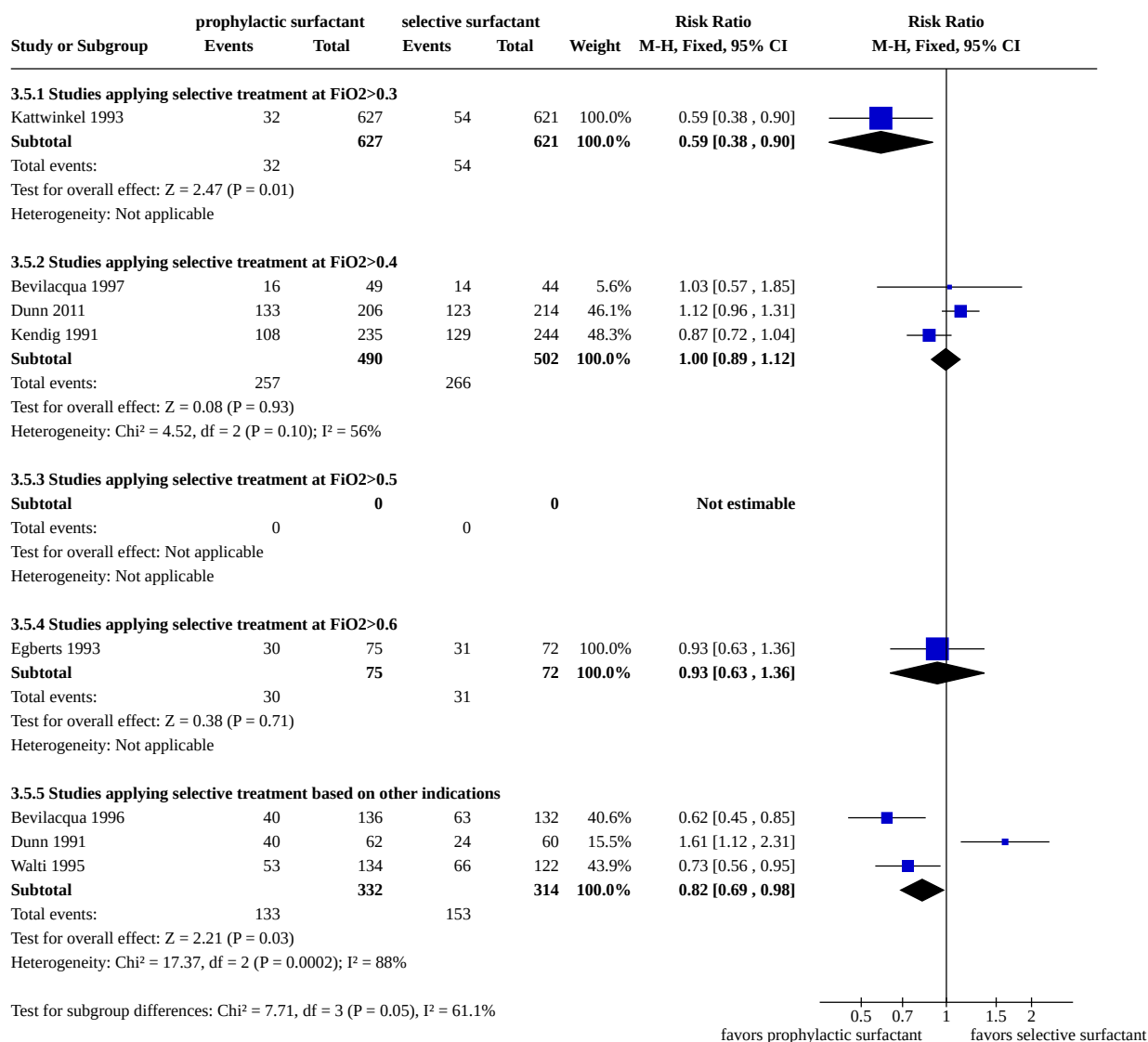
Analysis 3.3. Comparison 3: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed on the threshold used to apply selective treatment), Outcome 3: Pneumothorax**Risk of bias legend**

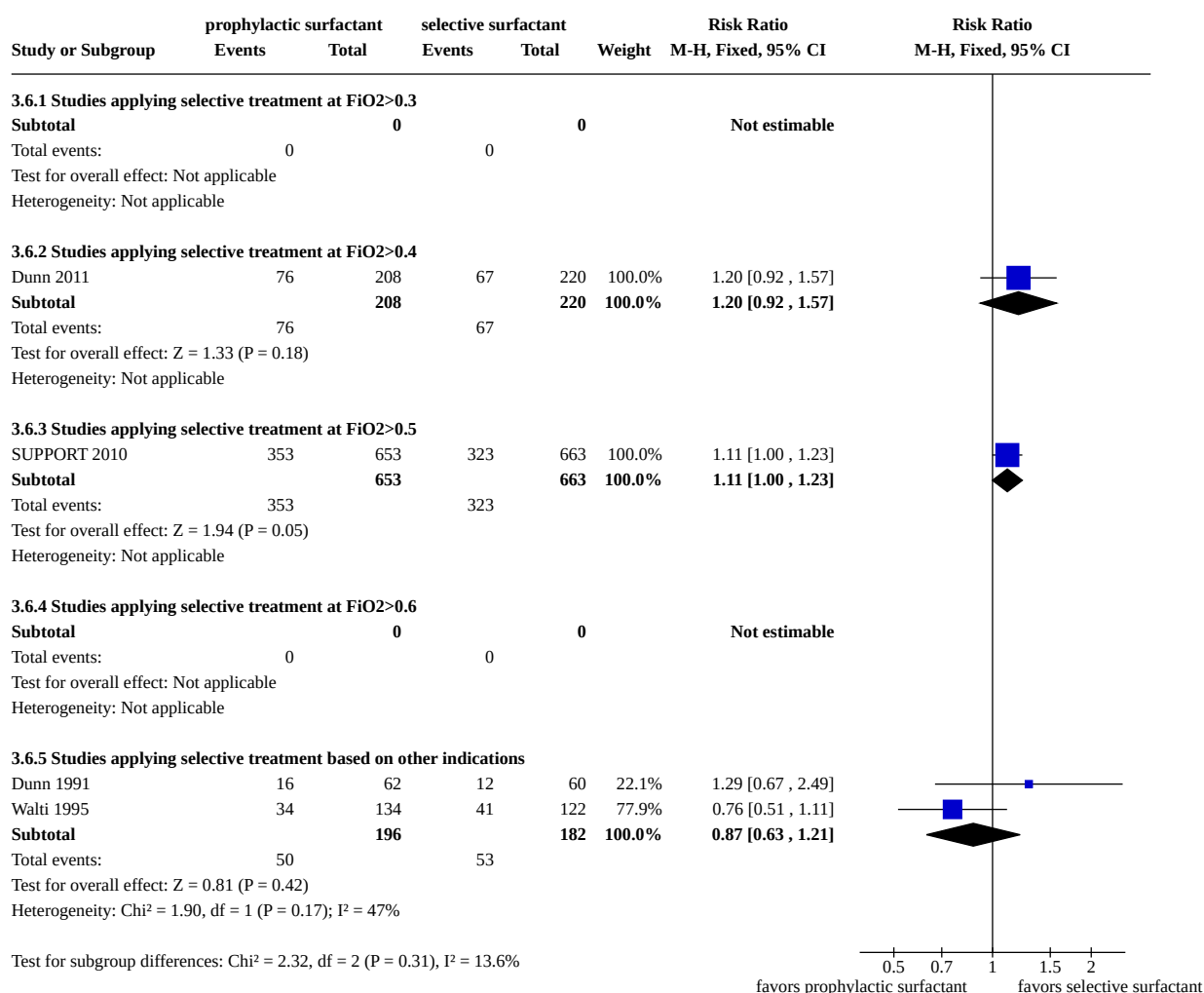
- (A) Random sequence generation (selection bias)
- (B) Allocation concealment (selection bias)
- (C) Blinding of participants and personnel (performance bias)
- (D) Incomplete outcome data (attrition bias)
- (E) Selective reporting (reporting bias)
- (F) Other bias

Analysis 3.4. Comparison 3: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed on the threshold used to apply selective treatment), Outcome 4: Bronchopulmonary dysplasia (oxygen requirement at 28 to 30 days of age)



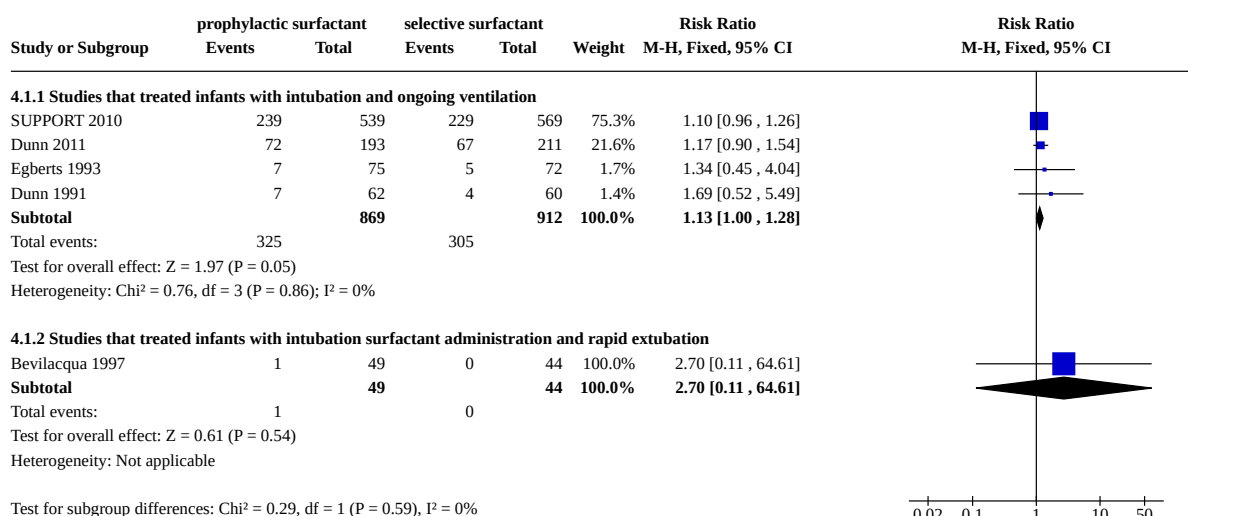
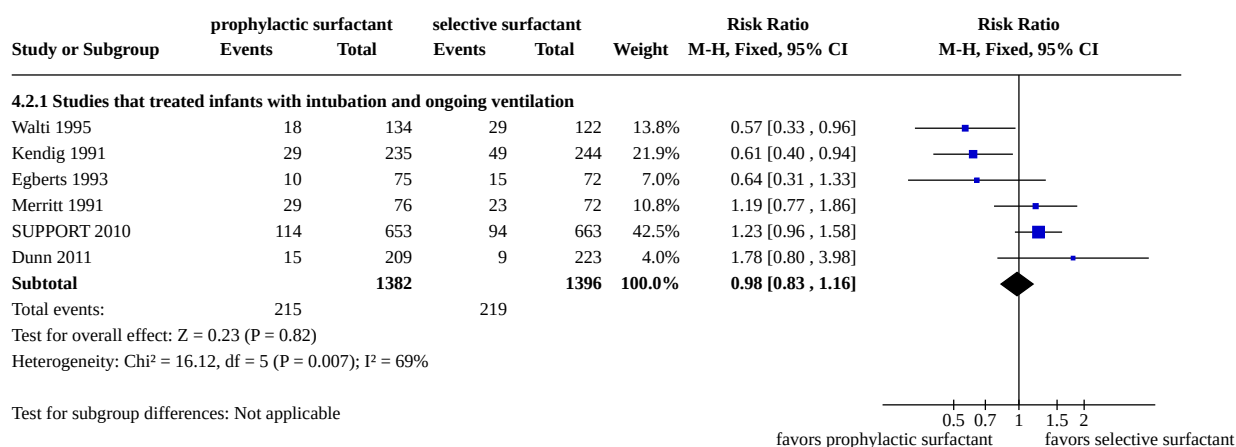
Analysis 3.5. Comparison 3: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed on the threshold used to apply selective treatment), Outcome 5: Bronchopulmonary dysplasia or death at 28 to 30 days of age

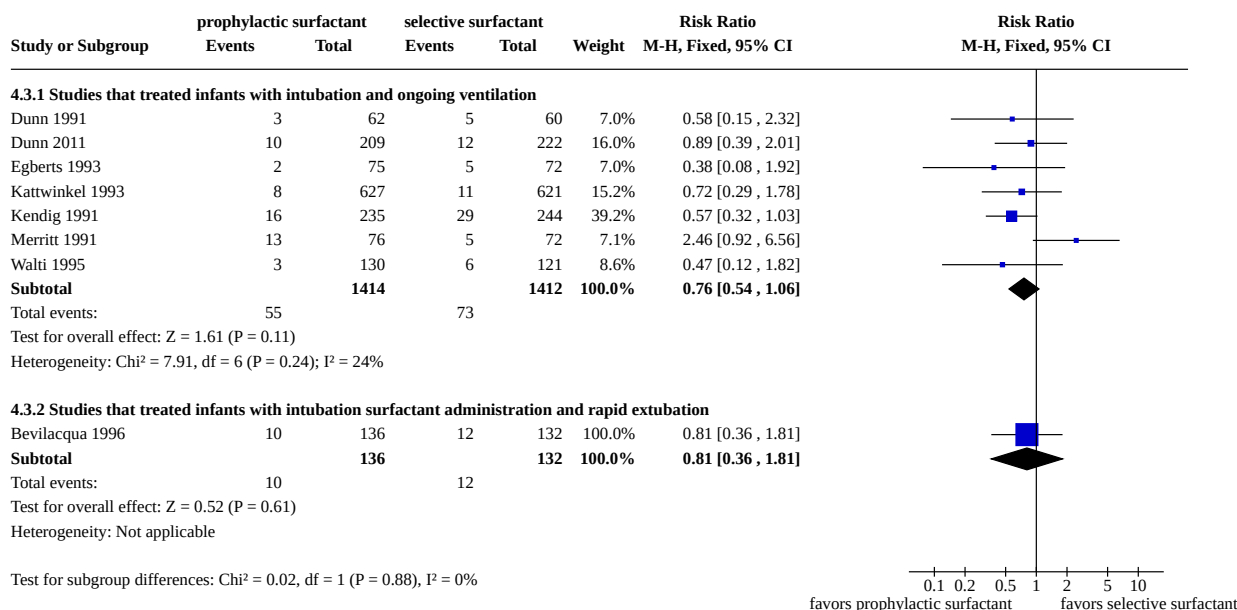
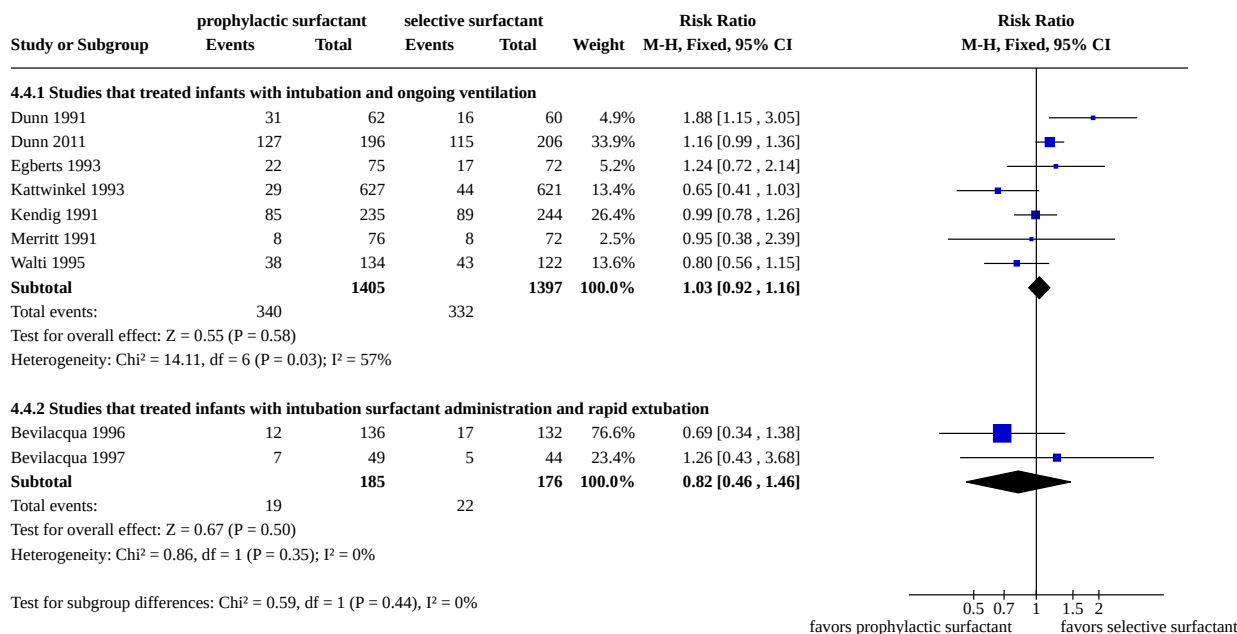


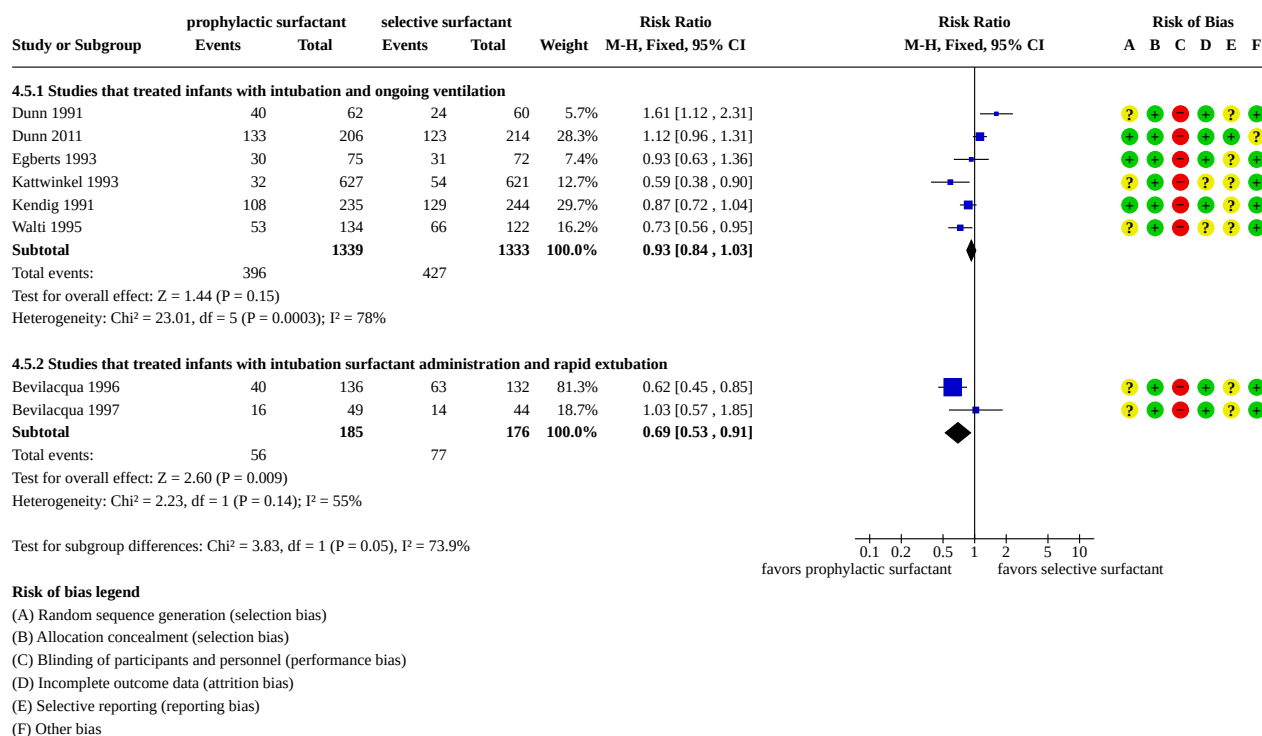
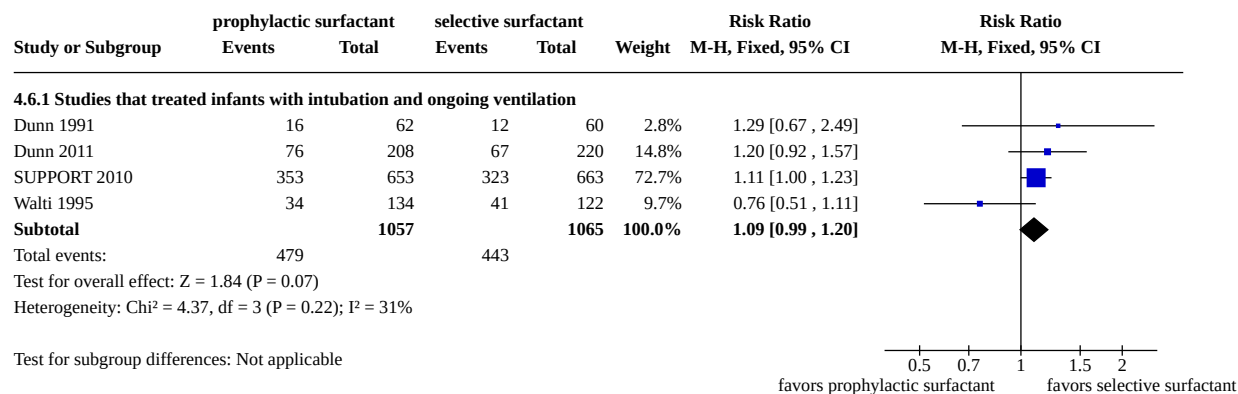
Analysis 3.6. Comparison 3: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed on the threshold used to apply selective treatment), Outcome 6: Chronic lung disease or death (at 36 weeks' PMA)**Comparison 4. Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by the method of surfactant administration)**

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
4.1 Chronic lung disease (supplemental oxygen at 36 weeks' PMA)	5		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
4.1.1 Studies that treated infants with intubation and ongoing ventilation	4	1781	Risk Ratio (M-H, Fixed, 95% CI)	1.13 [1.00, 1.28]
4.1.2 Studies that treated infants with intubation surfactant administration and rapid extubation	1	93	Risk Ratio (M-H, Fixed, 95% CI)	2.70 [0.11, 64.61]
4.2 Mortality at 36 weeks' PMA or prior to hospital discharge	6		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only

Outcome or subgroup title	No. of studies	No. of participants	Statistical method	Effect size
4.2.1 Studies that treated infants with intubation and ongoing ventilation	6	2778	Risk Ratio (M-H, Fixed, 95% CI)	0.98 [0.83, 1.16]
4.3 Pneumothorax	8		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
4.3.1 Studies that treated infants with intubation and ongoing ventilation	7	2826	Risk Ratio (M-H, Fixed, 95% CI)	0.76 [0.54, 1.06]
4.3.2 Studies that treated infants with intubation surfactant administration and rapid extubation	1	268	Risk Ratio (M-H, Fixed, 95% CI)	0.81 [0.36, 1.81]
4.4 Bronchopulmonary dysplasia (oxygen requirement at 28 to 30 days of age)	9		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
4.4.1 Studies that treated infants with intubation and ongoing ventilation	7	2802	Risk Ratio (M-H, Fixed, 95% CI)	1.03 [0.92, 1.16]
4.4.2 Studies that treated infants with intubation surfactant administration and rapid extubation	2	361	Risk Ratio (M-H, Fixed, 95% CI)	0.82 [0.46, 1.46]
4.5 Bronchopulmonary dysplasia or death at 28 to 30 days of age	8		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
4.5.1 Studies that treated infants with intubation and ongoing ventilation	6	2672	Risk Ratio (M-H, Fixed, 95% CI)	0.93 [0.84, 1.03]
4.5.2 Studies that treated infants with intubation surfactant administration and rapid extubation	2	361	Risk Ratio (M-H, Fixed, 95% CI)	0.69 [0.53, 0.91]
4.6 Chronic lung disease or death (at 36 weeks' PMA)	4		Risk Ratio (M-H, Fixed, 95% CI)	Subtotals only
4.6.1 Studies that treated infants with intubation and ongoing ventilation	4	2122	Risk Ratio (M-H, Fixed, 95% CI)	1.09 [0.99, 1.20]

Analysis 4.1. Comparison 4: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by the method of surfactant administration), Outcome 1: Chronic lung disease (supplemental oxygen at 36 weeks' PMA)**Analysis 4.2. Comparison 4: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by the method of surfactant administration), Outcome 2: Mortality at 36 weeks' PMA or prior to hospital discharge**

Analysis 4.3. Comparison 4: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by the method of surfactant administration), Outcome 3: Pneumothorax**Analysis 4.4. Comparison 4: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by the method of surfactant administration), Outcome 4: Bronchopulmonary dysplasia (oxygen requirement at 28 to 30 days of age)**

Analysis 4.5. Comparison 4: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by the method of surfactant administration), Outcome 5: Bronchopulmonary dysplasia or death at 28 to 30 days of age**Analysis 4.6. Comparison 4: Prophylactic surfactant administration compared to selective surfactant administration in preterm infants at risk of RDS (analyzed by the method of surfactant administration), Outcome 6: Chronic lung disease or death (at 36 weeks' PMA)****APPENDICES****Appendix 1. Definitions used****Gestational age**

Gestational age is defined as the duration of pregnancy measured in weeks and days from the first day of the mother's last menstrual period to the time of birth. According to the gestational age at birth, preterm infants are classified as follows:

- Late preterm infants – defined as infants 34⁰ to 37⁶ weeks gestational age
- Moderate preterm infants – defined as infants born at 32⁰ to 33⁶ weeks gestational age
- Very preterm infants – defined as infants born at 28⁰ to 31⁶ weeks gestational age
- Extremely preterm infants – defined as infants born at less than 28⁰ weeks gestational age

Birth weight

Given that studies in a variety of settings may not have reliable gestational age and may therefore use birth weight categories, we included infants categorized as:

- **Low birth weight (LBW) infants** – defined as infants with birth weight < 2500 g
- **Very low birth weight (VLBW) infants** – defined as infants with birth weight < 1500 g
- **Extremely low birth weight (ELBW) infants** – defined as infants with birth weight < 1000 g

Infants with birth weights 2000 to 2500 g were grouped with the late preterm infants. 1500 to 2000 g were grouped with the moderate preterm infants, infants 1000 to 1499 g were grouped with the very preterm infants, and infants less than 1000 g were grouped with the extremely preterm infants.

For those studies that reported on a broad population of infants (including both preterm infants < 34 weeks' gestation, late preterm infants, and term infants), we included studies that had a preponderance of preterm infants (defined as a mean gestational age of < 34 weeks or report greater than 80% of infants as preterm < 34 weeks gestational age).

Moderate-severe neurodevelopmental impairment (at 18 to 24 months and at 3 to 5 years)

Neurodevelopmental disability and each component of major neurodevelopmental disability is defined as cerebral palsy (CP) or gross motor disability (defined as $\geq$ level 2 according to the Gross Motor Function Classification System (GMFCS)) (Palisano 1997); developmental delay (Bayley or Griffith assessment > 2 SD below the mean) (Bayley 1993; Bayley 2006; Griffiths 1954); or intellectual impairment (IQ > 2 SD below the mean), blindness (vision < 6/60 in both eyes), or sensorineural deafness requiring amplification.

Appendix 2. Search strategies and history

Search narrative

We report the search strategy and results of the searches conducted for this update. As described in the review text, we updated the searches on two occasions. The most recent search covered the period from the date of the previous review update (2011) to 2025 to ensure that no relevant trials were missed.

We performed the searches without restrictions on publication type, publication status, or language.

Summary of search results

Date	Database	Date Limit	Results
January-31-2023	PubMed, Cochrane Information Network (CIN), Central, Embase, International Clinical Trials Registry Platform (ICTRP)	2011-2023	2745
February-02-2024	Medline	2010-2024	1590
February-02-2024	Embase	2010-2024	2059
February-02-2024	CINAHL	2010-2024	532
February-02-2024	CRS	2010-2024	692
January-31-2025	Medline	2024-2025	150
January-31-2025	Embase	2024-2025	92

(Continued)

January-31-2025	CINAHL	2024-2025	26
January-31-2025	CRS	2024-2025	21
January-31-2025	Trial registries	2010-2025	55
Gross			7962
Duplicates			5662
Net Screened			2300

2024 and 2025 searches

Ovid MEDLINE(R) All		
February-02-2024; January 31, 2025		
#	Searches	Results
1	exp Surface-Active Agents/	144129
2	exp Pulmonary Surfactants/	14297
3	(surfactant* or surface active agent* or Beractant or Poractant or Curosurf or Survanta or Exosurf or Lucinactant).ti,ab,kw,kf.	72877
4	or/1-3 [Surfactants]	196741
5	exp Infant, Newborn/ or Intensive Care, Neonatal/ or Intensive Care Units, Neonatal/ or Gestational Age/	732420
6	(babe or babes or baby* or babies or gestational age? or infant? or infantile or infancy or low birth weight or low birthweight or neonat* or neo-nat* or newborn* or new born? or newly born or premature or pre-mature or pre-matures or prematures or prematurity or pre-maturity or preterm or preterms or pre term? or premie or premies or premies or premie or VLBW or VLBWI or VLBW-I or VLBWs or LBW or LBWI or LBWs or ELBW or ELBWI or ELBWs or NICU or NICUs).ti,ab,kw,kf.	1069211
7	or/5-6 [Filter: Neonatal Population 04-2022-MEDLINE]	1381661
8	randomized controlled trial.pt.	607974
9	controlled clinical trial.pt.	95543
10	randomized.ti,ab.	689411
11	placebo.ti,ab.	252079
12	drug therapy.fs.	2663448
13	randomly.ti,ab.	427262

(Continued)

14	trial.ti,ab.	791219
15	groups.ti,ab.	2658885
16	or/8-15 [Cochrane HSSS-SM Filter; Box 6.4.a Cochrane Handbook]	5940181
17	(quasirandom* or quasi-random* or quasiexperiment* or quasi-experiment* or random*).ti,ab,kw,kf.	1509438
18	(control* adj2 (group? or trial? or study)).ti,ab,kw,kf.	1135598
19	or/17-18 [Additional terms to increase sensitivity]	2121488
20	exp animals/ not humans/	5193302
21	(or/16,19) not 20 [RCT Filter: Medline]	5710690
22	4 and 7 and 21	3473
23	22 and (201* or 202*).ed,yr. [Lines 1-23 represent the search as run, and number of results, in 2024]	1590
Update January 2025		
	23 and ("2024" or "2025").yr. [Above search rerun in 2025; these are unique references found]	150

Embase 1974 to 2024 February 01; Embase 1974 to 2025 January 30

February-02-2024; January 31, 2025

#	Searches	Results
1	exp lung surfactant/	14259
2	(surfactant* or surface active agent* or Beractant or Poractant or Curosurf or Survanta or Exosurf or Lucinactant).ti,ab,kw,kf.	88556
3	or/1-2 [Surfactant]	92318
4	newborn/ or prematurity/ or newborn intensive care/ or newborn care/ or gestational age/	811829
5	(babe or babes or baby* or babies or gestational age? or infant? or infantile or infancy or low birth weight or low birthweight or neonat* or neo-nat* or newborn* or new born? or newly born or premature or pre-mature or pre-matures or prematures or prematurity or pre-maturity or preterm or preterms or pre term? or premie or premies or premies or premie or VLBW or VLBWI or VLBW-I or VLBWs or LBW or LBWI or LBWs or ELBW or ELBWI or ELBWs or NICU or NICUs).ti,ab,kw,kf.	1282134
6	or/4-5 [Filter: Neonatal Population 03-2022-OVID EMBASE]	1557717

(Continued)

7	Randomized controlled trial/ or Controlled clinical study/	1000794
8	random\$.ti,ab,kw.	2035300
9	Randomization/	99100
10	placebo.ti,ab,kw.	372783
11	((double or single or doubly or singly) adj (blind or blinded or blind-ly)).ti,ab,kw.	278919
12	double blind procedure/	215597
13	(controlled adj7 (study or design or trial)).ti,ab,kw.	462648
14	parallel group\$1.ti,ab.	32991
15	(crossover or cross over).ti,ab.	126715
16	((assign\$ or match or matched or allocation) adj5 (alternate or group\$1 or intervention\$1 or patient\$1 or subject\$1 or participant\$1)).ti,ab.	425470
17	(open adj label).ti,ab.	112957
18	(quasirandom* or quasi-random* or random*).ti,ab,kw,kf.	2036539
19	(control* adj2 (group? or trial? or study)).ti,ab,kw,kf.	1583323
20	or/7-19 [Terms based on Cochrane Central strategy- How Central is Created]	3580057
21	(exp animals/ or exp invertebrate/ or animal experiment/ or animal model/ or animal tissue/ or animal cell/ or nonhuman/) and (human/ or normal human/ or human cell/)	26159008
22	exp animals/ or exp invertebrate/ or animal experiment/ or animal model/ or animal tissue/ or animal cell/ or nonhuman/	33508954
23	22 not 21 [Animal Exclusion- https://community-cochrane-org.ezproxy.uvm.edu/sites/default/files/uploads/inline-files/Embase%20animal%20filter.pdf]	7349946
24	20 not 23 [Filter: RCT-EMBASE]	3099156
25	3 and 6 and 24	2062
26	25 and (201* or 202*).yr,em. [Lines 1-25 represent the search as run, and number of results, in 2024]	2059
	Update January 2025	
	26 and ("2024" or "2025").yr. [Above search rerun in 2025; these are unique references found]	92

Cochrane CRS		
February-02-2024; January 31, 2025		
1	MESH DESCRIPTOR Infant, Newborn EXPLODE ALL AND CENTRAL:TARGET	22516
2	MESH DESCRIPTOR Intensive Care, Neonatal AND CENTRAL:TARGET	405
3	MESH DESCRIPTOR Intensive Care Units, Neonatal AND CENTRAL:TARGET	1190
4	MESH DESCRIPTOR Gestational Age AND CENTRAL:TARGET	5133
5	("babe" or "babes" or baby* or "babies" or "gestational age" or "gestational ages" or infant? or "infantile" or infancy or "low birth weight" OR "low birth weights" or "low birthweight" or "low birthweights" or neonat* or "neo-nat*" or newborn* or "new born?" or "newly born" or "premature" or "pre-mature" or "pre-matures" or prematures or prematurity or "pre-maturity" or "preterm" or "preterms" or "pre term?" or "preemie" or "preemies" or "premies" or "pre-mie" or "VLBW" or "VLBW" or "VLBW-I" or "VLBWs" or "LBW" or "LBWI" or "LBWs" or "ELBW" or "ELBWI" or "ELBWs" or "NICU" or "NICUs"):ti,ab,kw AND CENTRAL:TARGET	87425
6	#1 OR #2 OR #3 OR #4 OR #5	92042
7	MESH DESCRIPTOR Pulmonary Surfactants EXPLODE ALL AND CENTRAL:TARGET	750
8	MESH DESCRIPTOR Surface-Active Agents EXPLODE ALL AND CENTRAL:TARGET	5094
9	(surfactant* or Beractant OR Poractant OR Curosurf OR Survanta OR Exosurf OR Lucinactant):ti,ab,kw AND INSEGMENT	8763
10	#7 OR #8 OR #9	13919
11	#10 AND #6	1462
12	2010 TO 2024:YR AND CENTRAL:TARGET	1320872
13	#11 AND #12 [Lines 1-13 represent the search as run, and number of results, in 2024]	692
Update January 2025		
	#11 AND 2024 TO 2025:YR AND CENTRAL:TARGET [Above search rerun in 2025; these are unique references found]	21

CINAHL Complete		
February-02-2024; January 31, 2025		
#	Query	Results

(Continued)

1	(MH "Pulmonary Surfactants")	1,783
2	TI (surfactant* OR "surface active agent* OR Beractant OR Poractant OR Curosurf OR Survanta OR Exosurf OR Lucinactant) OR AB (surfactant* OR "surface active agent* OR Beractant OR Poractant OR Curosurf OR Survanta OR Exosurf OR Lucinactant)	3,333
3	(MH "Surface-Active Agents+")	5,597
4	S1 OR S2 OR S3	8,790
5	(MH "Infant, Newborn+") OR (MH "Infant, Large for Gestational Age") OR (MH "Infant, Low Birth Weight+") OR (MH "Infant, Postmature") OR (MH "Infant, Premature") OR (MH "Intensive Care, Neonatal+") OR (MH "Intensive Care Units, Neonatal") OR (MH "Gestational Age")	177,747
6	TI ((babe or babes or baby* or babies or gestational age# or infant# or infantile or infancy or low birth weight or low birthweight or neonat* or neo-nat* or newborn* or new born# or newly born or premature or pre-mature or pre-matures or prematures or prematurity or pre-maturity or preterm or preterms or pre term# or preemie or preemies or premies or premie or VLBW or VLBWI or VLBW-I or VLBWs or LBW or LBWI or LBWs or ELBW or ELBWI or ELBWs or NICU or NICUs) OR AB (babe or babes or baby* or babies or gestational age# or infant# or infantile or infancy or low birth weight or low birthweight or neonat* or neo-nat* or newborn* or new born# or newly born or premature or pre-mature or pre-matures or prematures or prematurity or pre-maturity or preterm or preterms or pre term# or preemie or preemies or premies or premie or VLBW or VLBWI or VLBW-I or VLBWs or LBW or LBWI or LBWs or ELBW or ELBWI or ELBWs or NICU or NICUs)	268,359
7	S5 OR S6	333,725
8	(MH "Single-Blind Studies") OR (MH "Triple-Blind Studies") OR (MH "Randomized Controlled Trials+") OR (MH "Double-Blind Studies")	183,069
9	(MH "Clinical Trials+")	356,932
10	TI (randomi#ed or randomly) OR AB (randomi#ed or randomly) OR AB (randomi#ed or randomly) OR AB (randomi#ed or randomly)	387,239
11	AB randomly	111,425
12	AB placebo	66,920
13	AB (trial)	344,324
14	AB groups	929,772
15	TI (quasirandom* or quasi-random*) OR AB (quasirandom* or quasi-random*)	2,266
16	S8 OR S9 OR S10 OR S11 OR S12 OR S13 OR S14 OR S15	1,356,400
17	(MH "Animal Studies")	157,087
18	(MH "Human")	2,784,025
19	S17 NOT S18	131,401

(Continued)

20	S16 NOT S19	1,314,367
21	S4 AND S7 AND S20	857
22	S4 AND S7 AND S20 [2010-2024] [Lines 1-22 represent the search as run, and number of results, in 2024]	532
Update January 2025		
S22 AND Limiters - Publication Date: 20240101-20251231 [Above search rerun in 2025; these are unique references found]		26

Trial registries

Date	Source	Terms	Results
Jan-31-2025	ICTRP	Intervention: Pulmonary Surfactant	5
Jan-31-2025	ICTRP	Intervention: surfactant or Beractant or salbutamol or Poractant or Curosurf or Survanta or Exosurf or Lucinactant AND Title: Infant or neonate or newborn	11
Jan-31-2025	ICTRP	Homepage search box: (intubation or intubated or bolus) AND (surfactant or Beractant or salbutamol or Poractant or Curosurf or Survanta or Exosurf or Lucinactant) AND (Infant or neonate or newborn)	37
Jan-31-2025	ctgov	prophylactic surfactant	2
Total			55

2023 searches

The searches documented below were run in January 2023. The PRISMA diagram in presubmission version 7.36 of this manuscript provided the following data:

2732 records database searching; 2 other sources; 11 previous version of review; 1951 records after duplicates removed.

The review search methods stated the following.

"For this update, we conducted a comprehensive search including Cochrane Central Register of Controlled Trials (CENTRAL, Issue 1 of 12, January 2023) in The Cochrane Library; MEDLINE via PubMed (2011 to Jan 2023); EMBASE (2011 to 2021); and CINAHL (2011 to Jan 2023). We used Cochrane Neonatal's search strategy for neonates and RCTs (see [Appendix 2](#) for full search strategies for each database). See [Appendix 3](#) for the search method used for the last published version of this review (Rojas 2011). We applied no language restrictions.

To avoid publication bias, We searched clinical trial databases, conference proceedings, and reference lists of retrieved articles for RCTs and quasi-RCTs.

To identify unpublished studies from the Society for Pediatric Research (USA), abstracts for 2000 to 2023 were searched electronically through the PAS website (abstracts online). For abstract books that do not include keywords, the search was limited to relevant sections such as pulmonary and neonatology.

We searched clinical trial registries for ongoing or recently completed trials (clinicaltrials.gov; the World Health Organization's International Trials Registry and Platform www.who.int/ictpr/search/en/, and the [ISRCTN Registry](http://www.isrctn.com))."

PubMed

31/01/2023

- #1 "Infant, Newborn"[Mesh] 665,752
- #2 babies[tiab] 41,414
- #3 baby*[tiab] 47,444
- #4 infan*[tiab] 542,009
- #5 LBW[tiab] 4,891
- #6 VLBW[tiab] 4,297
- #7 low birth weight[tiab] 31,682
- #8 low birthweight[tiab] 8,446
- #9 neonat*[tiab] 307,647
- #10 new born[tiab] 4,388
- #11 new born*[tiab] 4,943
- #12 newly born[tiab] 1,236
- #13 newborn*[tiab] 197,551
- #14 pre term[tiab] 2,933
- #15 preterm[tiab] 92,29
- #16 premature[tiab] 138,908
- #17 prematurity[tiab] 26,305
- #18 #1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 1,293,204
- #19 "Surface-Active Agents"[MeSH Terms] 54,289
- #20 "Pulmonary Surfactants"[MeSH Terms] 11,347
- #21 surfactant*[tiab] 68,09
- #22 surface active agent*[tiab] 1,859
- #23 Beractant[tiab] 114
- #24 Poractant[tiab] 150
- #25 Curosurf[tiab] 364
- #26 Survanta[tiab] 289
- #27 Exosurf[tiab] 216
- #28 Lucinactant[tiab] 37
- #29 "pulmonary surfactants"[Pharmacological Action] 14,099
- #30 #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 107,086
- #31 #18 AND #30 8,313
- #32 (randomized controlled trial[pt] OR controlled clinical trial[pt] OR randomized[tiab] OR placebo[tiab] OR drug therapy[sh] OR randomly[tiab] OR trial[tiab] OR groups[tiab]) NOT (animals [mh] NOT humans [mh]) 4,927,075

#33 #31 AND #32 2,928

#34 #31 AND #32 AND (2021/01/01:3000/12/12[pdat]) 286

EMBASE

embase.com

31/01/2023

#1 "infant'/exp" 796726

#2 "newborn'/exp" 410545

#3 "(infant*:ti,ab OR newborn*:ti,ab OR neonat*:ti,ab OR baby*:ti,ab OR babies:ti,ab OR prematur*:ti,ab OR preterm:ti,ab OR preterms:ti,ab OR premies:ti,ab OR vlbw:ti,ab OR lbw:ti,ab OR elbw:ti,ab OR nicu:ti,ab)" 966171

#4 "((new NEXT/2 born):ti,ab)" 4468

#5 "((new NEXT/2 borns):ti,ab)" 1049

#6 "((newly NEXT/2 born):ti,ab)" 1287

#7 "((pre NEXT/1 term):ti,ab)" 4288

#8 "((low NEXT/2 birth NEXT/2 weight):ti,ab)" 33892

#9 "((low NEXT/2 birthweight):ti,ab)" 8919

#10 "#1 OR #2 OR #3 OR #4 OR #5 OR #6 OR #7 OR #8 OR #9" 1284840

#11 "lung surfactant'/exp" 11598

#12 "surfactant*:ti,ab" 64894

#13 "((surface NEXT/2 active NEXT/2 agent*):ti,ab)" 782

#14 "(beractant:ti,ab OR poractant:ti,ab OR curosurf:ti,ab OR survanta:ti,ab OR exosurf:ti,ab OR lucinactant:ti,ab)" 1062

#15 "#11 OR #12 OR #13 OR #14" 68496

#16 "#10 AND #15" 9919

#17 "randomized controlled trial'/de" 611392

#18 "controlled clinical trial'/de" 357609

#19 "random*:ti,ab,tt" 1574222

#20 "randomization'/de" 62216

#21 "intermethod comparison'/de" 290854

#22 "placebo:ti,ab,tt" 326639

#23 "(compare:ti,tt OR compared:ti,tt OR comparison:ti,tt)" 491746

#24 "(evaluated:ab OR evaluate:ab OR evaluating:ab OR assessed:ab OR assess:ab) AND (compare:ab OR compared:ab OR comparing:ab OR comparison:ab)" 2298278

#25 "((open NEXT/1 label):ti,ab,tt)" 99575

#26 "(((double OR single OR doubly OR singly) NEXT/1 (blind OR blinded OR blindly)):ti,ab,tt)" 245015

#27 "double blind procedure'/de" 183164

#28 "((parallel NEXT/1 group*):ti,ab,tt)" 27879

#29 "(crossover:ti,ab,tt OR 'cross over':ti,ab,tt)" 102805

Prophylactic versus selective use of surfactant for preventing morbidity and mortality in preterm infants at risk of respiratory distress syndrome (Review)

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#30 “(((assign* OR match OR matched OR allocation) NEAR/6 (alternate OR group OR groups OR intervention OR interventions OR patient OR patients OR subject OR subjects OR participant OR participants)):ti,ab,tt)” 382731

#31 “(assigned:ti,ab,tt OR allocated:ti,ab,tt)” 374628

#32 “((controlled NEAR/8 (study OR design OR trial)):ti,ab,tt)” 379702

#33 “(volunteer:ti,ab,tt OR volunteers:ti,ab,tt)” 240906

#34 “‘human experiment’/de” 575363

#35 “trial:ti,tt” 333514

#36 “#17 OR #18 OR #19 OR #20 OR #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35” 5200228

#37 “((random* NEXT/1 sampl* NEAR/8 ('cross section*' OR questionnaire* OR survey OR surveys OR database OR databases)):ti,ab,tt) NOT ('comparative study'/de OR 'controlled study'/de OR 'randomised controlled':ti,ab,tt OR 'randomized controlled':ti,ab,tt OR 'randomly assigned':ti,ab,tt)” 2068

#38 “‘cross-sectional study’ NOT ('randomized controlled trial'/de OR 'controlled clinical study'/de OR 'controlled study'/de OR 'randomised controlled':ti,ab,tt OR 'randomized controlled':ti,ab,tt OR 'control group':ti,ab,tt OR 'control groups':ti,ab,tt)” 215485

#39 “‘case control’:ti,ab,tt AND random*:ti,ab,tt NOT ('randomised controlled':ti,ab,tt OR 'randomized controlled':ti,ab,tt)” 18511

#40 “‘systematic review’:ti,tt NOT (trial:ti,tt OR study:ti,tt)” 192783

#41 “nonrandom*:ti,ab,tt NOT random*:ti,ab,tt” 15658

#42 “‘random field’:ti,ab,tt” 1644

#43 “((‘random cluster’ NEAR/4 sampl*):ti,ab,tt)” 1204

#44 “review:ab AND review:it NOT trial:ti,tt” 861771

#45 “‘we searched’:ab AND (review:ti,tt OR review:it)” 36753

#46 “‘update review’:ab” 103

#47 “((databases NEAR/5 searched):ab)” 48882

#48 “(rat:ti,tt OR rats:ti,tt OR mouse:ti,tt OR mice:ti,tt OR swine:ti,tt OR porcine:ti,tt OR murine:ti,tt OR sheep:ti,tt OR lambs:ti,tt OR pigs:ti,tt OR piglets:ti,tt OR rabbit:ti,tt OR rabbits:ti,tt OR cat:ti,tt OR cats:ti,tt OR dog:ti,tt OR dogs:ti,tt OR cattle:ti,tt OR bovine:ti,tt OR monkey:ti,tt OR monkeys:ti,tt OR trout:ti,tt OR marmoset*:ti,tt) AND ‘animal experiment’/de” 1182076

#49 “‘animal experiment’/de NOT ('human experiment’/de OR 'human’/de)” 2477742

#50 “#37 OR #38 OR #39 OR #40 OR #41 OR #42 OR #43 OR #44 OR #45 OR #46 OR #47 OR #48 OR #49” 3780296

#51 “#36 NOT #50” 4567292

#52 “#16 AND #51” 2095

#53 “#16 AND #51 AND [2021-2023]/py” 234

CINAHL Complete

EBSCOhost Research Databases

31/01/2023

S1 T1 babies OR baby OR “newly born” OR “new born” OR newborn* OR neonat* OR “pre term” OR preterm OR prematurity OR premature OR “low birth weight” OR “low birthweight” OR VLBW OR LBW or infan* 145,977

S2 AB babies OR baby OR “newly born” OR “new born” OR newborn* OR neonat* OR “pre term” OR preterm OR prematurity OR premature OR “low birth weight” OR “low birthweight” OR VLBW OR LBW or infan* 195,387

S3 S1 OR S2 259,211

Prophylactic versus selective use of surfactant for preventing morbidity and mortality in preterm infants at risk of respiratory distress syndrome (Review)

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S4 (MH "Infant, Newborn+") 158,886

S5 S3 OR S4 318,7

S6 TI surfactant* OR "surface active agent*" OR Beractant OR Poractant OR Curosurf OR Survanta OR Exosurf OR Lucinactant 1,46

S7 AB surfactant* OR "surface active agent*" OR Beractant OR Poractant OR Curosurf OR Survanta OR Exosurf OR Lucinactant 2,74

S8 (MH "Pulmonary Surfactants") OR (MH "Surface-Active Agents+") 7,21

S9 S6 OR S7 OR S8 8,703

S10 (S6 OR S7 OR S8) AND (S5 AND S9) 2,205

S11 PT clinical trial 114,546

S12 TI randomized controlled trial OR controlled clinical trial OR randomized OR placebo OR randomly OR trial 201,042

S13 AB randomized controlled trial OR controlled clinical trial OR randomized OR placebo OR randomly OR trial 485,113

S14 S11 OR S12 OR S13 608,691

S15 S10 AND S14 555

S16 S10 AND S14 publication date 20210101-20221231 58

Cochrane Central Register of Controlled Trials

Issue 1 of 12, January 2023

The Cochrane Library

31/01/2023

#1 MeSH descriptor: [Pulmonary Surfactants] explode all trees 625

#2 surfactant*:ti,ab 2213

#3 (Beractant OR Poractant OR Curosurf OR Survanta OR Exosurf OR Lucinactant):ti,ab 424

#4 #1 OR #2 OR #3 2314

#5 MeSH descriptor: [Infant, Newborn] explode all trees 20174

#6 (infant OR infants OR infant's OR "infant s" OR infantile OR infancy OR newborn* OR "new born" OR "new borns" OR "newly born" OR neonat* OR baby* OR babies OR premature OR prematures OR prematurity OR preterm OR preterms OR "pre term" OR premies OR "low birth weight" OR "low birthweight" OR VLBW OR LBW OR ELBW OR NICU):ti,ab 76278

#7 #5 OR #6 80078

#8 #4 AND #7 1560

#9 #4 AND #7 with Publication Year from 2021 to 2023, in Trials 139

ISRCTN registry

<https://www.isrctn.com/>

31/01/2023

Interventions: Surfactant* AND Participant age range: Neonate 8

Appendix 3. Risk of bias assessment

We will use the standard methods of Cochrane and Cochrane Neonatal to assess the risk of bias in the trials. For each trial, we will seek information regarding the method of randomization, blinding, and reporting of all outcomes of all the infants enrolled in the trial. We will assess each criterion as being at a low, high, or unclear risk of bias. Two review authors will separately assess each study. Any disagreements

will be resolved by discussion. We will add this information to the 'Characteristics of included studies' table. We will evaluate the following issues and enter the findings into the 'Risk of bias' table.

1. Sequence generation (checking for possible selection bias). Was the allocation sequence adequately generated?

For each included study, we categorized the method used to generate the allocation sequence as:

- low risk (any truly random process, e.g. random number table; computer random number generator);
- high risk (any non-random process, e.g. odd or even date of birth; hospital or clinic record number); or
- unclear risk.

2. Allocation concealment (checking for possible selection bias). Was allocation adequately concealed?

For each included study, we categorized the method used to conceal the allocation sequence as:

- low risk (e.g. telephone or central randomization; consecutively numbered, sealed, opaque envelopes);
- high risk (open random allocation; unsealed or non-opaque envelopes, alternation; date of birth); or
- unclear risk

3. Blinding of participants and personnel (checking for possible performance bias). Was knowledge of the allocated intervention adequately prevented during the study?

For each included study, we categorized the methods used to blind study participants and personnel from knowledge of which intervention a participant received. We assessed blinding separately for different outcomes or class of outcomes. We categorized the methods as:

- low risk, high risk, or unclear risk for participants; and
- low risk, high risk, or unclear risk for personnel.

4. Blinding of outcome assessment (checking for possible detection bias). Was knowledge of the allocated intervention adequately prevented at the time of outcome assessment?

For each included study, we categorized the methods used to blind outcome assessment. We assessed blinding separately for different outcomes or class of outcomes. We categorized the methods as:

- low risk for outcome assessors;
- high risk for outcome assessors; or
- unclear risk for outcome assessors.

5. Incomplete outcome data (checking for possible attrition bias through withdrawals, dropouts, protocol deviations). Were incomplete outcome data adequately addressed?

For each included study and for each outcome, we described the completeness of data, including attrition and exclusions from the analysis. We noted whether attrition and exclusions were reported, the numbers included in the analysis at each stage (compared with the total randomized participants), reasons for attrition or exclusion where reported, and whether missing data were balanced across groups or were related to outcomes. Where sufficient information is reported or supplied by the trial authors, we re-included missing data in the analyses. We categorized the methods as:

- low risk (< 20% missing data);
- high risk ($\geq$ 20% missing data); or
- unclear risk.

6. Selective reporting bias. Are reports of the study free of the suggestion of selective outcome reporting?

For each included study, we described how we investigated the possibility of selective outcome reporting bias and what we found. For studies in which study protocols were published in advance, we compared prespecified outcomes versus outcomes eventually reported in the published results. If the study protocol was not published in advance, we contacted study authors to gain access to the study protocol. We assessed the methods as:

- low risk (where it is clear that all of the study's prespecified outcomes and all expected outcomes of interest to the review have been reported);
- high risk (where not all the study's prespecified outcomes have been reported; one or more reported primary outcomes were not prespecified outcomes of interest and are reported incompletely and so cannot be used; the study fails to include results of a key outcome that would have been expected to have been reported); or
- unclear risk.

7. Other sources of bias. Was the study apparently free of other problems that could put it at high risk of bias?

For each included study, we described any important concerns we had about other possible sources of bias (e.g. whether there was a potential source of bias related to the specific study design or whether the trial was stopped early due to some data-dependent process). We assessed whether each study was free of other problems that could put it at risk of bias as:

- low risk;
- high risk;
- unclear risk.

If required, we planned to explore the impact of the level of bias by undertaking sensitivity analyses.

Appendix 4. Additional outcomes

1.1 Cerebral palsy

One study reported on cerebral palsy in infants enrolled in [SUPPORT 2010](#) at 18 to 22 months of corrected age ([Vaucher 2012](#)). Infants in this study were routinely placed on CPAP. Cerebral palsy was observed in 19 (4.0%) of the 479 toddlers of the prophylactic group and in 21 (4.1%) of the 511 toddlers of the selective group. This suggests that prophylactic surfactant administration does not reduce the risk of cerebral palsy compared with selective surfactant administration (RR 0.93, 95% CI 0.51 to 1.72; test for heterogeneity not applicable). The certainty of evidence was low, downgraded for risk of bias and important imprecision on the effect estimate. No subgroup analysis could be performed.

1.2 Significant mental developmental delay

One study reported on significant mental developmental delay in infants enrolled in [SUPPORT 2010](#) at 18 to 22 months of corrected age ([Vaucher 2012](#)). Infants in this study were routinely placed on CPAP. Significant mental developmental delay was observed in 36 (7.6%) of the 472 toddlers of the prophylactic group and in 36 (7.2%) of the 502 toddlers of the selective group. This suggests that prophylactic surfactant administration does not reduce the risk of significant mental developmental delay compared with selective surfactant administration (RR 0.95, 95% CI 0.61 to 1.50; test for heterogeneity not applicable). The certainty of evidence was low, downgraded for risk of bias and important imprecision on the effect estimate. No subgroup analysis could be performed.

1.3 Legal blindness

One study reported on legal blindness in infants enrolled in [SUPPORT 2010](#) at 18 to 22 months of corrected age ([Vaucher 2012](#)). Infants in this study were routinely placed on CPAP. Legal blindness was observed in 7 (1.5%) of the 479 toddlers of the prophylactic group and in 4 (0.8%) of the 511 toddlers of the selective group. This suggests that prophylactic surfactant administration does not reduce the risk of legal blindness compared with selective surfactant administration (RR 0.53, 95% CI 0.16 to 1.78; test for heterogeneity not applicable). The certainty of evidence was low, downgraded for risk of bias and important imprecision on the effect estimate. No subgroup analysis could be performed.

1.4 Hearing deficit

One study reported on hearing deficit in infants enrolled in [SUPPORT 2010](#) at 18 to 22 months of corrected age ([Vaucher 2012](#)). Infants in this study were routinely placed on CPAP. Hearing deficit was observed in 7 (1.5%) of the 479 toddlers of the prophylactic group and in 17 (3.3%) of the 511 toddlers of the selective group. This suggests that prophylactic surfactant administration results in a slight increased risk of hearing deficit compared with selective surfactant administration (RR 2.27, 95% CI 0.96 to 5.37; test for heterogeneity not applicable). The certainty of evidence was low, downgraded for risk of bias and important imprecision on the effect estimate. No subgroup analysis could be performed.

WHAT'S NEW

Date	Event	Description
22 July 2026	New search has been performed	We ran the searches most recently on 31 January 2025. We found no new studies. We identified one follow-up report of an already included study.
22 July 2026	New citation required but conclusions have not changed	The previous version of the review included 11 studies (Ro-jas-Reyes 2012). We excluded one previously included study (Iarŭkova 1999). Iarŭkova 1999 is a retrospective study comparing the use of prophylactic curosurf (200 mg/kg) with rescue curosurf treatment (200 mg/kg) if indicated. We excluded it because it is not a randomized controlled trial.

Date	Event	Description
		We made the following changes in this update: 1. we changed the contact author and added new authors; 2. a variety of methods of surfactant administration are now eligible for inclusion; 3. we changed the inclusion criteria so that the surfactant could be administered via endotracheal tube, intubation, surfactant administration and rapid extubation (InSuRE) thin catheter administration, laryngeal mask airway, and hypopharyngeal deposition, and we excluded trials of aerosolized or nebulized surfactant; 4. we added new subgroup analyses exploring the key factors that may affect the effect of interventions; 5. we created a summary of findings table and assessed the certainty of the evidence using GRADEpro GDT; and 6. we removed a study that had mistakenly been included in the previous version as it was not an RCT but was a retrospective study comparing the use of prophylactic curosurf versus rescue curosurf treatment.

HISTORY

Protocol first published: Issue 4, 1997

Review first published: Issue 4, 1997

Date	Event	Description
14 August 2012	Amended	Edit to institution name in 'Acknowledgements' section.
13 March 2012	Amended	Minor text copy edits.
20 December 2011	New search has been performed	This updates the review "Prophylactic versus selective use of surfactant in preventing morbidity and mortality in preterm infants" published in the Cochrane Database of Systematic Reviews (Soll 2001). Search updated in December 14, 2011
20 December 2011	New citation required and conclusions have changed	New co-author - Rojas-Reyes MX New studies and subgroup analyses added to address routine stabilization of infants on positive airway pressure. Change in conclusions: Although the early trials of prophylactic surfactant administration to infants judged to be at risk of developing RDS compared to selective use of surfactant in infants with established RDS demonstrated a decreased risk of air leak and mortality, recent large trials that reflect current practice (including greater utilization of maternal steroids and routine post delivery stabilization on CPAP) do not support these differences and demonstrate less risk of chronic lung disease or death when using early stabilization on CPAP with selective surfactant administration to infants requiring intubation.
26 February 2010	Amended	Converted to new review format.

Date	Event	Description
7 February 2001	New citation required but conclusions have not changed	Substantive amendment
7 February 2001	New search has been performed	Systematic search for new trials through December, 2000. Additional study of Bevilacqua (1997) found and added to the review.

CONTRIBUTIONS OF AUTHORS

Professor MX Rojas and Dr R Soll conceived the idea for the review and developed the protocol for its update. They made significant contributions to the data analysis, edited the presentation of results, and drew conclusions. Professor MX Rojas led the conduct the systematic review in all stages.

Dr R Beijers and Sergio Alonso-Fernández screened search results, selected trials, extracted data, conducted meta-analyses, and drafted the text of the review.

Professor MX Rojas and Dr R Beijers performed the GRADE certainty assessment of the evidence.

All authors checked the final version of the review prior to its submission.

We thank Dr CJ Morley for his contribution to the previous versions of the review ([Rojas-Reyes 2012](#); [Soll 2001](#)). Dr Morley selected trials, extracted data, and reviewed the text for the 2002 version.

DECLARATIONS OF INTEREST

Rosanne JHCG Beijers has no interests to declare.

Sergio Alonso-Fernández has no interests to declare.

Roger Soll is Co-ordinating Editor for Cochrane Neonatal; however, he did not participate in the acceptance of this review or its editorial processing. Dr Soll was co-principal investigator of a study included in the review ([Dunn 2011](#)). The study was funded by the Vermont Oxford Network, a not-for-profit collaboration of NICUs worldwide. Two independent review authors (either RB and SA, or MXR and RB) undertook the review tasks that involved this study: screening and selection; data extraction; risk of bias assessment; and assessment of the certainty of evidence.

Maria X Rojas-Reyes has no interests to declare.

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Internal sources

- Institut d'Recerca Hospital de la Santa Creu i Sant Pau (IR Sat Pau), Spain

Protected the time dedicated by the affiliated author Dr MX Rojas

- Evaluation Unit, Canary Island Health Service (SESCS), Spain

Protected the time dedicated by the affiliated author Dr MX Rojas

- European Respiratory Society, Spain

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External sources

- Vermont Oxford Network, USA

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DIFFERENCES BETWEEN PROTOCOL AND REVIEW

The authorship team has changed in this 2026 update of the review.

In updating the review, we made the following changes to the protocol methods. We have indicated whether the changes were made in the 2012 version of the review ([Rojas-Reyes 2012](#)) or this 2026 update. Most of these changes were introduced to reflect current developments in Cochrane methodology and to comply with Cochrane quality standards.

1. In order to identify all potentially relevant studies, we searched for quasi-RCTs or cluster trials in addition to RCTs (2012 version).
2. We used a broader definition of eligible infants (all preterm infants) since, with newer administration techniques, the threshold for eligibility has changed (2012 version).
3. We included two subgroups: one according to whether mothers received antenatal steroids and the other according to the use of CPAP for early stabilisation of infants after birth (2012 version).
4. We included two additional subgroups: one on the FiO₂ threshold used to apply selective treatment and one on the method of surfactant administration (2026 update).
5. Searches were translated from original interfaces (Pubmed; Embase.com; CENTRAL via Cochrane Library) to OVID and Cochrane CRS. We did not conduct independent searches of conference abstracts, and relied upon those found in Embase and CENTRAL (2026 update).
6. Primary outcomes were limited to chronic lung disease, mortality prior to hospital discharge, moderate-severe neurodevelopmental impairment and pneumothorax; and the remaining outcomes were classified as secondary (2026 update).
7. BPD at 28 days has been superseded by the current definition of BPD at 28 to 30 days of age (2026 update).
8. We include the summary of findings and the assessment of the certainty of the evidence for each outcome following the GRADE approach. A summary of findings table for the main comparison was included using [GRADEpro GDT](#) (2026 update).

To align with current Cochrane recommendations, we revisited and clarified the following review sections in this update.

1. Methods for data collection and analysis
2. Description of the unit-of-analysis issue.
3. Information on how we dealt with missing data
4. Assessment of reporting biases
5. Plans for sensitivity analysis